Pontificia Universidad Católica del Perú
Facultad de Ciencias e Ingeniería
Departamento Académico de Ingeniería

1INF52 Deep Learning

Principios de arquitecturas convolucionales para la visión: más allá de la clasificación

Prof. Pablo Fonseca

Outline

- Repaso
- Global Average Pooling
- Skip Connections: Resnets vs DenseNets
- Segmentación Semántica y U-net
- Detección de objetos

Pregunta: ¿Cuál es el shape resultante?

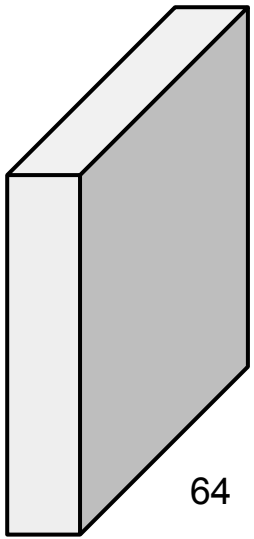


Imagen de entrada

```
Conv2D(16, 1,
activation='relu',
padding='same')
```

A) (64, 64, 1)

B) (64, 64, 8)

C) (64, 64, 16)

D) (64, 64, 8, 16)

Pregunta: ¿Cuál es el shape resultante?

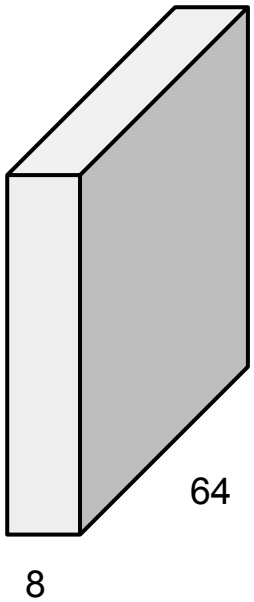


Imagen de entrada

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activation='relu',
padding='same')
```

A) (64, 64, 1)

B) (64, 64, 8)

C) (64, 64, 16)

D) (64, 64, 8, 16)

Pregunta: ¿Cuál es el shape de (cada) filtro?

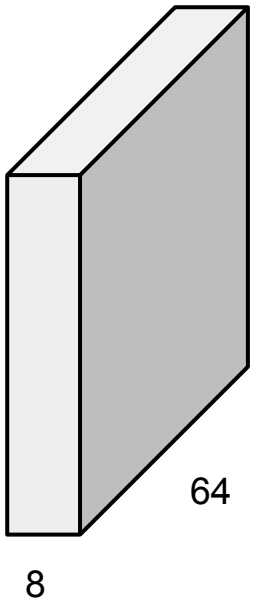


Imagen de entrada

```
Conv2D(16, 1,
activation='relu',
padding='same')
```

A) (1, 1, 8)

B) (1, 1, 16)

C) (8, 8, 16)

D) (64, 64, 1)

Pregunta: ¿Cuál es el shape de (cada) filtro?

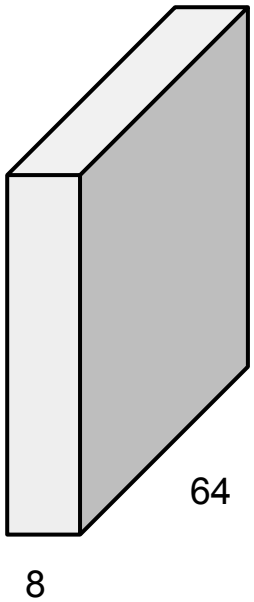


Imagen de entrada

```
Conv2D(16, 1,
activation='relu',
padding='same')
```

A) (1, 1, 8)

B) (1, 1, 16)

C) (8, 8, 16)

D) (64, 64, 1)

Pregunta: ¿Cuántos filtros 1x1x8 se están aplicando?

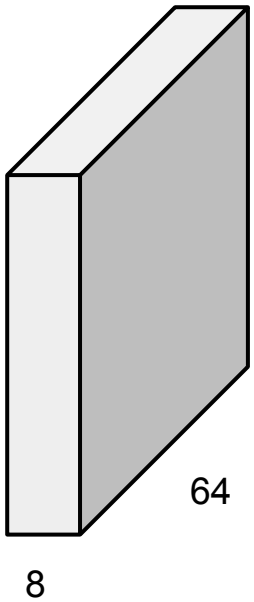


Imagen de entrada

```
Conv2D(16, 1,
activation='relu',
padding='same')
```

A) 1

B) 8

C) 16

D) 64

Pregunta: ¿Cuántos filtros 1x1x8 se están aplicando?

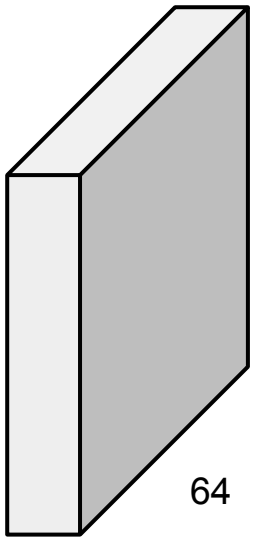


Imagen de entrada

```
Conv2D(16, 1,
activation='relu',
padding='same')
```

A) 1

B) 8

C) 16

D) 64

Pregunta: ¿Cuál es el shape resultante?

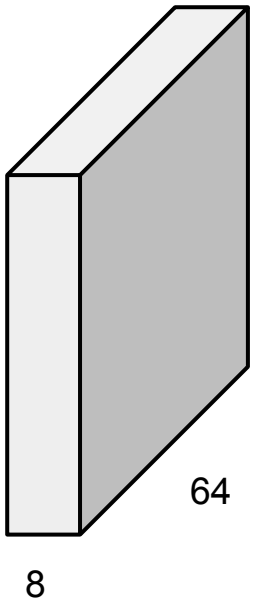


Imagen de entrada

`MaxPooling2D(2)`

A) (1, 1, 8)

B) (64, 64, 4)

C) (32, 32, 8)

D) (32, 32, 4)

Pregunta: ¿Cuál es el shape resultante?

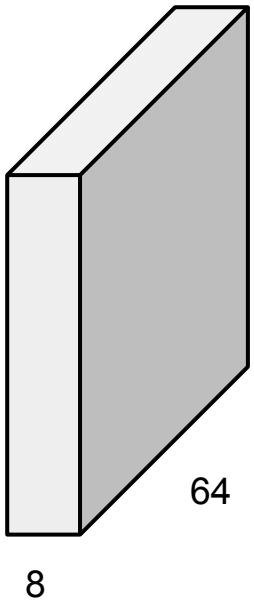


Imagen de entrada

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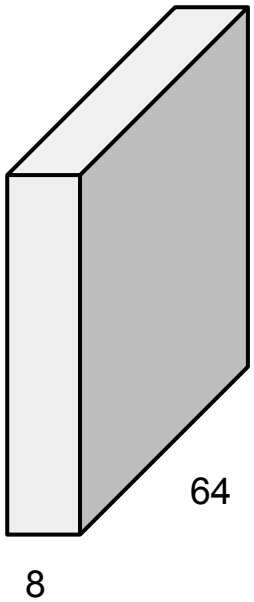


Imagen de entrada

`AvgPooling2D(2)`

A) (1, 1, 8)

B) (64, 64, 4)

C) (32, 32, 8)

D) (32, 32, 4)

Pregunta: ¿Cuál es el shape resultante?

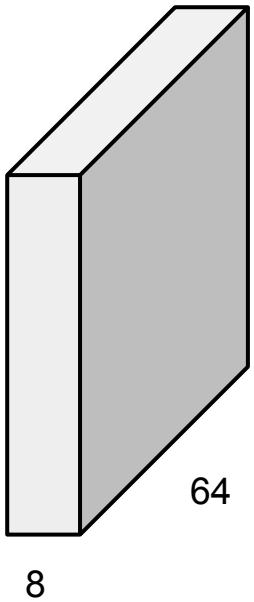


Imagen de entrada

`AvgPooling2D(2)`

A) (1, 1, 8)

B) (64, 64, 4)

C) (32, 32, 8)

D) (32, 32, 4)

Pregunta: ¿Cuál es el shape resultante?

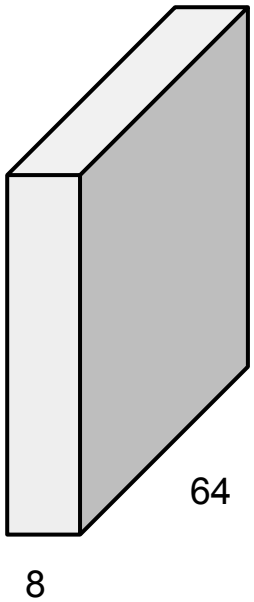


Imagen de entrada

`AvgPooling2D(64)`

A) (1, 1, 8)

B) (64, 64, 4)

C) (32, 32, 8)

D) (32, 32, 4)

Pregunta: ¿Cuál es el shape resultante?

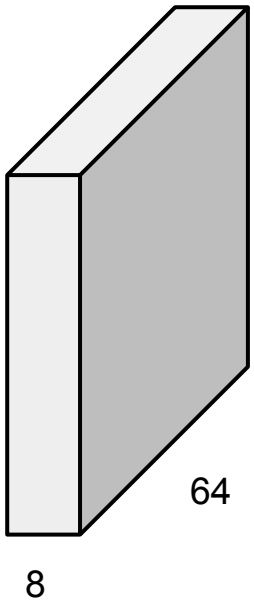


Imagen de entrada

`AvgPooling2D(64)`

A) (1, 1, 8)

B) (64, 64, 4)

C) (32, 32, 8)

D) (32, 32, 4)

Pregunta: ¿Cuál es el shape resultante?

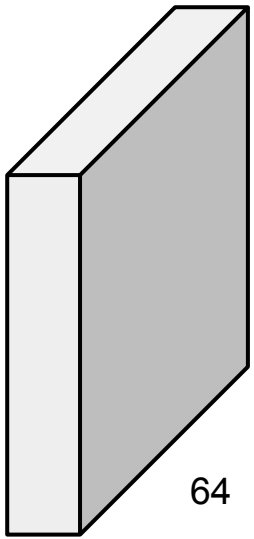


Imagen de entrada

`UpSampling2D(64)`

A) (64, 64, 16)

B) (32, 32, 8)

C) (128, 128, 1)

D) (128, 128, 8)

Pregunta: ¿Cuál es el shape resultante?

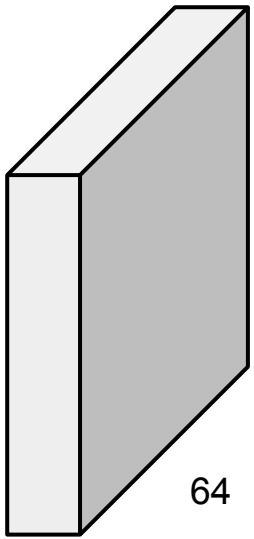


Imagen de entrada

`UpSampling2D(64)`

A) (64, 64, 16)

B) (32, 32, 8)

C) (128, 128, 1)

D) (128, 128, 8)

Pregunta: ¿C

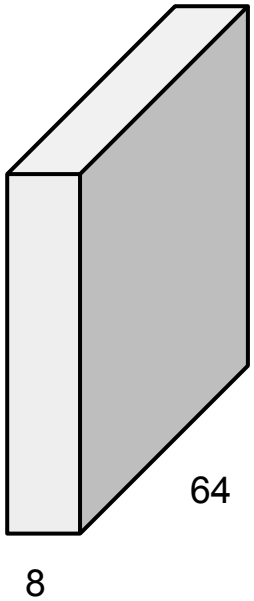


Imagen de entrada

▶ Keras 3 API documentation / Layers API / Reshaping layers / UpSampling2D layer

UpSampling2D layer

UpSampling2D class [source]

```
keras.layers.UpSampling2D(
    size=(2, 2), data_format=None, interpolation="nearest", **kwargs
)
```

Upsampling layer for 2D inputs.

The implementation uses interpolative resizing, given the resize method (specified by the `interpolation` argument). Use `interpolation=nearest` to repeat the rows and columns of the data.

Example

```
>>> input_shape = (2, 2, 1, 3)
>>> x = np.arange(np.prod(input_shape)).reshape(input_shape)
>>> print(x)
[[[[ 0  1  2]]
  [[ 3  4  5]]]
 [[[ 6  7  8]]
  [[ 9 10 11]]]]
>>> y = keras.layers.UpSampling2D(size=(1, 2))(x)
>>> print(y)
[[[[ 0  1  2]
   [ 0  1  2]]
  [[ 3  4  5]
   [ 3  4  5]]]
 [[[ 6  7  8]
   [ 6  7  8]]
  [[ 9 10 11]
   [ 9 10 11]]]]
```

Arguments

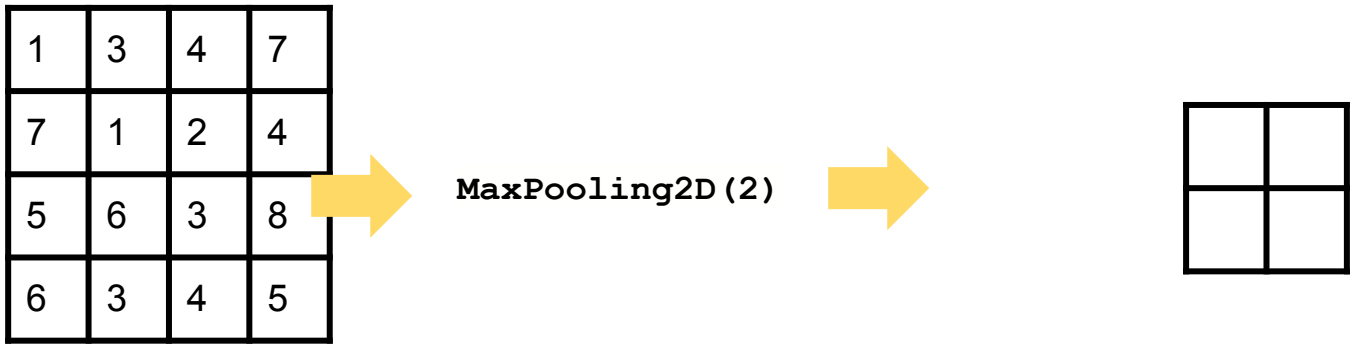
, 16)

, 8)

28, 1)

28, 8)

Pregunta: ¿Cuál es la capa resultante?



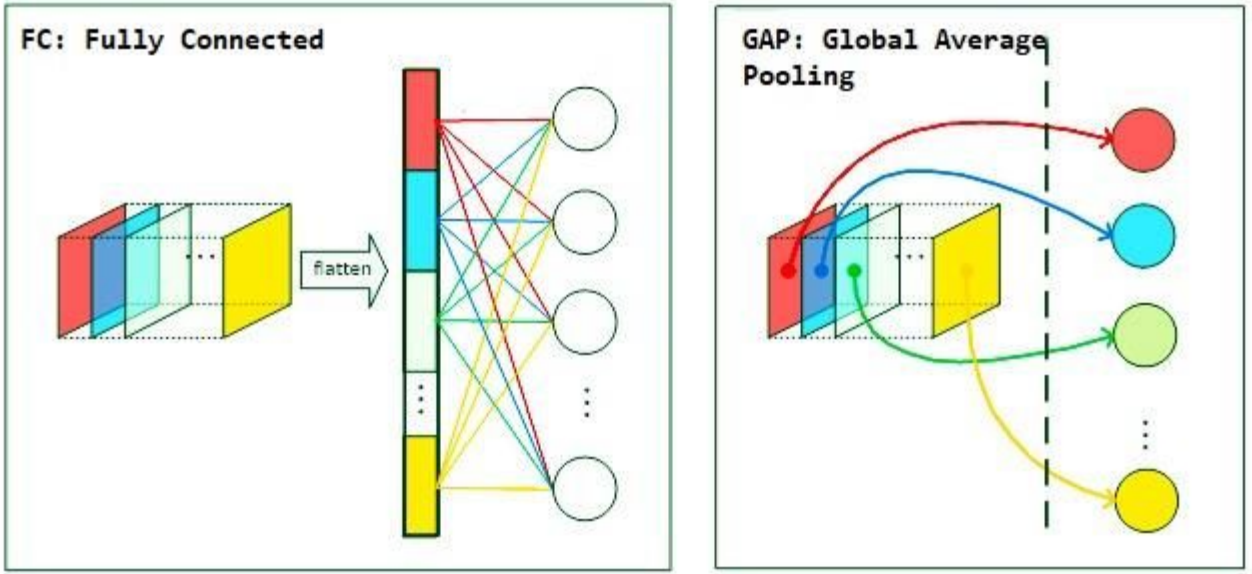
Pregunta: ¿Cuál es la capa resultante?



Pregunta: ¿Cómo hacemos clasificación sin una capa totalmente conectada?

Pregunta: ¿Cuáles son las ventajas de una red totalmente convolucional?

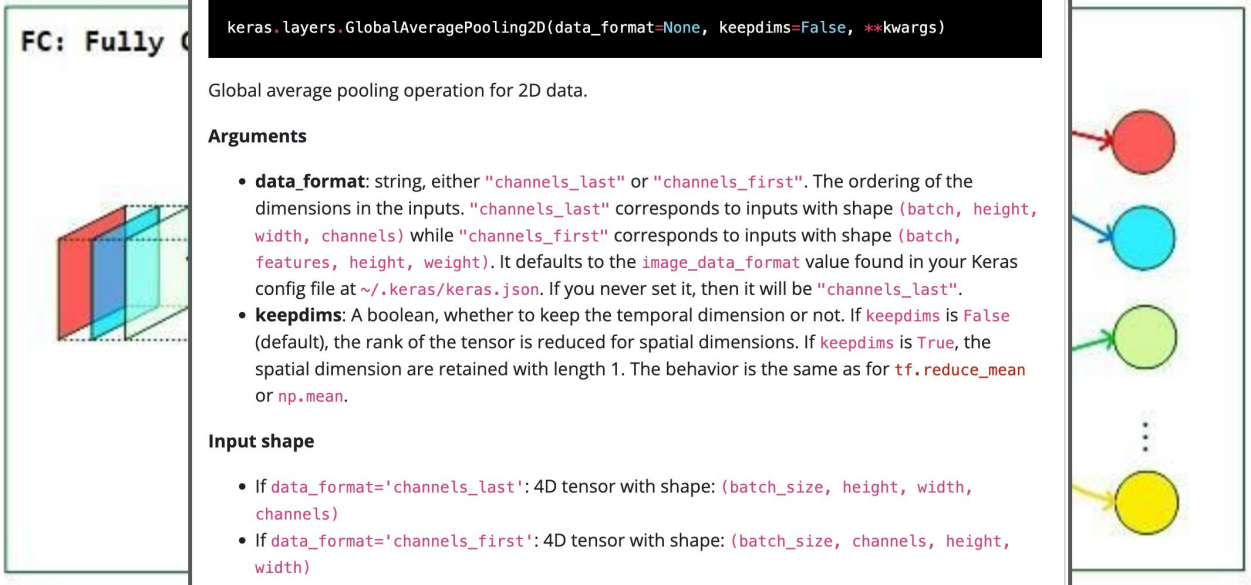
Global average pooling layer



https://www.programmersought.com/article/38525223979/

Lin, M., Chen, Q., & Yan, S. (2013). Network in network. *arXiv preprint arXiv:1312.4400*. https://arxiv.org/abs/1312.4400

Global averag



▶ Keras 3 API documentation / Layers API / Pooling layers / GlobalAveragePooling2D layer

GlobalAveragePooling2D layer

`GlobalAveragePooling2D` class [source]

```
keras.layers.GlobalAveragePooling2D(data_format=None, keepdims=False, **kwargs)
```

Global average pooling operation for 2D data.

Arguments

- **data_format**: string, either `"channels_last"` or `"channels_first"`. The ordering of the dimensions in the inputs. `"channels_last"` corresponds to inputs with shape `(batch, height, width, channels)` while `"channels_first"` corresponds to inputs with shape `(batch, features, height, weight)`. It defaults to the `image_data_format` value found in your Keras config file at `~/.keras/keras.json`. If you never set it, then it will be `"channels_last"`.
- **keepdims**: A boolean, whether to keep the temporal dimension or not. If `keepdims` is `False` (default), the rank of the tensor is reduced for spatial dimensions. If `keepdims` is `True`, the spatial dimension are retained with length 1. The behavior is the same as for `tf.reduce_mean` or `np.mean`.

Input shape

- If `data_format='channels_last'`: 4D tensor with shape: `(batch_size, height, width, channels)`
- If `data_format='channels_first'`: 4D tensor with shape: `(batch_size, channels, height, width)`

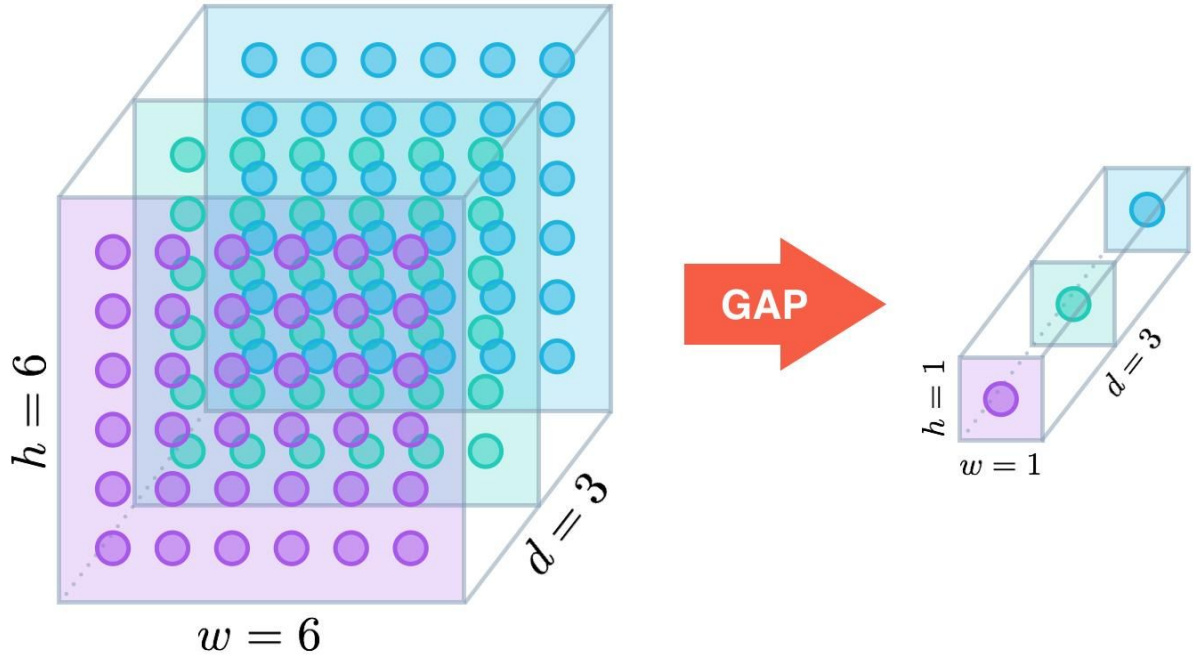
Output shape

- If `keepdims=False`: 2D tensor with shape `(batch_size, channels)`.
- If `keepdims=True`: - If `data_format="channels_last"`: 4D tensor with shape `(batch_size, 1, 1, channels)` - If `data_format="channels_first"`: 4D tensor with shape `(batch_size, channels, 1, 1)`

https://www.programmersought.com/article/38525223979/

Lin, M., Chen, Q., & Yan, S. (20 ps/1312.4400

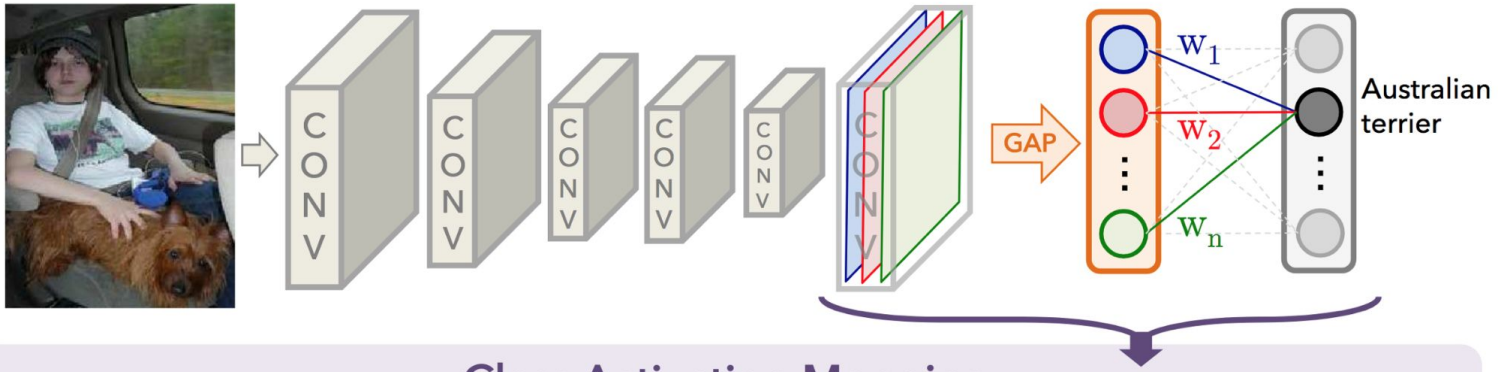
Global average pooling layer

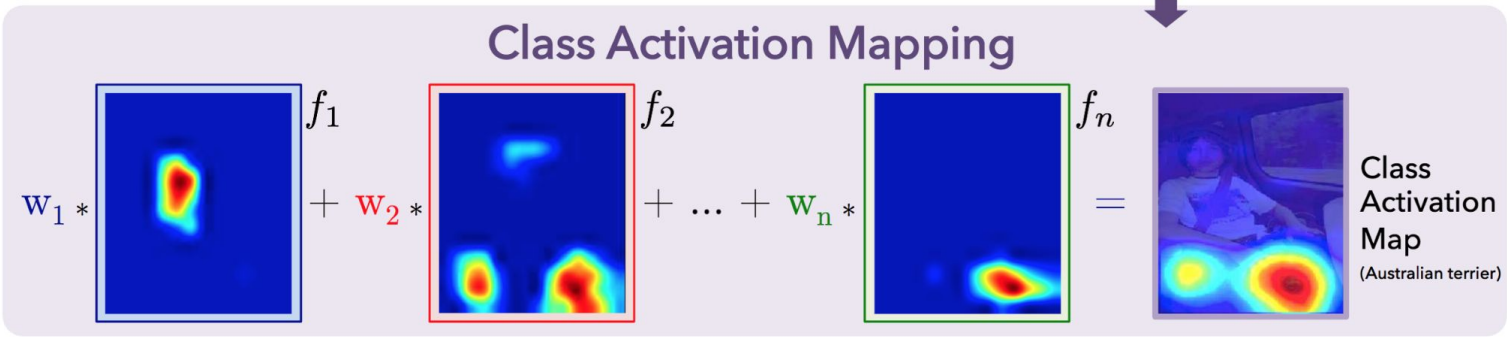


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Lin, M., Chen, Q., & Yan, S. (2013). Network in network. *arXiv preprint arXiv:1312.4400*. https://arxiv.org/abs/1312.4400

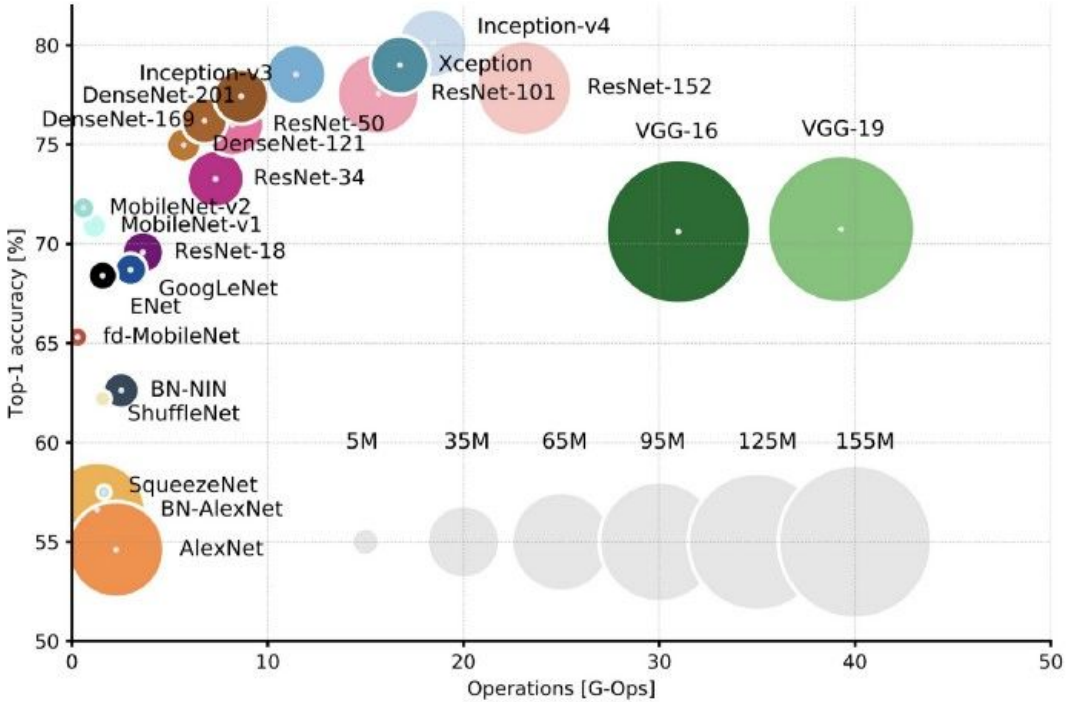
Global average pooling layer





https://alexisbcook.github.io/2017/global-average-pooling-layers-for-object-localization/

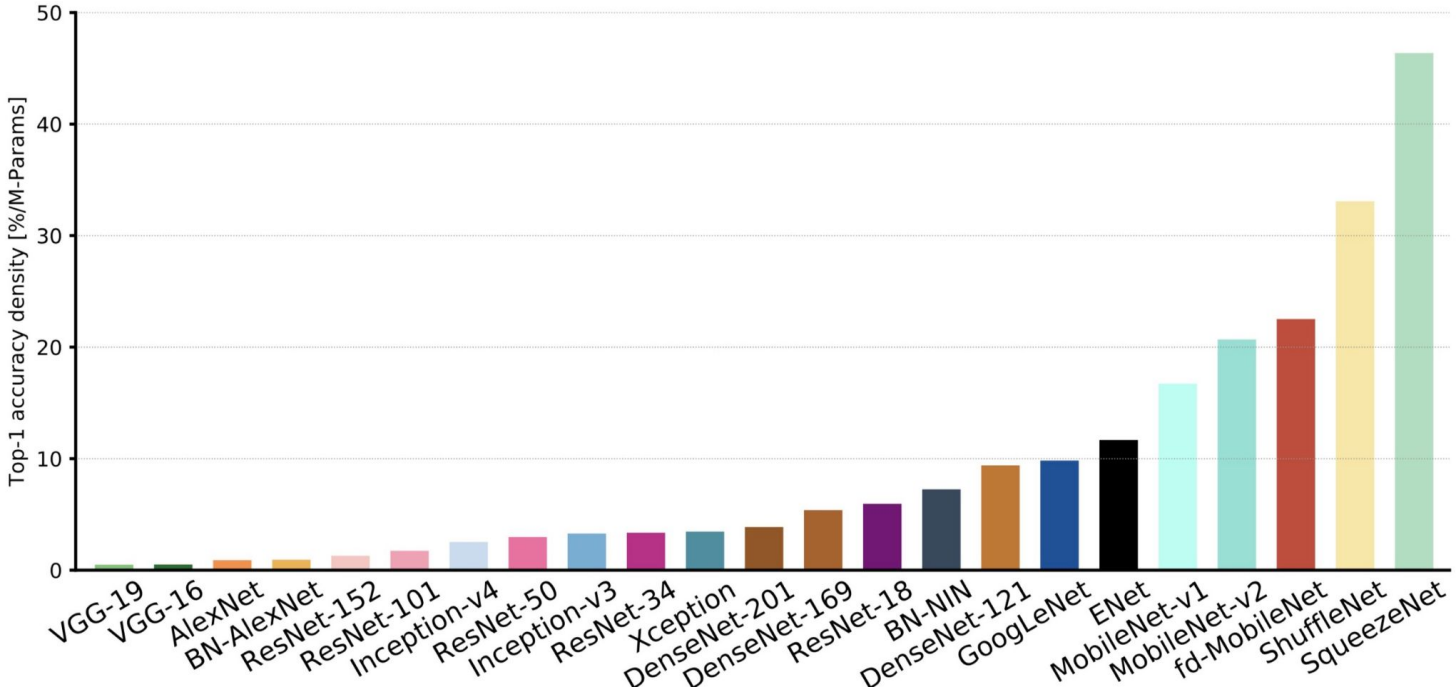
Comparación de arquitecturas



Canziani, A., Paszke, A., & Culurciello, E. (2016). An analysis of deep neural network models for practical applications. *arXiv preprint arXiv:1605.07678*.

https://culurciello.medium.com/analysis-of-deep-neural-networks-dcf398e71aae updated Figure 2: Top1 vs. operations, size ∝ parameters. Top-1 one-crop accuracy versus amount of operations required for a single forward pass. The size of the blobs is proportional to the number of network parameters; a legend is reported in the bottom right corner, spanning from 5×10^6 to 155×10^6 params. Both these figures share the same y-axis, and the grey dots highlight the centre of the blobs.

Eficiencia de las arquitecturas



Canziani, A., Paszke, A., & Culurciello, E. (2016). An analysis of deep neural network models for practical applications. *arXiv preprint arXiv:1605.07678*.

https://culurciello.medium.com/analysis-of-deep-neural-networks-dcf398e71aae updated Figure 10: Accuracy per parameter vs. network. Information density (accuracy per parameters) is an efficiency metric that highlight that capacity of a specific architecture to better utilise its parametric space. Models like VGG and AlexNet are clearly oversized, and do not take fully advantage of their potential learning ability. On the far right, ResNet-18, BN-NIN, GoogLeNet and ENet (marked by grey arrows) do a better job at “squeezing” all their neurons to learn the given task, and are the winners of this section.

Modelos pre-entrenados

Ver más:
https://keras.io/api/applications/

Model	Size	Top-1 Accuracy	Top-5 Accuracy	Parameters	Depth
Xception	88 MB	0.790	0.945	22,910,480	126
VGG16	528 MB	0.713	0.901	138,357,544	23
VGG19	549 MB	0.713	0.900	143,667,240	26
ResNet50	98 MB	0.749	0.921	25,636,712	-
ResNet101	171 MB	0.764	0.928	44,707,176	-
ResNet152	232 MB	0.766	0.931	60,419,944	-
ResNet50V2	98 MB	0.760	0.930	25,613,800	-
ResNet101V2	171 MB	0.772	0.938	44,675,560	-
ResNet152V2	232 MB	0.780	0.942	60,380,648	-
InceptionV3	92 MB	0.779	0.937	23,851,784	159
InceptionResNetV2	215 MB	0.803	0.953	55,873,736	572
MobileNet	16 MB	0.704	0.895	4,253,864	88
MobileNetV2	14 MB	0.713	0.901	3,538,984	88
DenseNet121	33 MB	0.750	0.923	8,062,504	121
DenseNet169	57 MB	0.762	0.932	14,307,880	169
DenseNet201	80 MB	0.773	0.936	20,242,984	201
NASNetMobile	23 MB	0.744	0.919	5,326,716	-
NASNetLarge	343 MB	0.825	0.960	88,949,818	-
EfficientNetB0	29 MB	-	-	5,330,571	-
EfficientNetB1	31 MB	-	-	7,856,239	-
EfficientNetB2	36 MB	-	-	9,177,569	-
EfficientNetB3	48 MB	-	-	12,320,535	-
EfficientNetB4	75 MB	-	-	19,466,823	-
EfficientNetB5	118 MB	-	-	30,562,527	-
EfficientNetB6	166 MB	-	-	43,265,143	-
EfficientNetB7	256 MB	-	-	66,658,687	-

Modelos pre-entrenados

Ver más:
https://pytorch.org/vision/stable/models.html

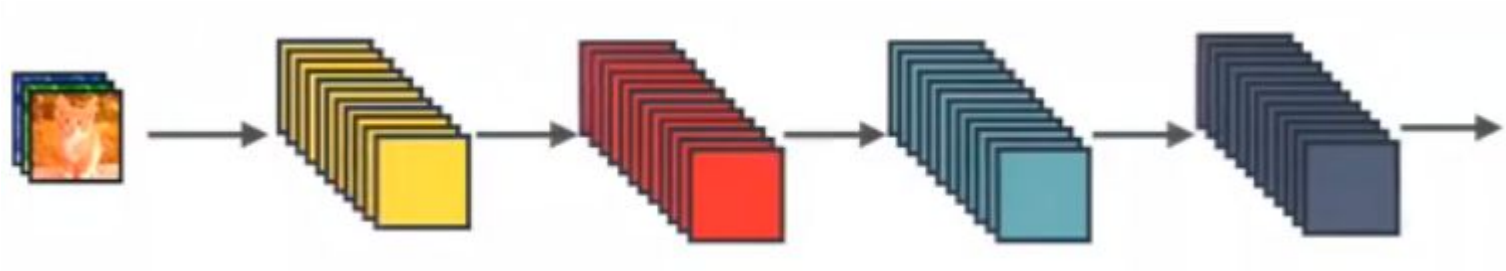
Table of all available classification weights

Accuracies are reported on ImageNet-1K using single crops:

Weight	Acc@1	Acc@5	Params	GFLOPS	Recipe
AlexNet_Weights.IMAGENET1K_V1	56.522	79.066	61.1M	0.71	link
ConvNeXt_Base_Weights.IMAGENET1K_V1	84.062	96.87	88.6M	15.36	link
ConvNeXt_Large_Weights.IMAGENET1K_V1	84.414	96.976	197.8M	34.36	link
ConvNeXt_Small_Weights.IMAGENET1K_V1	83.616	96.65	50.2M	8.68	link
ConvNeXt_Tiny_Weights.IMAGENET1K_V1	82.52	96.146	28.6M	4.46	link
DenseNet121_Weights.IMAGENET1K_V1	74.434	91.972	8.0M	2.83	link
DenseNet161_Weights.IMAGENET1K_V1	77.138	93.56	28.7M	7.73	link
DenseNet169_Weights.IMAGENET1K_V1	75.6	92.806	14.1M	3.36	link
DenseNet201_Weights.IMAGENET1K_V1	76.896	93.37	20.0M	4.29	link
EfficientNet_B0_Weights.IMAGENET1K_V1	77.692	93.532	5.3M	0.39	link
EfficientNet_B1_Weights.IMAGENET1K_V1	78.642	94.186	7.8M	0.69	link
EfficientNet_B1_Weights.IMAGENET1K_V2	79.838	94.934	7.8M	0.69	link
EfficientNet_B2_Weights.IMAGENET1K_V1	80.608	95.31	9.1M	1.09	link
EfficientNet_B3_Weights.IMAGENET1K_V1	82.008	96.054	12.2M	1.83	link
EfficientNet_B4_Weights.IMAGENET1K_V1	83.384	96.594	19.3M	4.39	link
EfficientNet_B5_Weights.IMAGENET1K_V1	83.444	96.628	30.4M	10.27	link
EfficientNet_B6_Weights.IMAGENET1K_V1	84.008	96.916	43.0M	19.07	link
EfficientNet_B7_Weights.IMAGENET1K_V1	84.122	96.908	66.3M	37.75	link
EfficientNet_V2_L_Weights.IMAGENET1K_V1	85.808	97.788	118.5M	56.08	link
EfficientNet_V2_M_Weights.IMAGENET1K_V1	85.112	97.156	54.1M	24.58	link
EfficientNet_V2_S_Weights.IMAGENET1K_V1	84.228	96.878	21.5M	8.37	link
GoogLeNet_Weights.IMAGENET1K_V1	69.778	89.53	6.6M	1.5	link
Inception_V3_Weights.IMAGENET1K_V1	77.294	93.45	27.2M	5.71	link
MNASNet0_5_Weights.IMAGENET1K_V1	67.734	87.49	2.2M	0.1	link
MNASNet0_75_Weights.IMAGENET1K_V1	71.18	90.496	3.2M	0.21	link
MNASNet1_0_Weights.IMAGENET1K_V1	73.456	91.51	4.4M	0.31	link
MNASNet1_3_Weights.IMAGENET1K_V1	76.506	93.522	6.3M	0.53	link
MaxVit_T_Weights.IMAGENET1K_V1	83.7	96.722	30.9M	5.56	link
MobileNet_V2_Weights.IMAGENET1K_V1	71.878	90.286	3.5M	0.3	link
MobileNet_V2_Weights.IMAGENET1K_V2	72.154	90.822	3.5M	0.3	link
MobileNet_V3_Large_Weights.IMAGENET1K_V1	74.042	91.34	5.5M	0.22	link
MobileNet_V3_Large_Weights.IMAGENET1K_V2	75.274	92.566	5.5M	0.22	link
MobileNet_V3_Small_Weights.IMAGENET1K_V1	67.668	87.402	2.5M	0.06	link
RegNet_X_16GF_Weights.IMAGENET1K_V1	80.058	94.944	54.3M	15.94	link
RegNet_X_16GF_Weights.IMAGENET1K_V2	82.716	96.196	54.3M	15.94	link
RegNet_X_1_6GF_Weights.IMAGENET1K_V1	77.04	93.44	9.2M	1.6	link
RegNet_X_1_6GF_Weights.IMAGENET1K_V2	79.668	94.922	9.2M	1.6	link
RegNet_X_32GF_Weights.IMAGENET1K_V1	80.622	95.248	107.8M	31.74	link

Skip connections

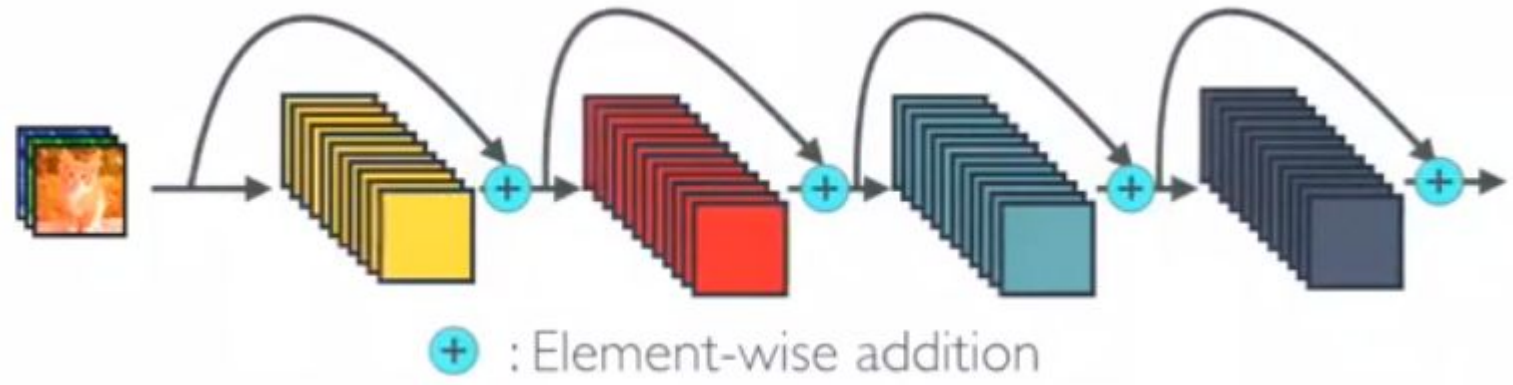
- En una red tradicional con L capas existen L conexiones, de una capa con la siguiente.



https://towardsdatascience.com/review-densenet-image-classification-b6631a8ef803

Skip connections

- En una ResNet tenemos skip connections aditivos



https://towardsdatascience.com/review-densenet-image-classification-b6631a8ef803

Skip connections: DenseNet

- En una DenseNet se tienen L(L+1)/2 conexiones, de una capa con todas sus subsecuentes

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 4700-4708).

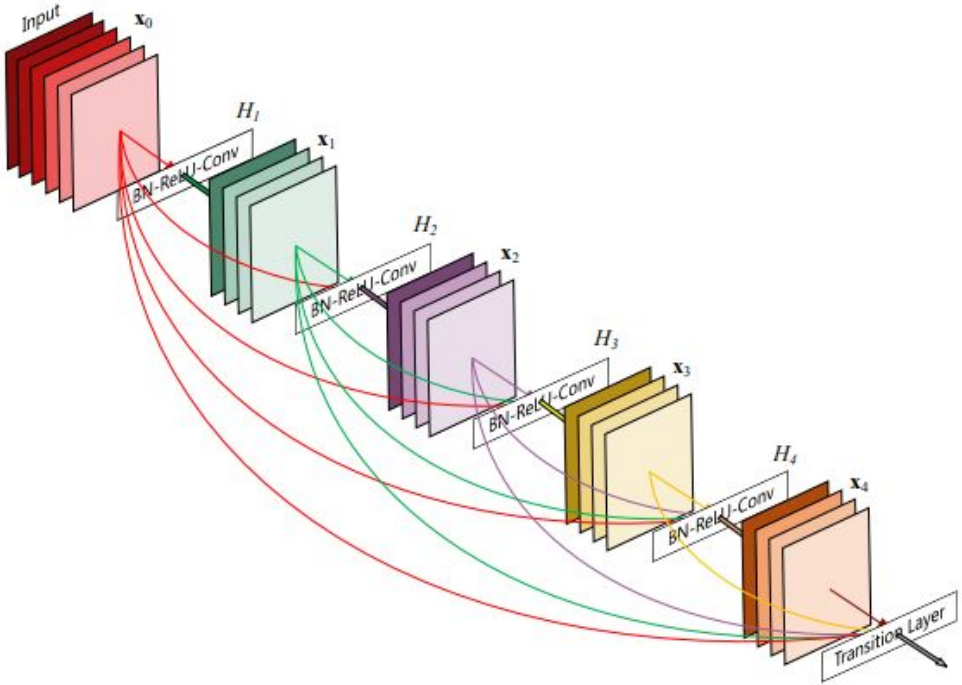
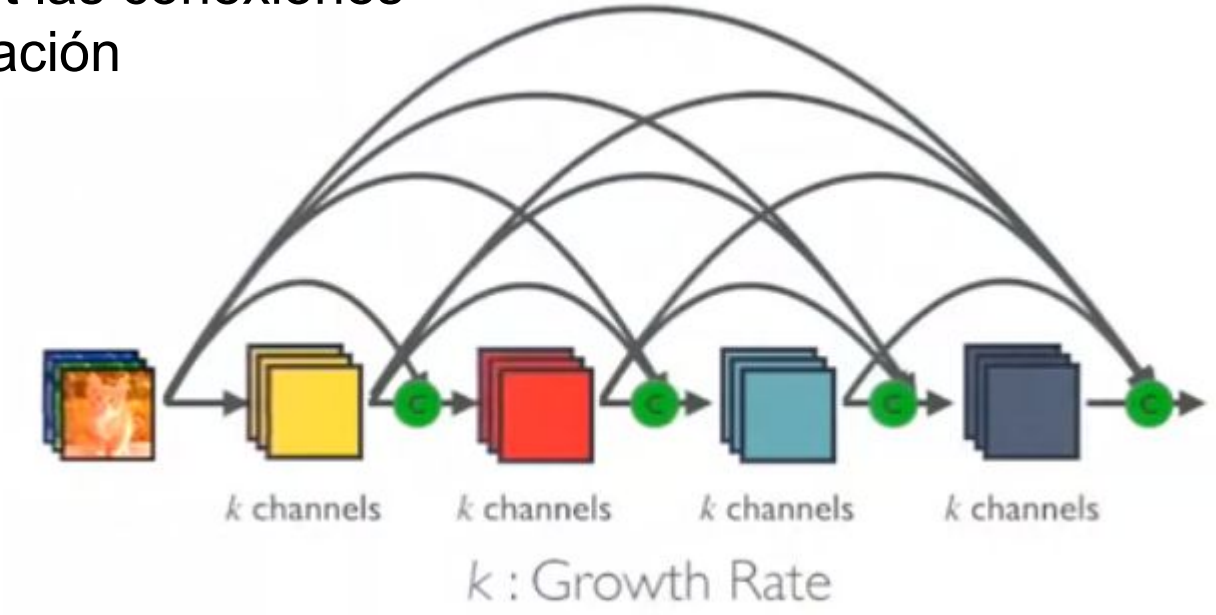


Figure 1. A 5-layer dense block with a growth rate of $k = 4$. Each layer takes all preceding feature-maps as input.

Skip Connections de concatenación

- En una DenseNet las conexiones son de concatenación



https://towardsdatascience.com/review-densenet-image-classification-b6631a8ef803

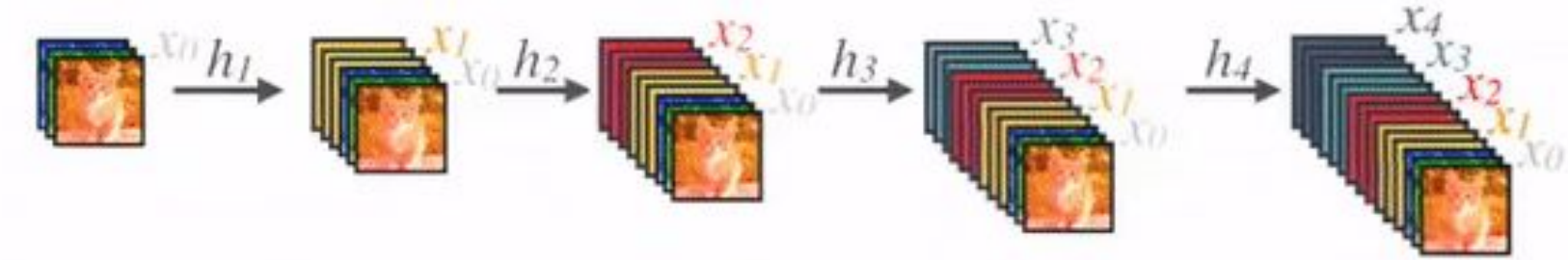
Skip Connections de concatenación

- En una DenseNet las conexiones son de concatenación



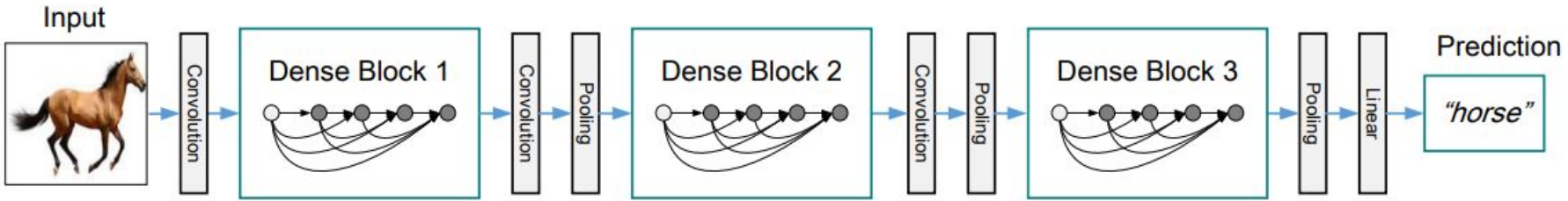
https://towardsdatascience.com/review-densenet-image-classification-b6631a8ef803

Arquitectura de las DenseNets



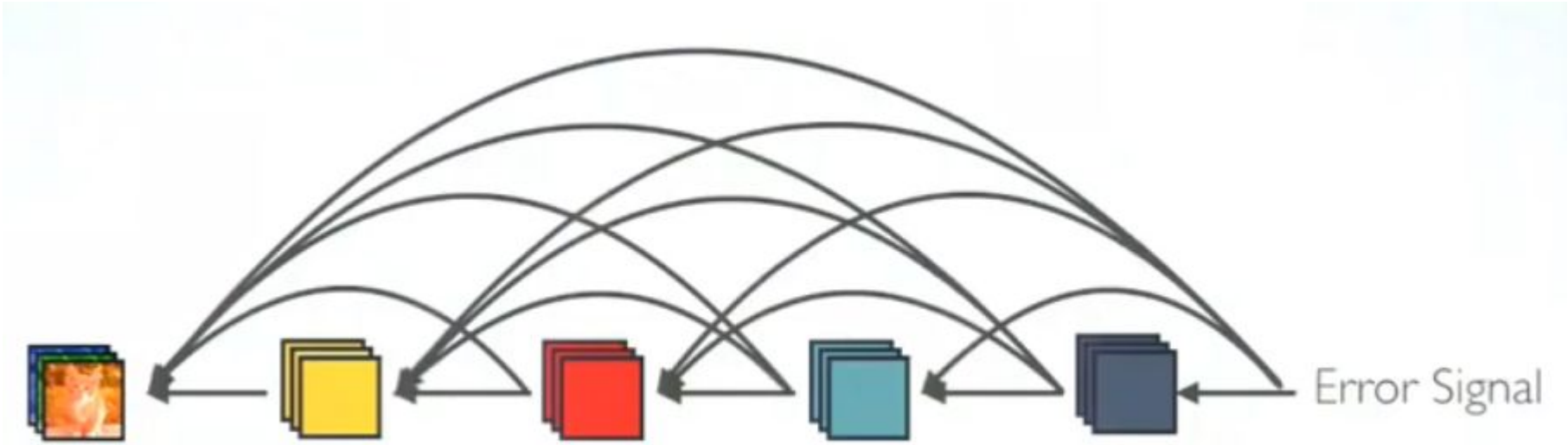
https://towardsdatascience.com/review-densenet-image-classification-b6631a8ef803

Arquitectura de las DenseNets



Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 4700-4708).

Gradient Flow



https://towardsdatascience.com/review-densenet-image-classification-b6631a8ef803

ResNet vs DenseNet

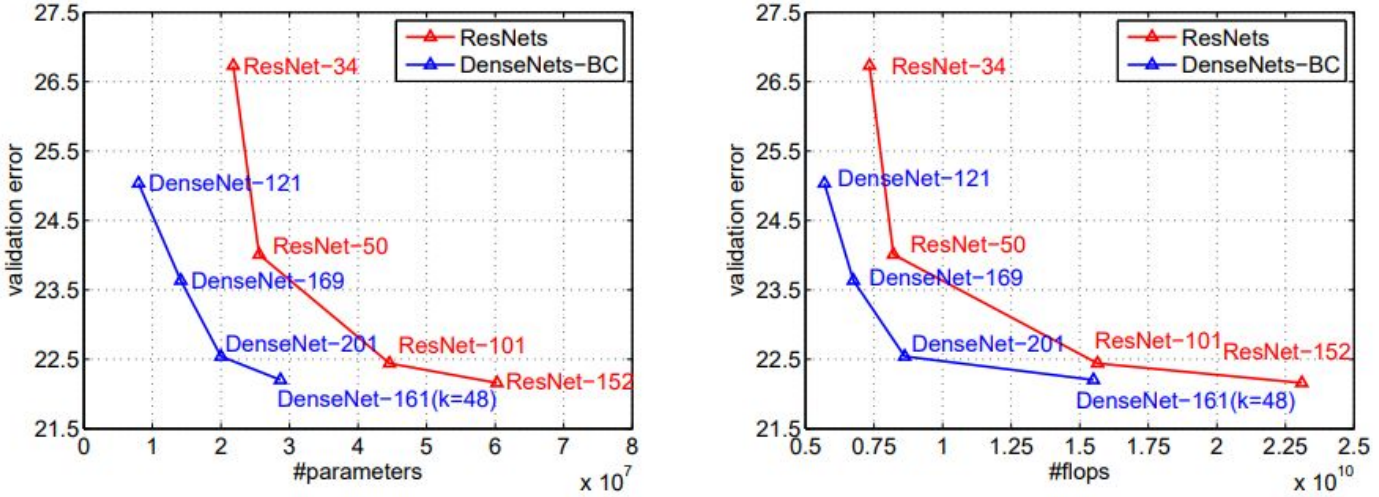
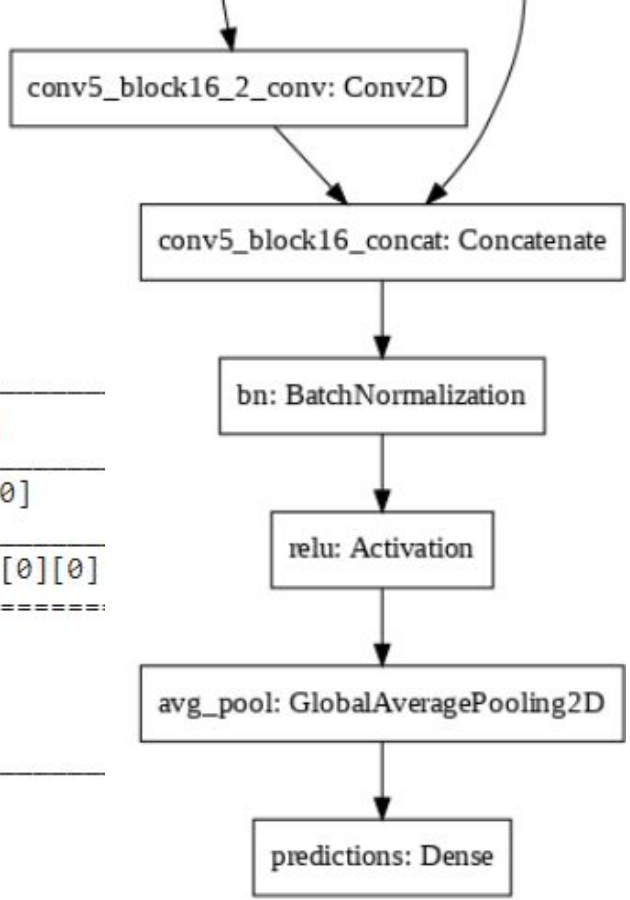


Figure 3. Comparison of the DenseNet and ResNet Top-1 (single model and single-crop) error rates on the ImageNet classification dataset as a function of learned parameters (*left*) and flops during test-time (*right*).

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 4700-4708).

Salida en DenseNet

```
relu (Activation)                (None, 7, 7, 1024)    0           bn[0][0]
avg_pool (GlobalAveragePooling2  (None, 1024)          0           relu[0][0]
predictions (Dense)              (None, 1000)          1025000     avg_pool[0][0]
==========================================================================
Total params: 8,062,504
Trainable params: 7,978,856
Non-trainable params: 83,648
```



Salida en DenseNet 121

TensorFlow + Keras



	1	2
0	African_elephant	0.679755
1	tusker	0.138115
2	Indian_elephant	0.110896
3	bison	0.067035
4	brown_bear	0.002510
5	warthog	0.000506
6	triceratops	0.000302
7	wild_boar	0.000181
8	American_black_bear	0.000110
9	Arabian_camel	0.000106

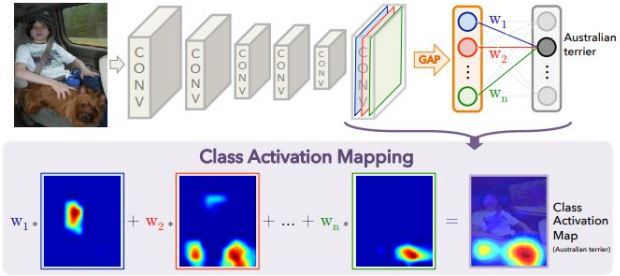
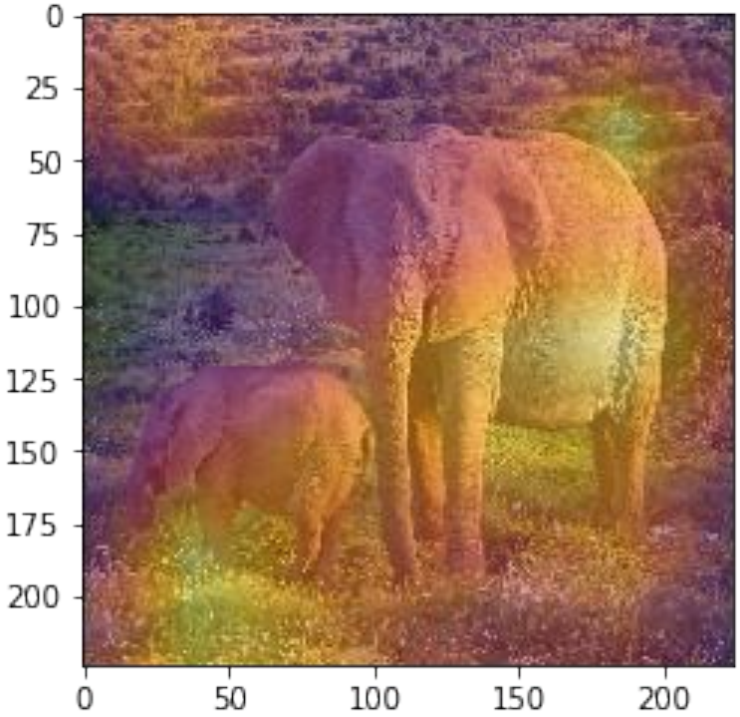


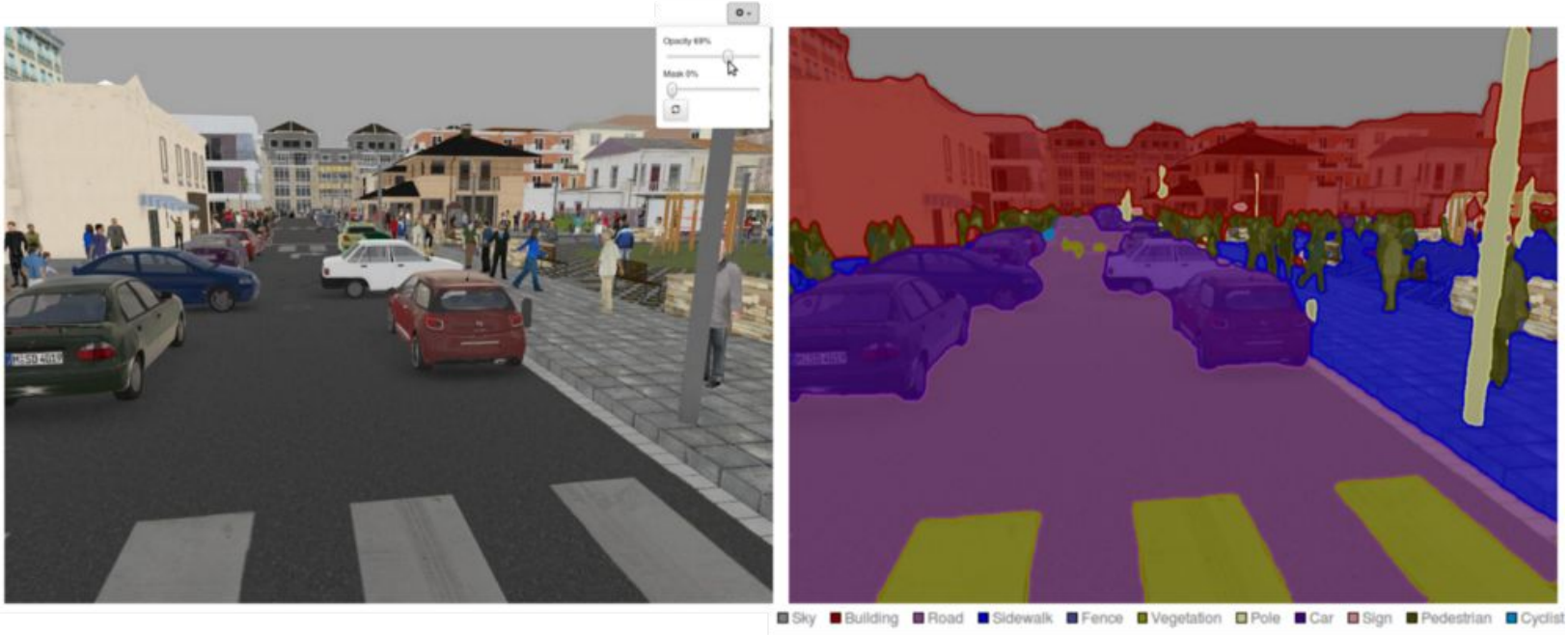
Figure 2. Class Activation Mapping: the predicted class score is mapped back to the previous convolutional layer to generate the class activation maps (CAMs). The CAM highlights the class-specific discriminative regions.



¿Qué es la segmentación semántica?

- Es un problema de clasificación de cada píxel de una imagen en una serie de clases
- Supervisado
- Imagen (entrada) -> Imagen (salida)

¿Qué es la segmentación semántica?



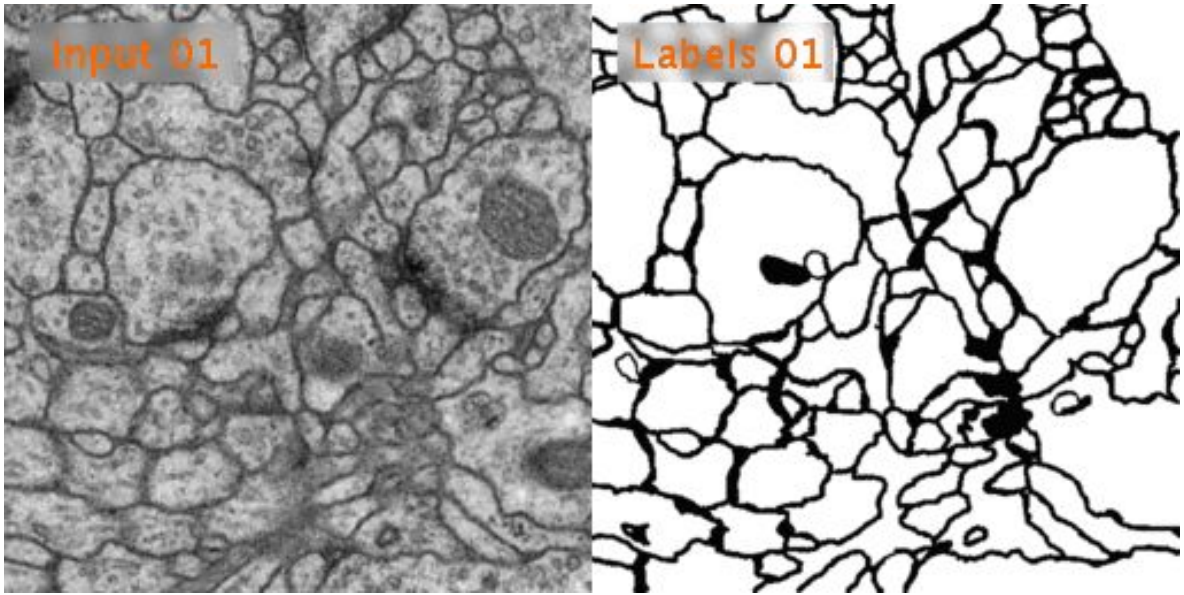
Fuente:https://blog.goodaudience.com/using-convolutional-neural-networks-for-image-segmentation-a-quick-intro-75bd68779225)

¿Qué es la segmentación semántica?



Fuente:https://www.youtube.com/watch?v=PNzQ4PNZSzc

¿Qué es la segmentación semántica?



Fuente:https://towardsdatascience.com/review-u-net-biomedical-image-segmentation-d02bf06ca760

¿Qué es la segmentación semántica?





Fuente:
Propia

Otras tareas relacionadas:

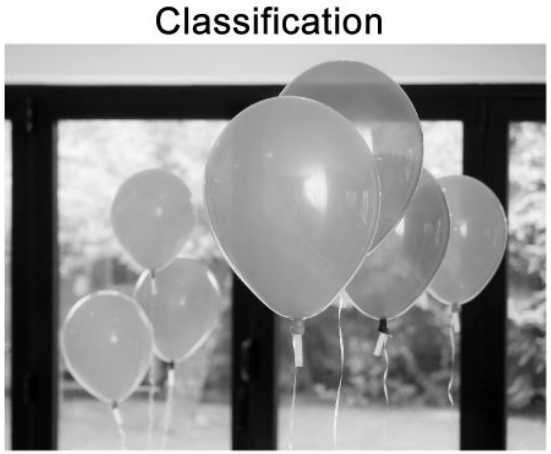
- Object Detection
- Instance Object Segmentation

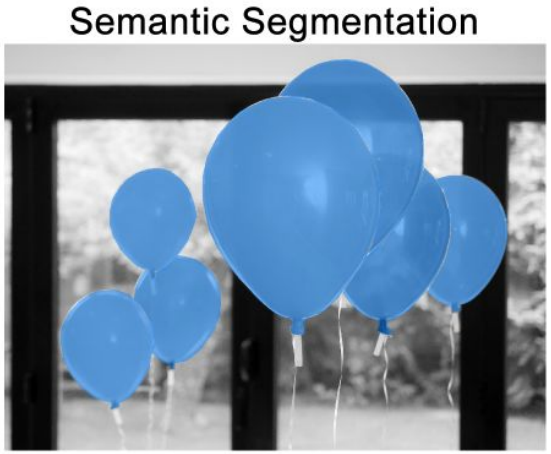
Otras tareas relacionadas:

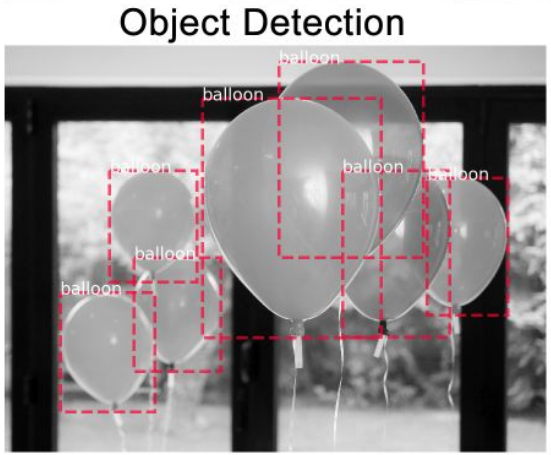
- Object Detection
- Instance Object Segmentation

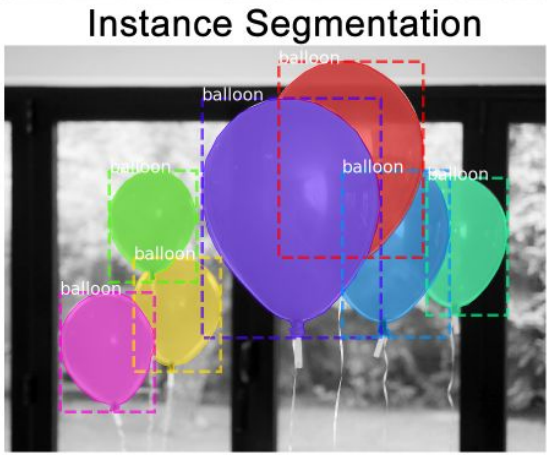
$$\frac{\text{Object Detection + Semantic Segmentation}}{\text{Instance Object Segmentation}}$$

Otras tareas relacionadas:



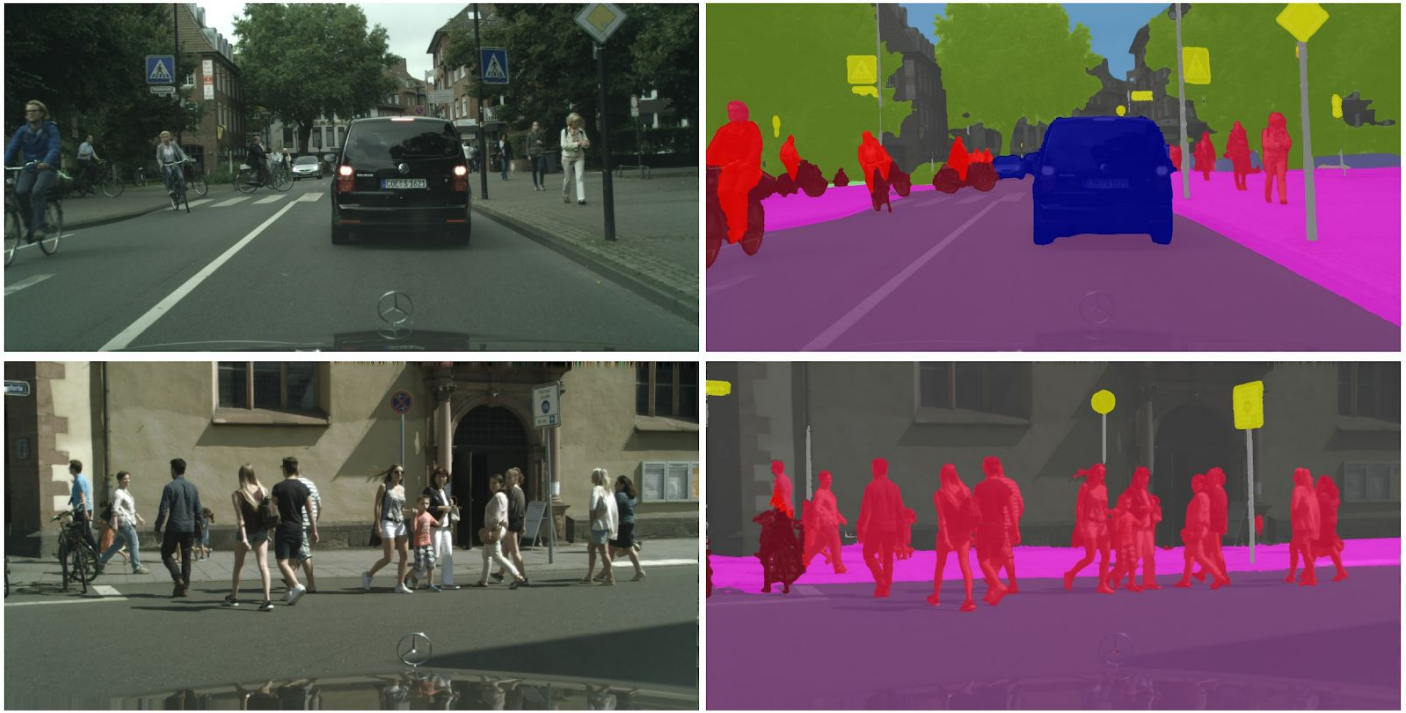






Fuente:https://engineering.matterport.com/splash-of-color-instance-segmentation-with-mask-r-cnn-and-tensorflow-7c761e238b46

Aplicaciones principales: Scene understanding



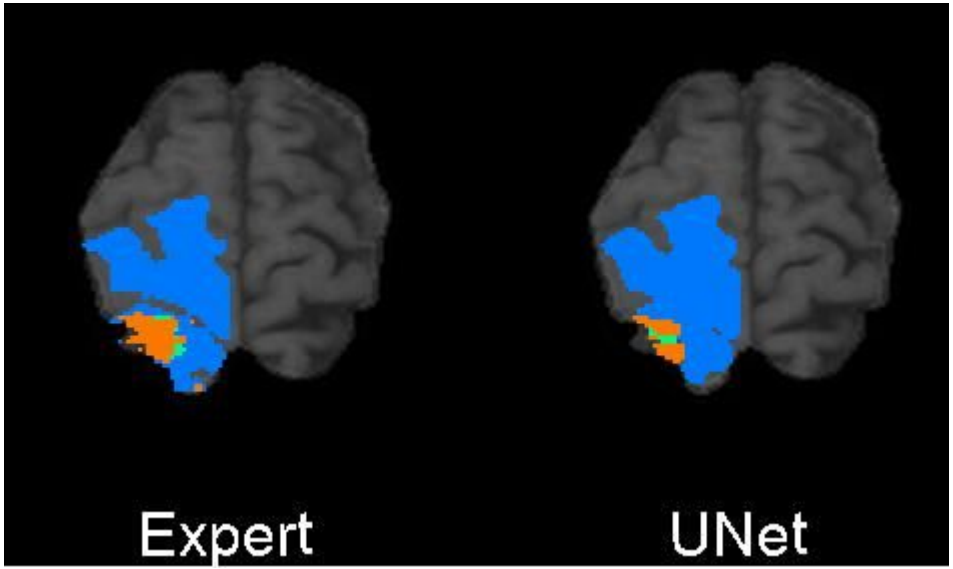
Fuente:
http://vladlen.info/projects/scene-understanding/fso-3/

Aplicaciones principales: Remote Sensing



Fuente:
https://deepsense.ai/deep-learning-for-satellite-imagery-via-image-segmentation/

Aplicaciones principales: Imágenes Médicas

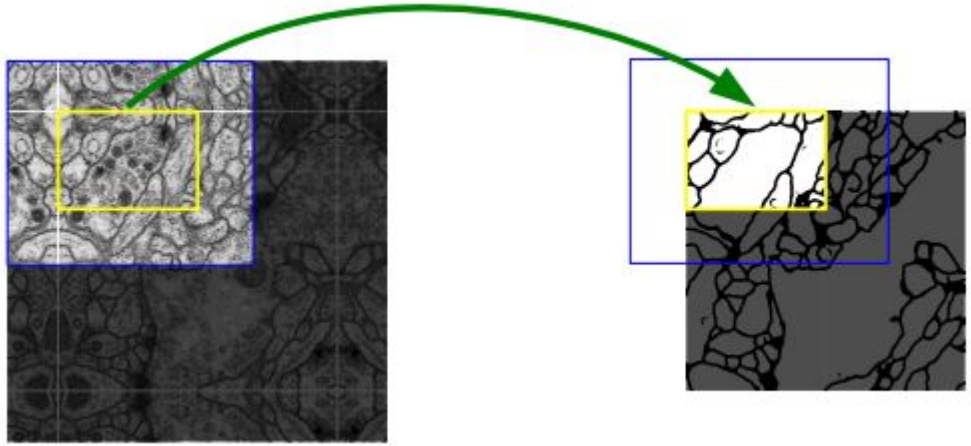


Fuente:https://pythonawesome.com/keras-3d-u-net-convolution-neural-network-designed-for-medical-image-segmentation/

U-net

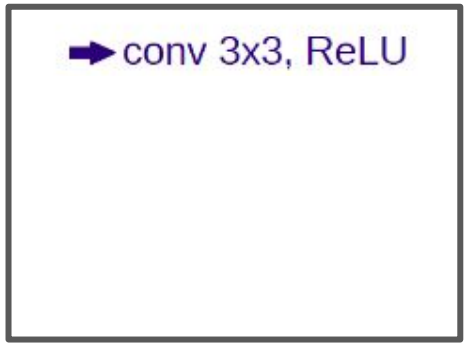
conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

- U-net es una arquitectura propuesta en el 2015
- Ideada para imágenes biomédicas
- Ganó un concurso ISBI



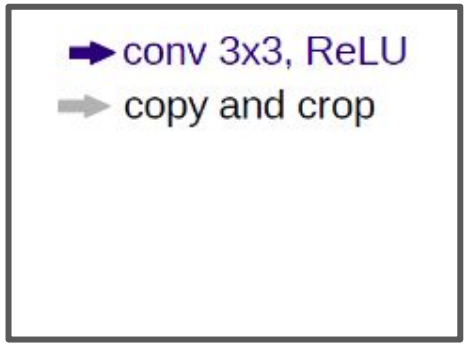
Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

U-net: Las operaciones básicas



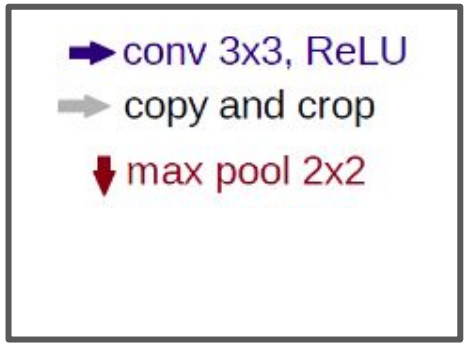
Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

U-net: Las operaciones básicas



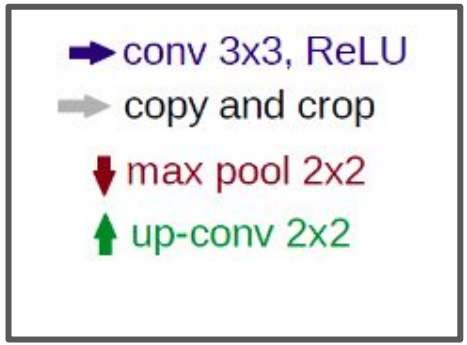
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U-net: Las operaciones básicas



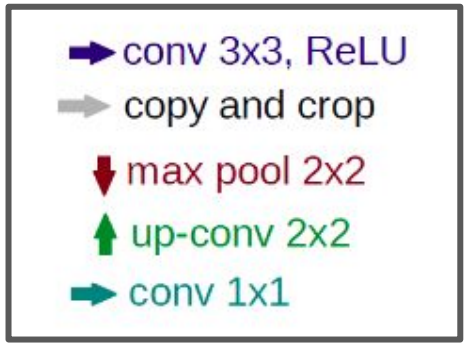
Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

U-net: Las operaciones básicas



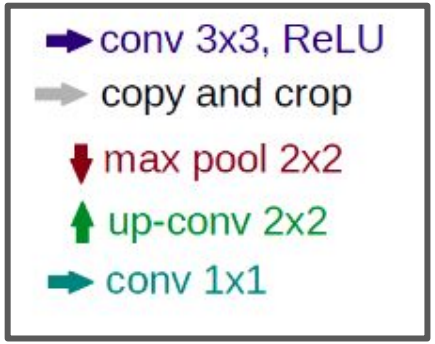
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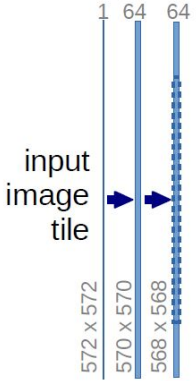
U-net: Las operaciones básicas



Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

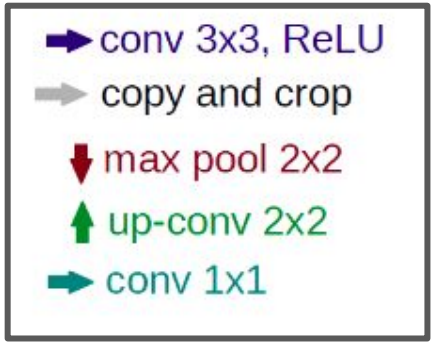
U-net

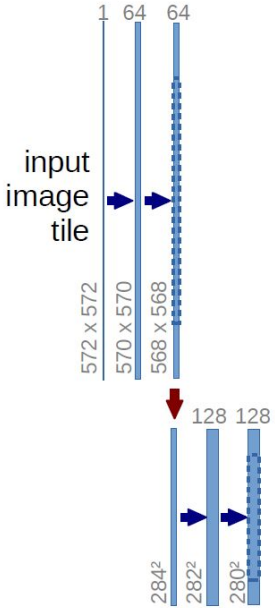




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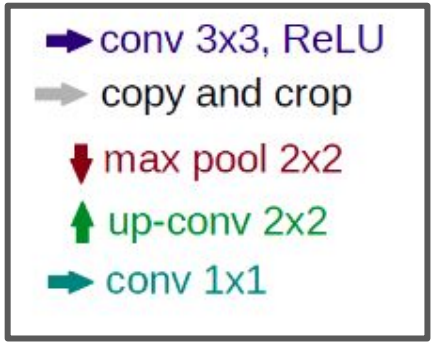
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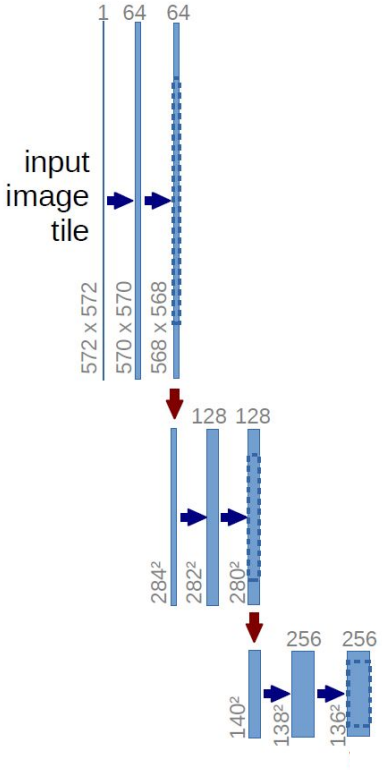




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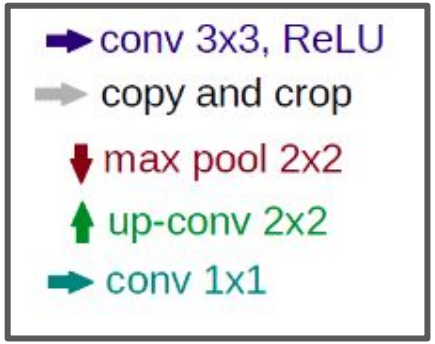
U-net

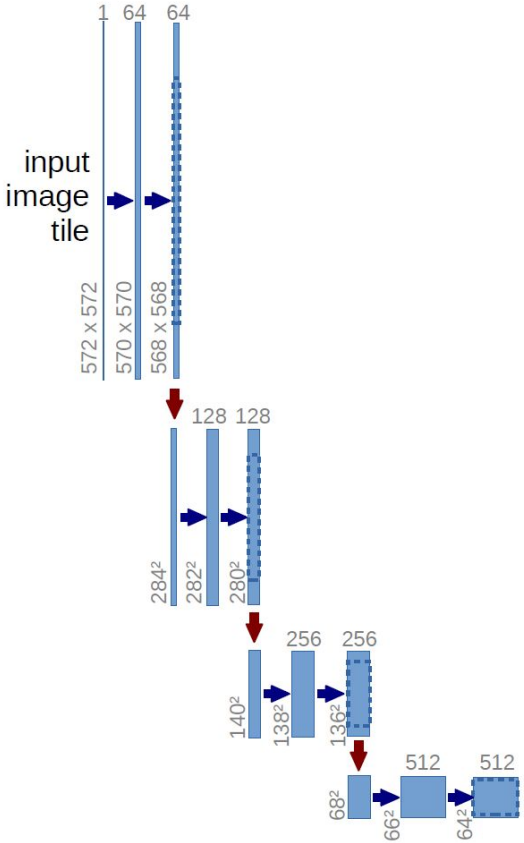




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

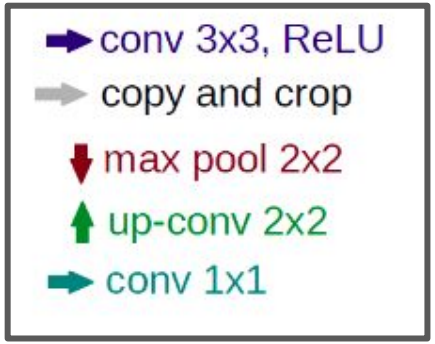
U-net

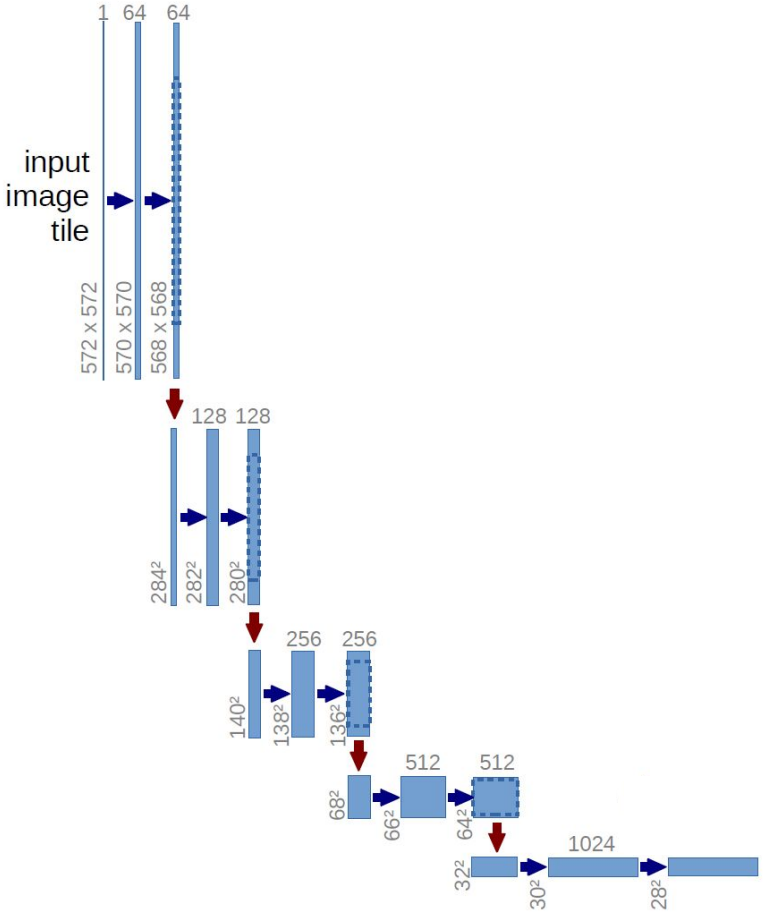




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

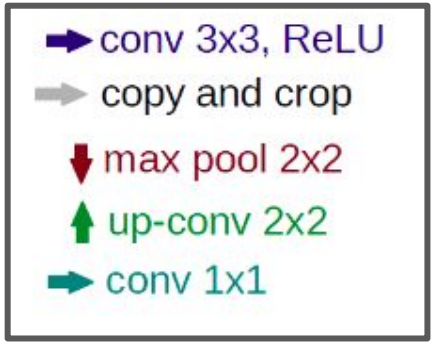
U-net

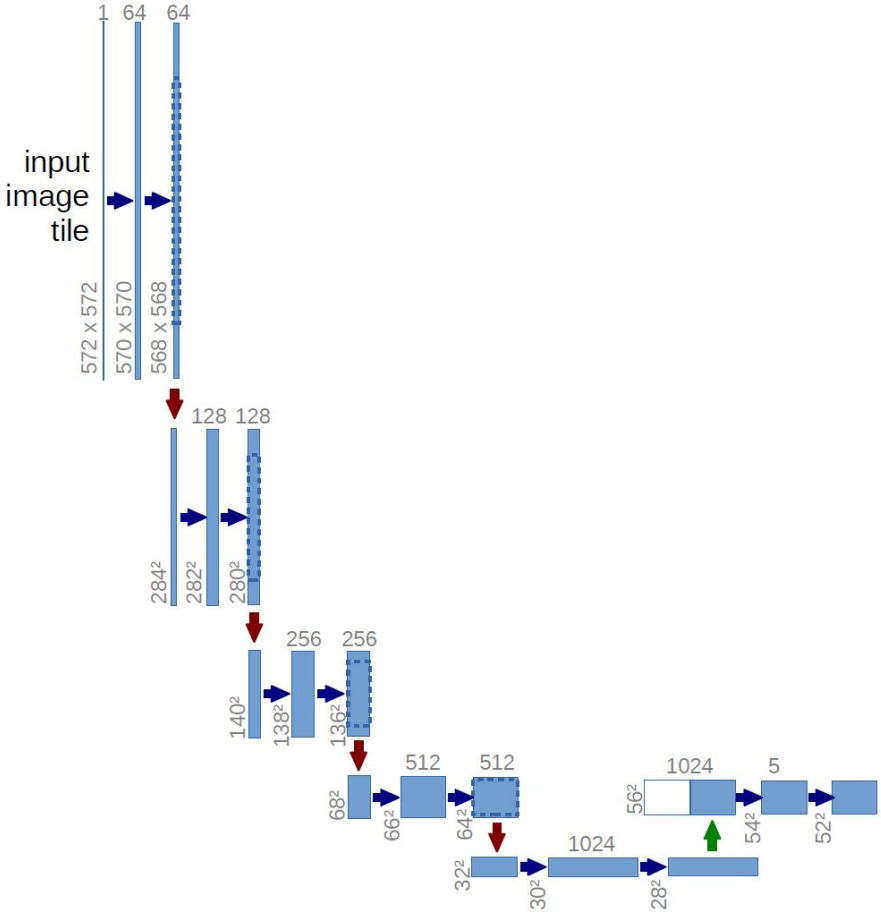




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

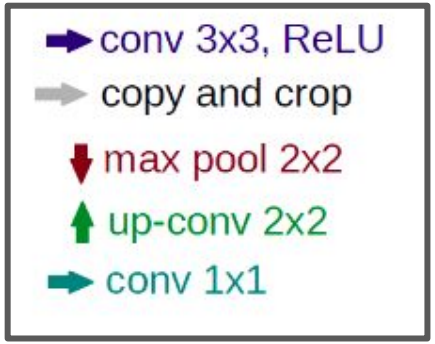
U-net

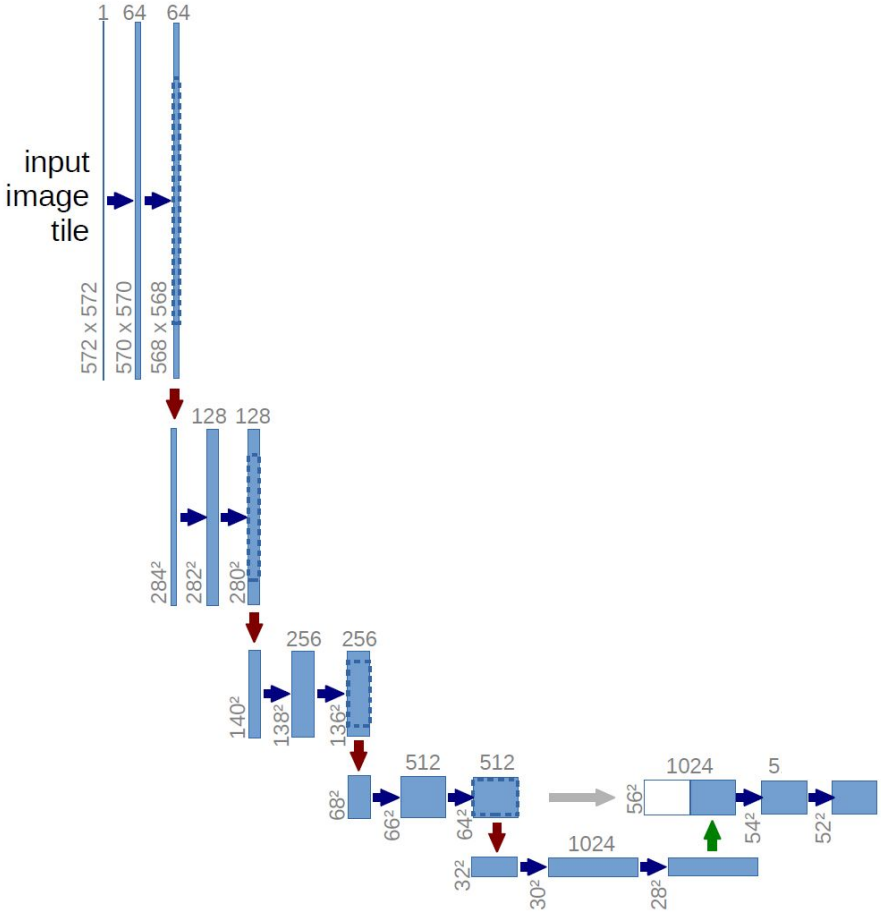




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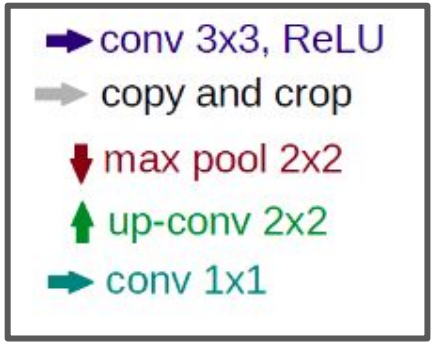
U-net

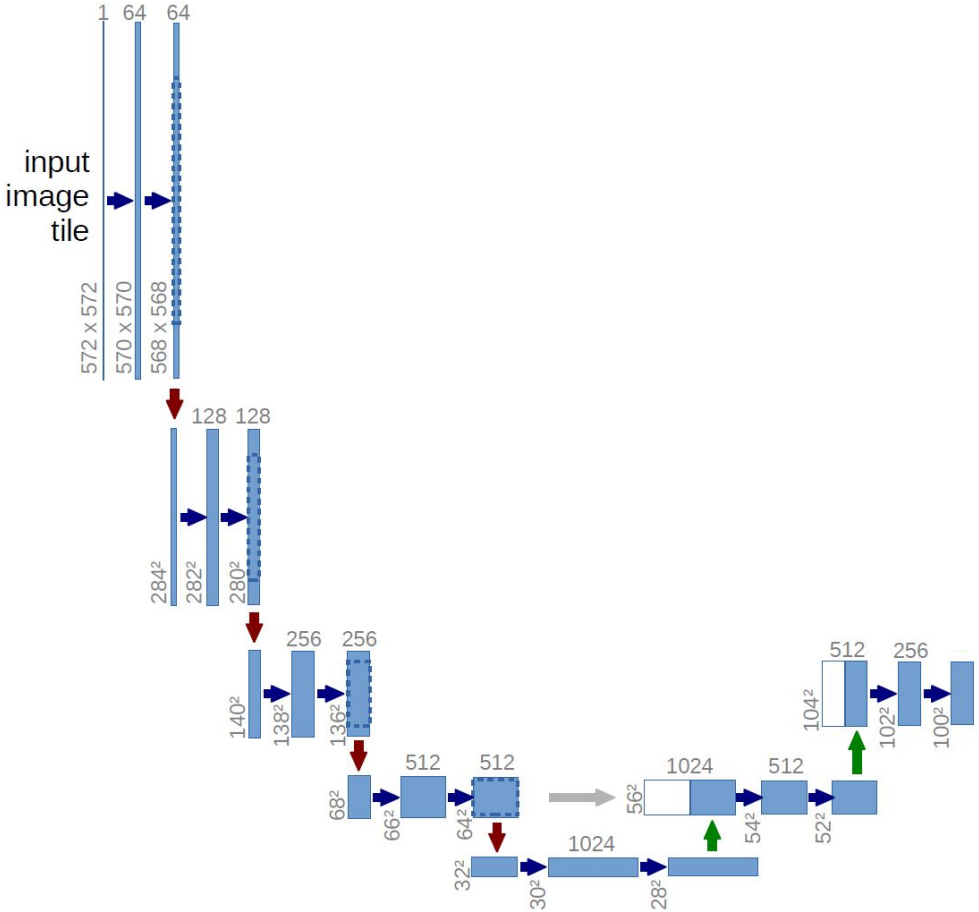




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

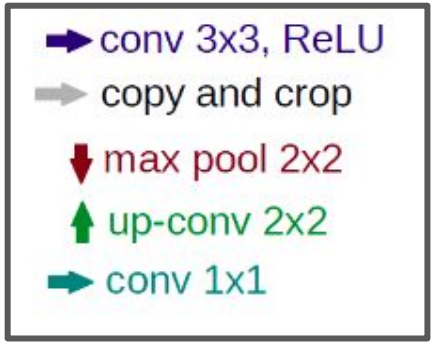
U-net

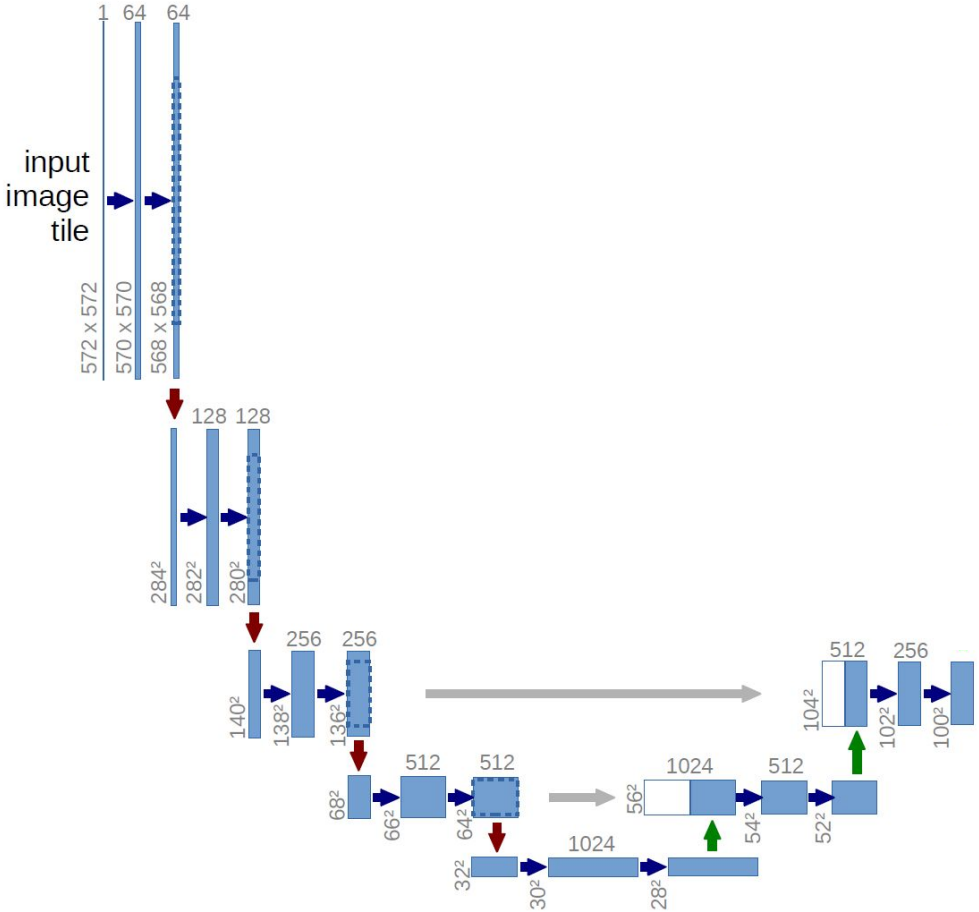




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

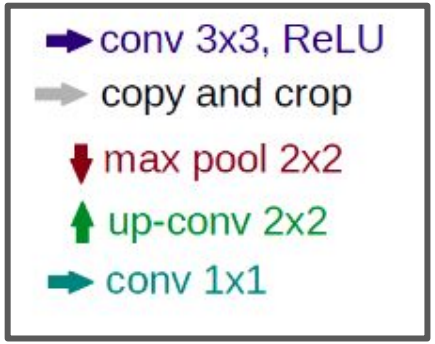
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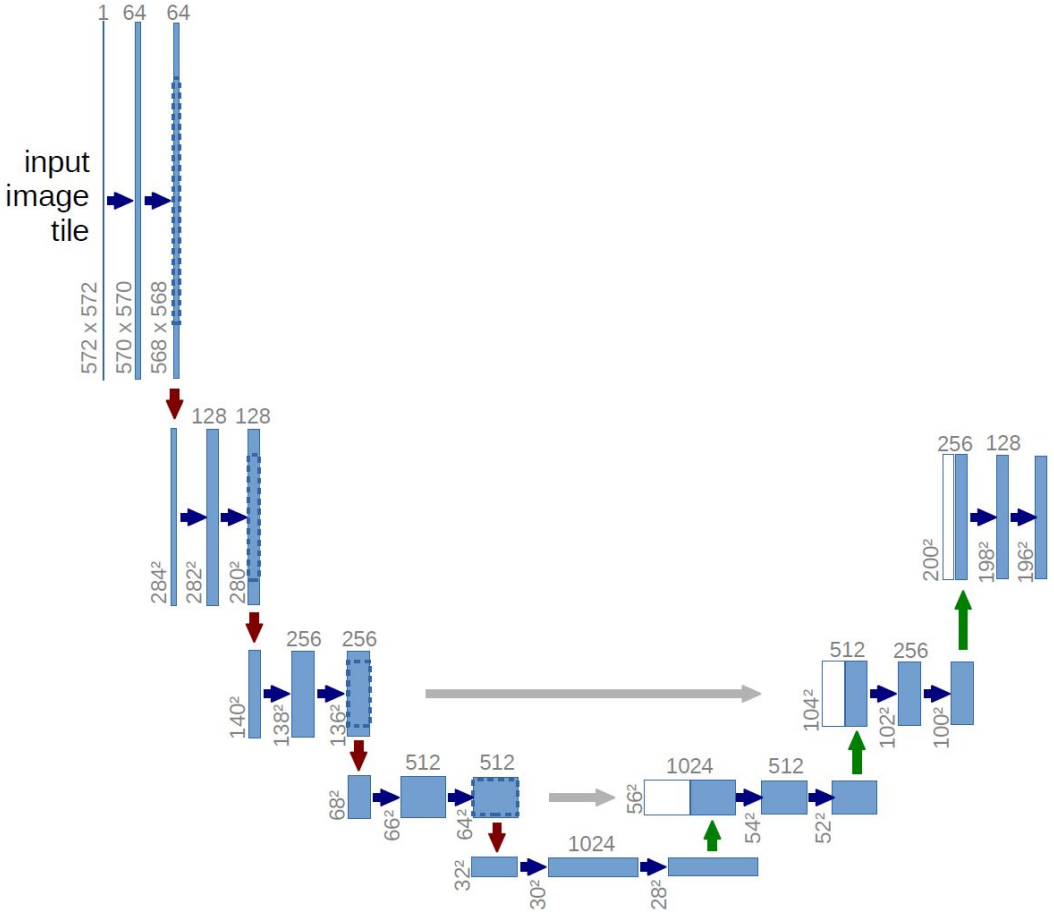




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

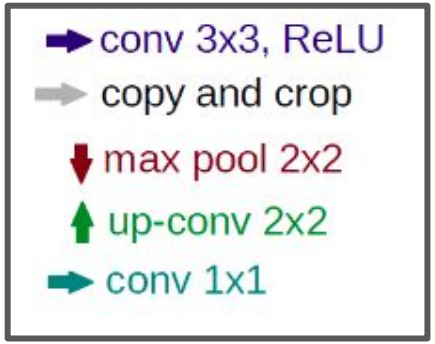
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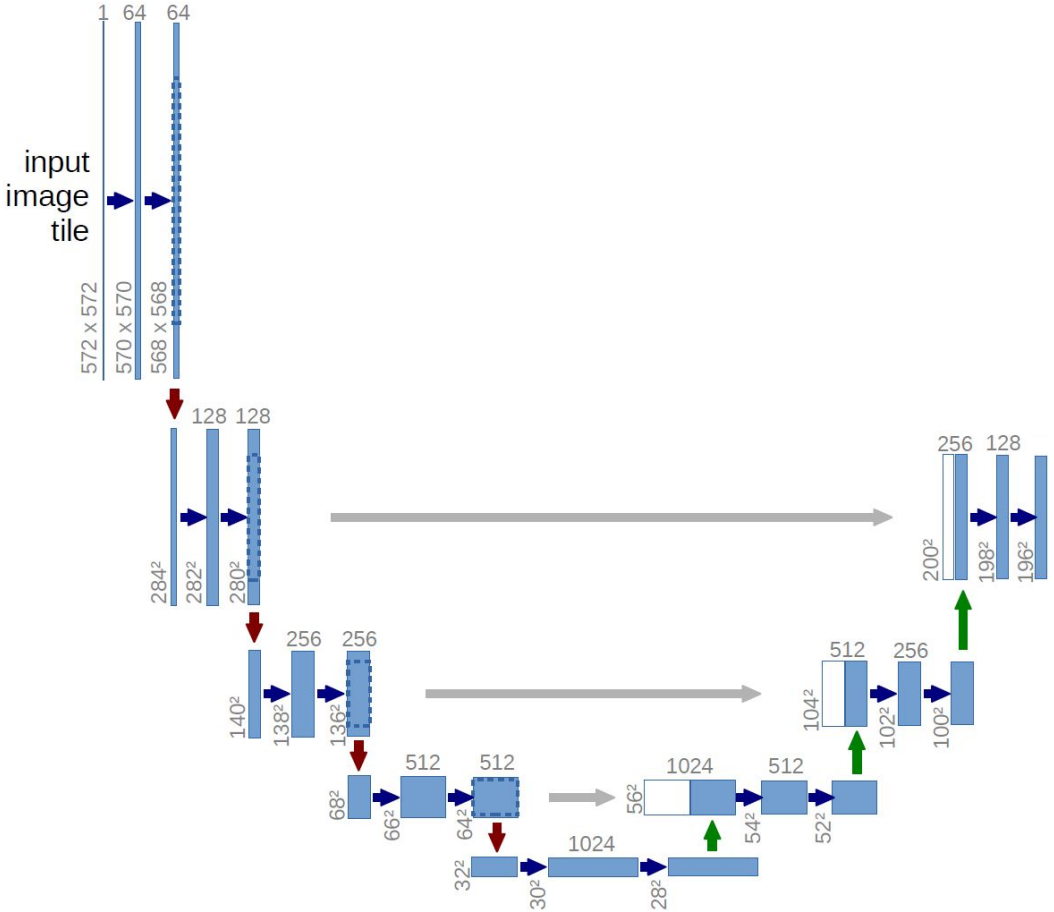




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

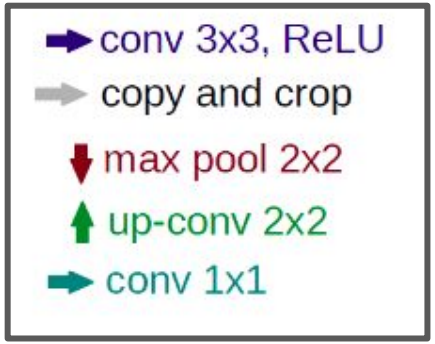
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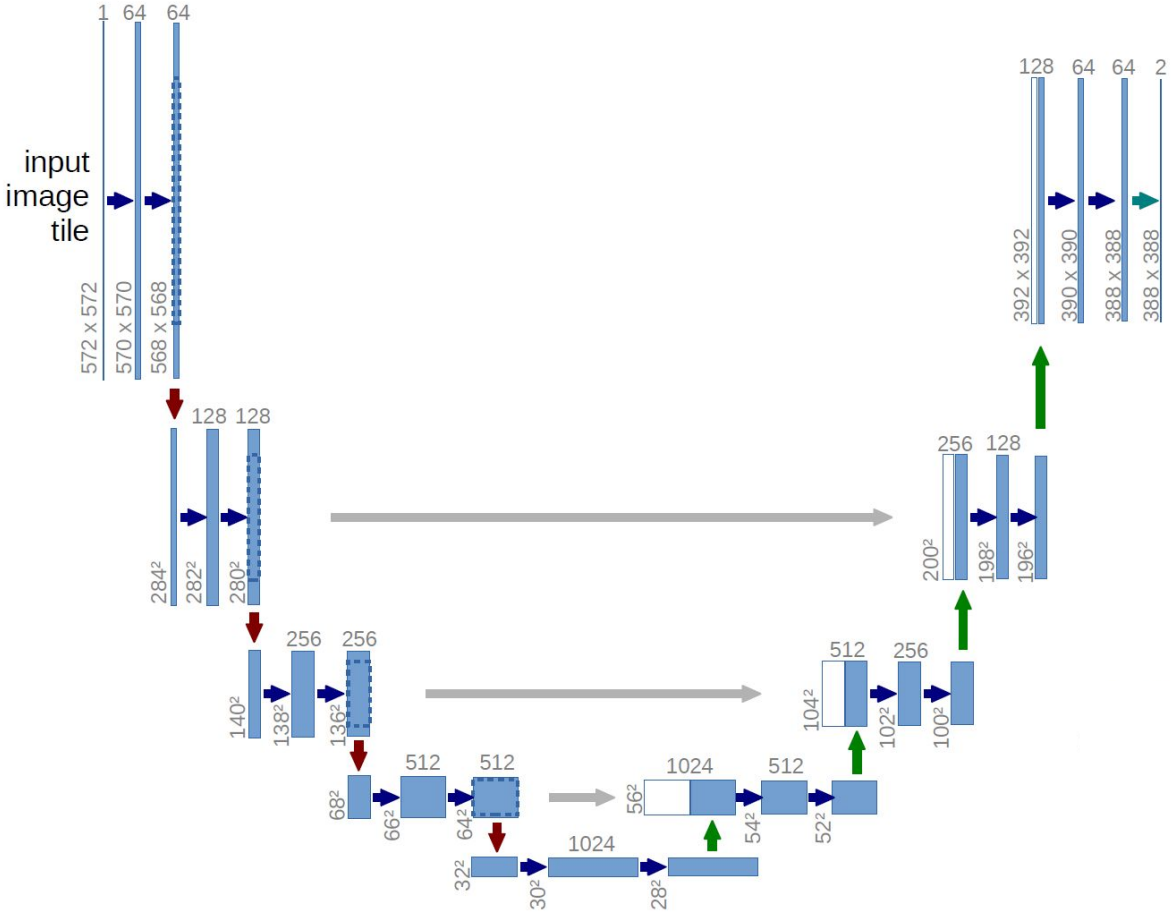




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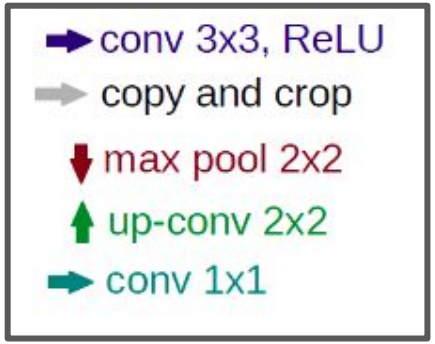
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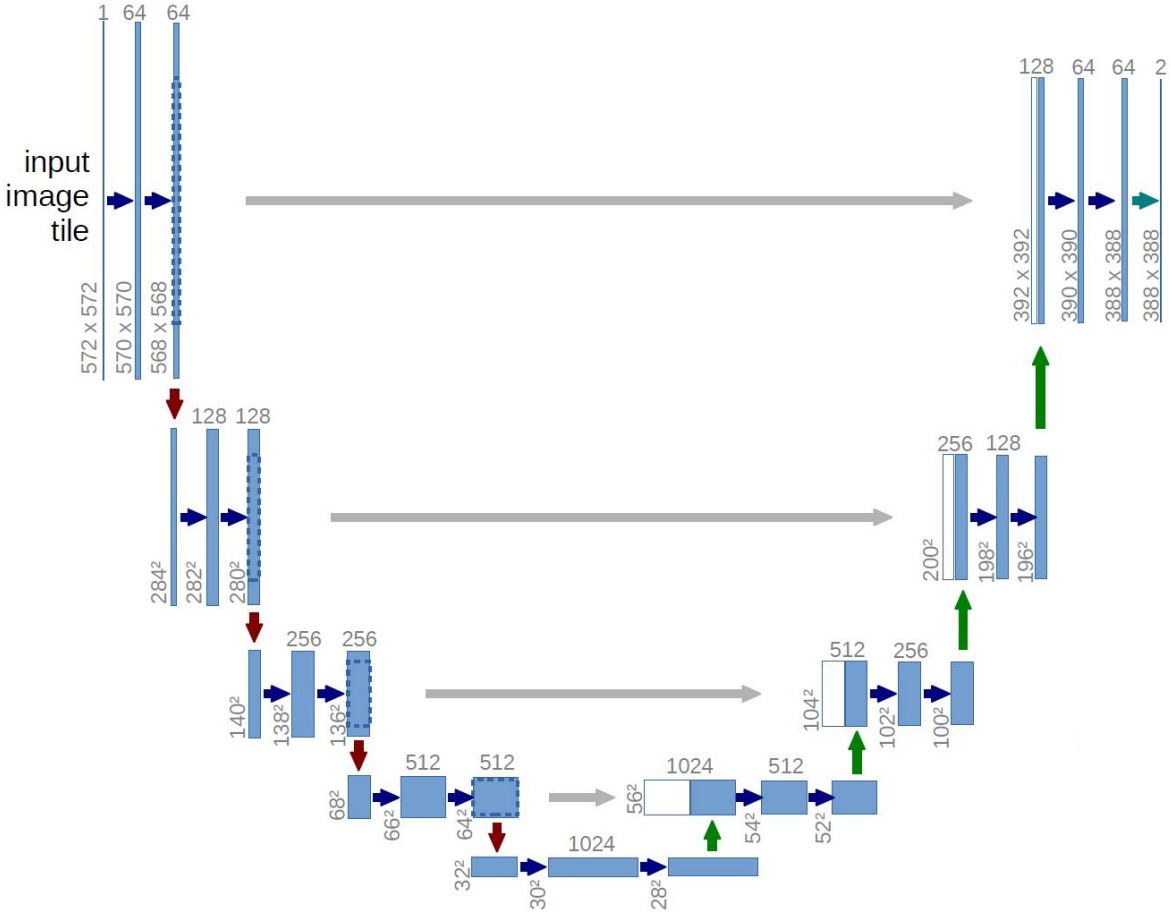




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

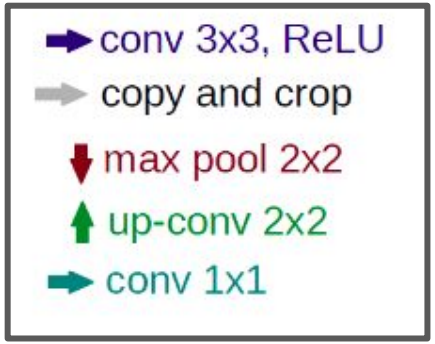
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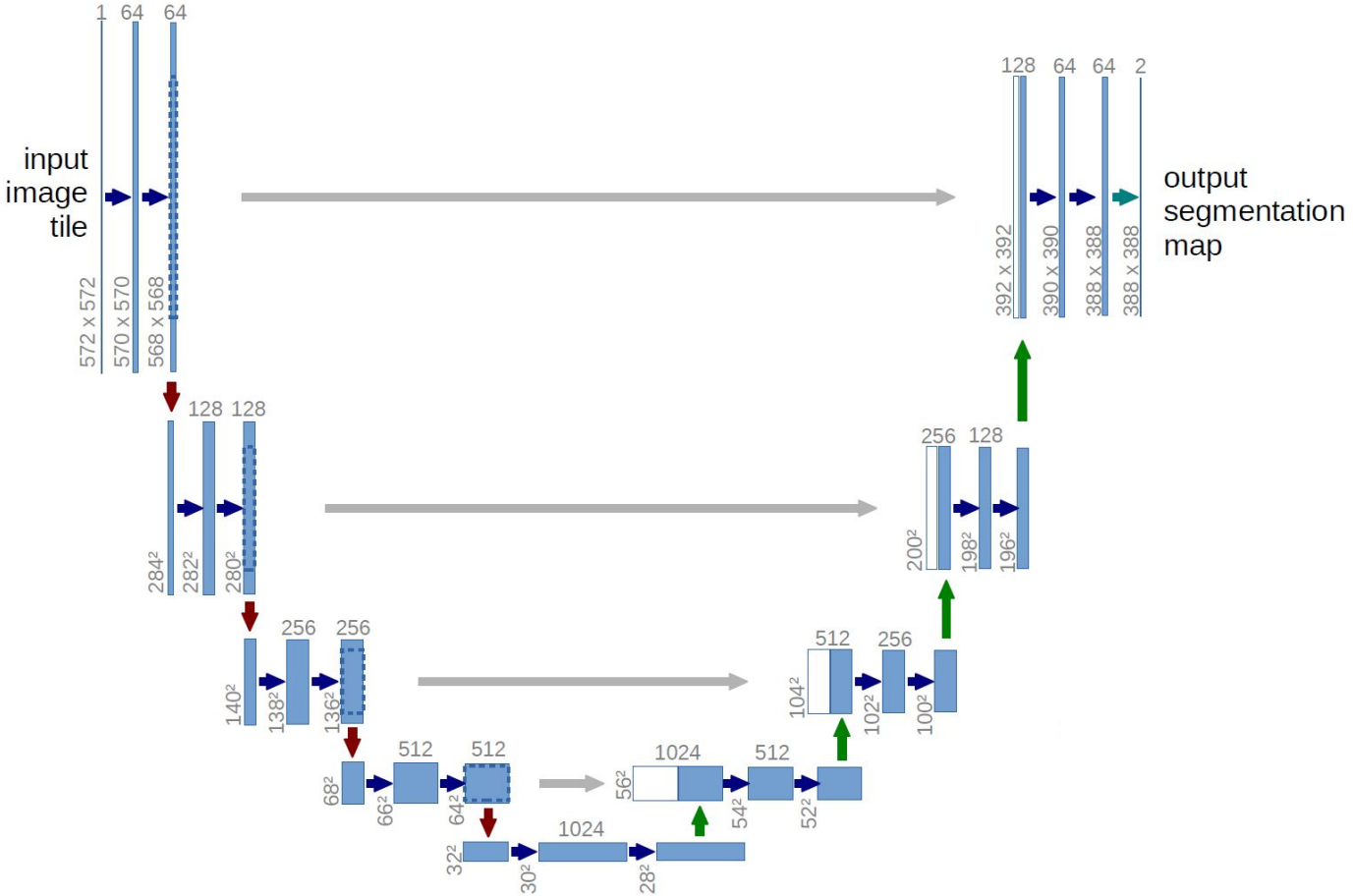




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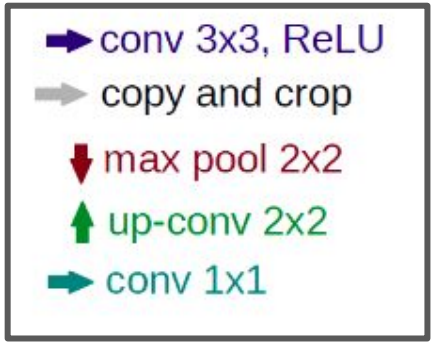
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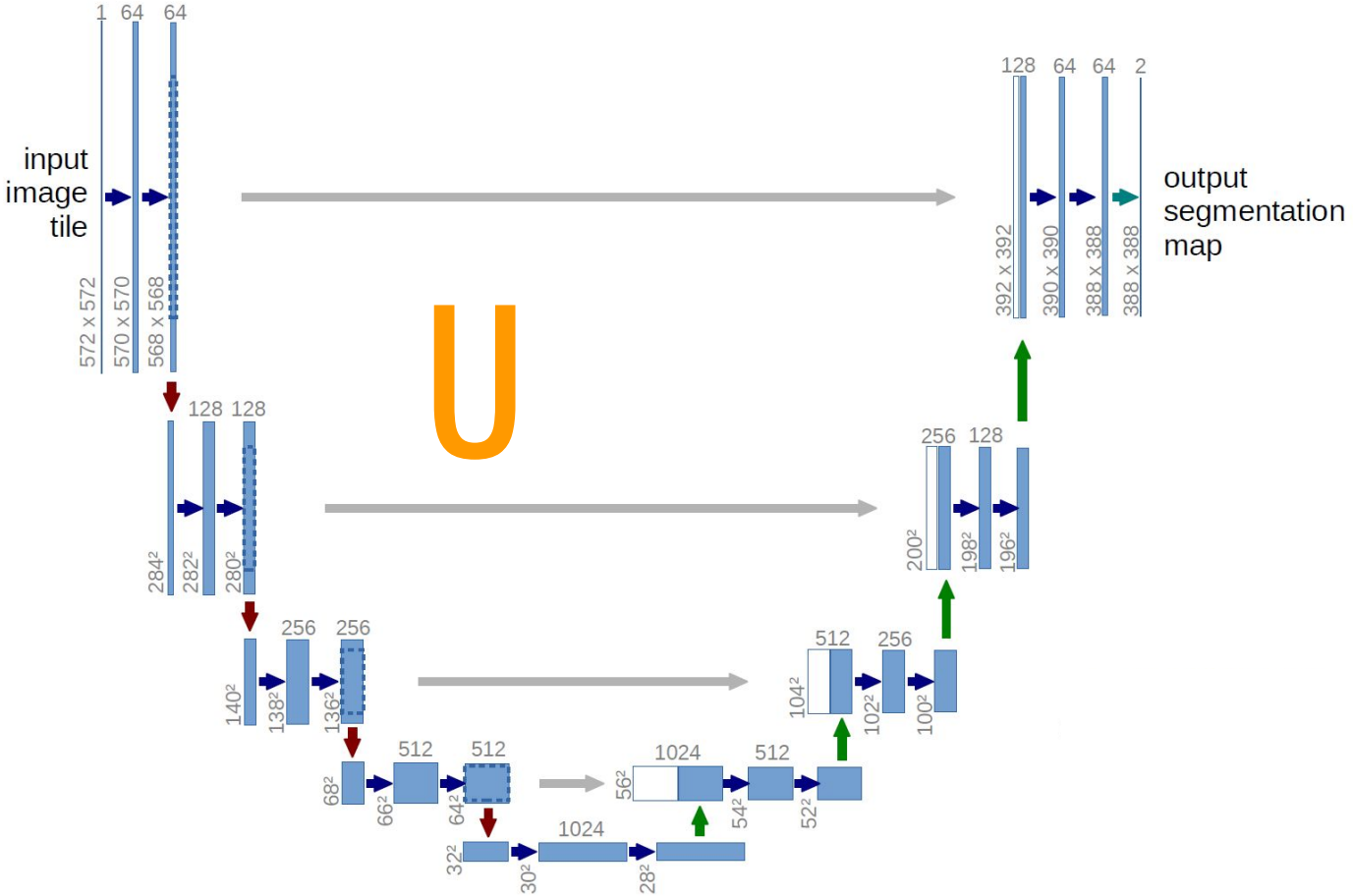




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

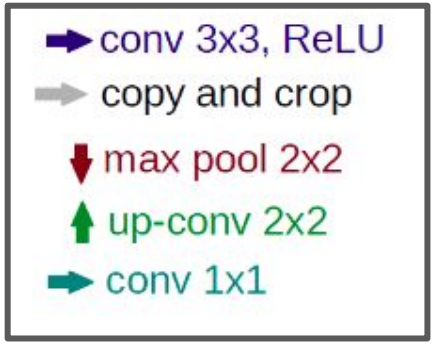
U-net

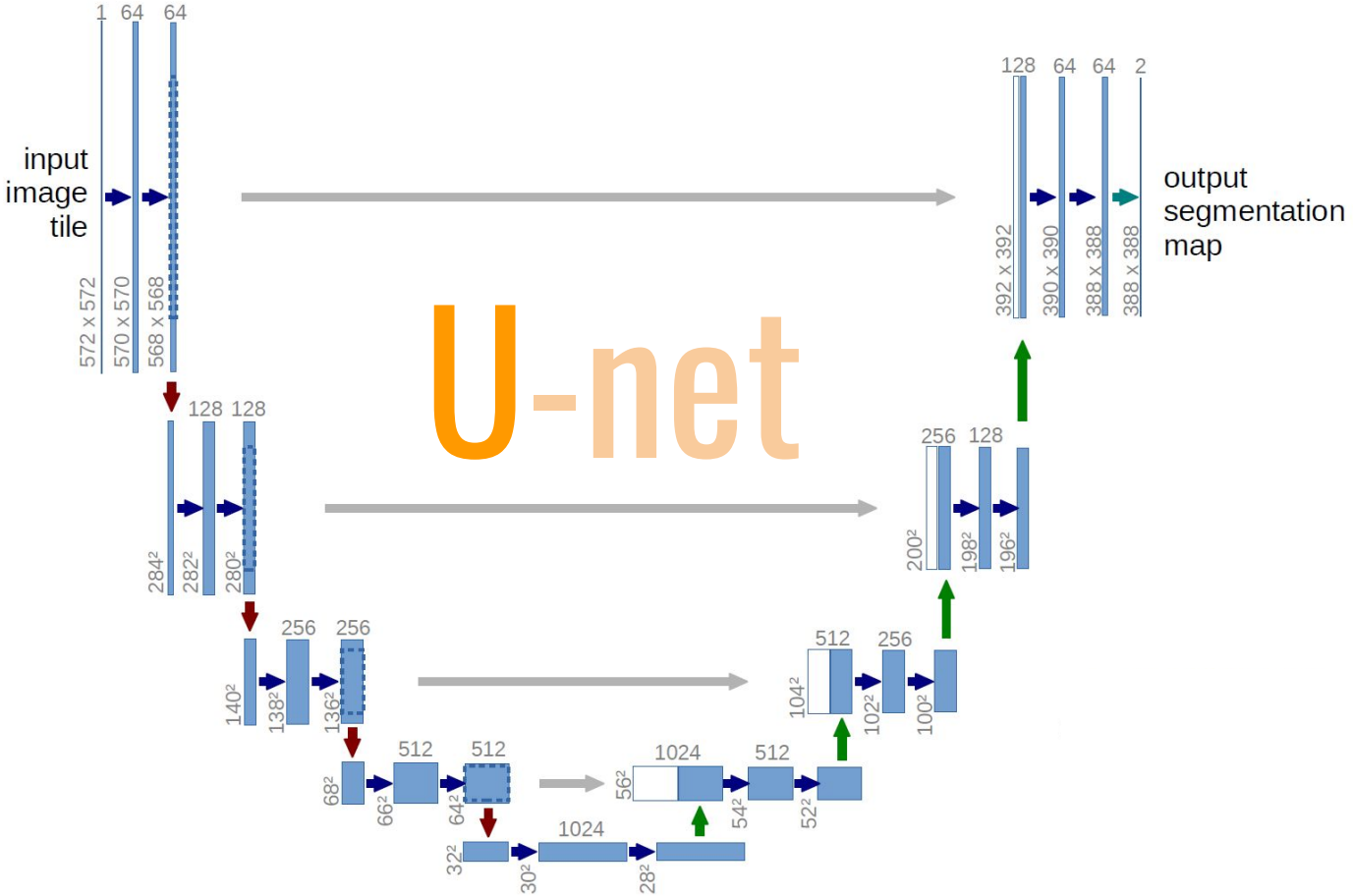




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

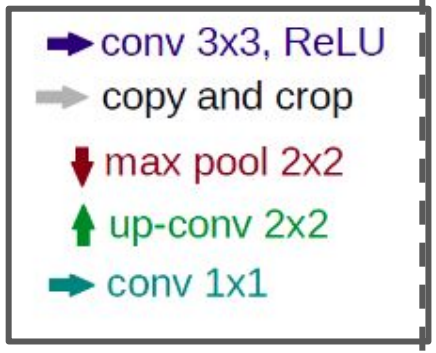
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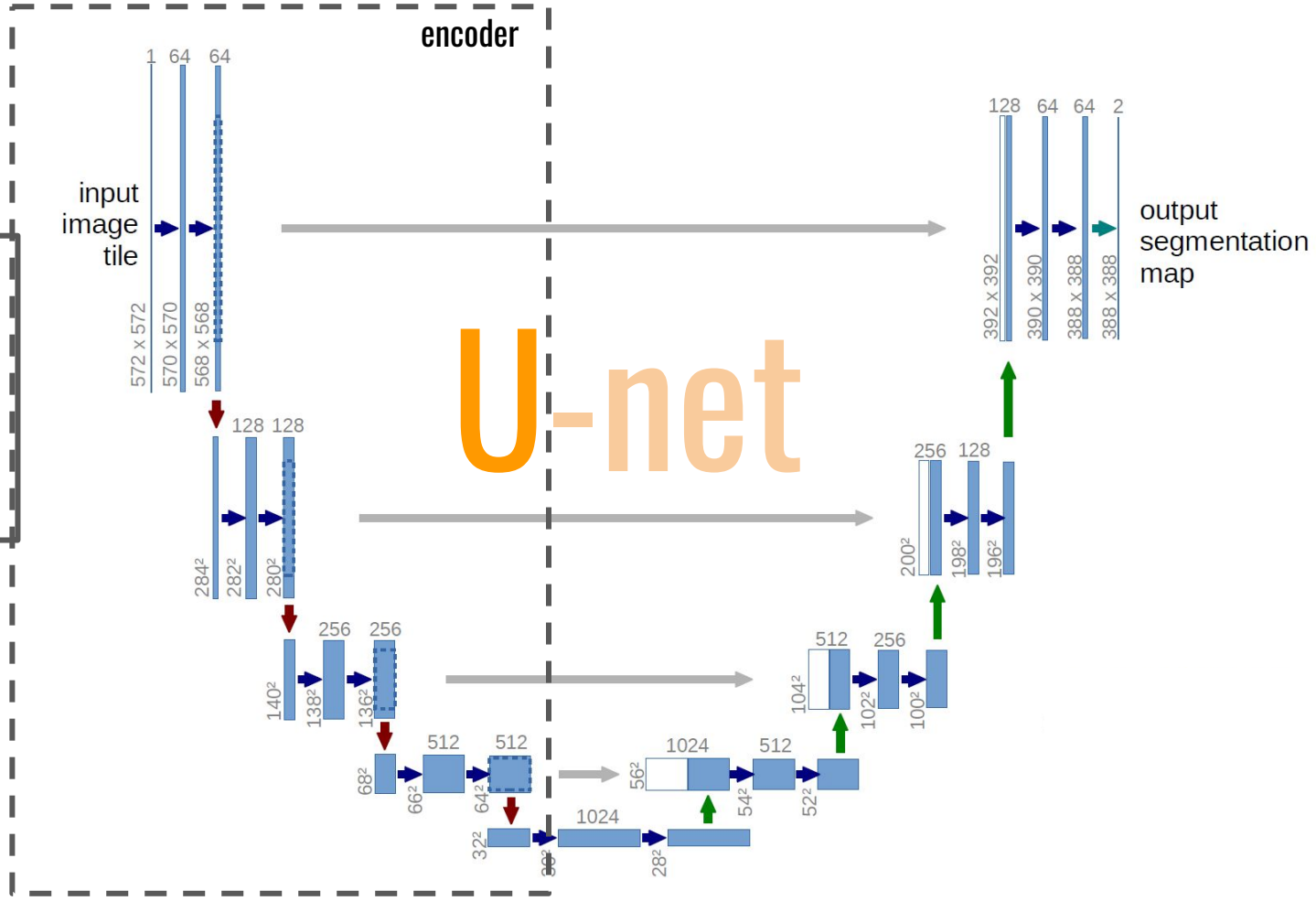




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

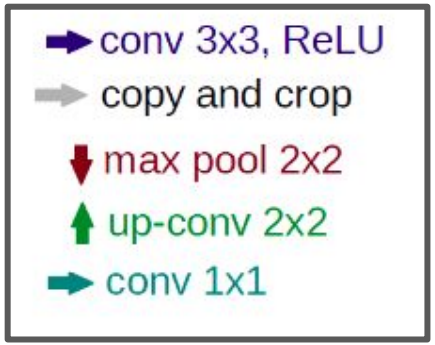
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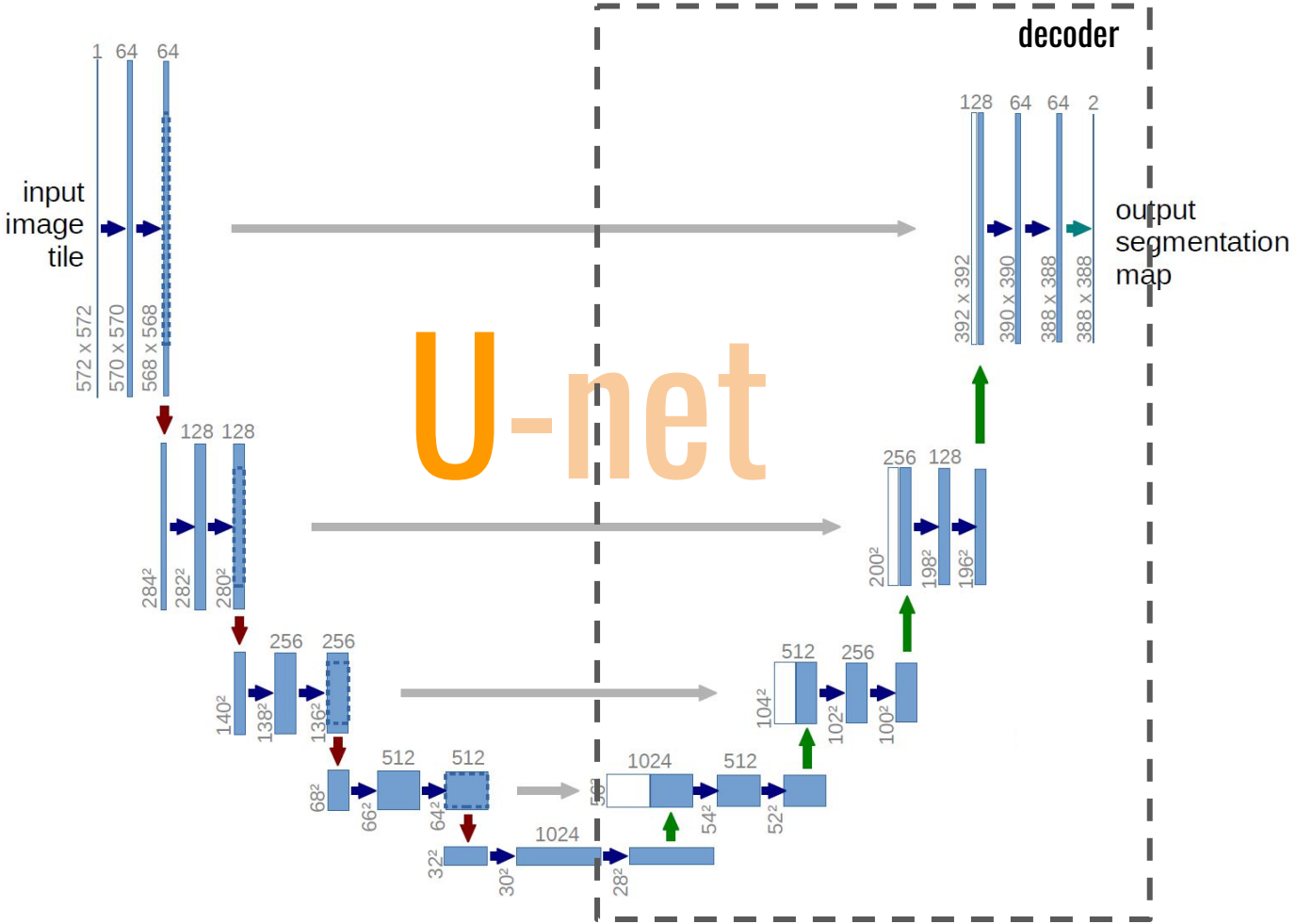




Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

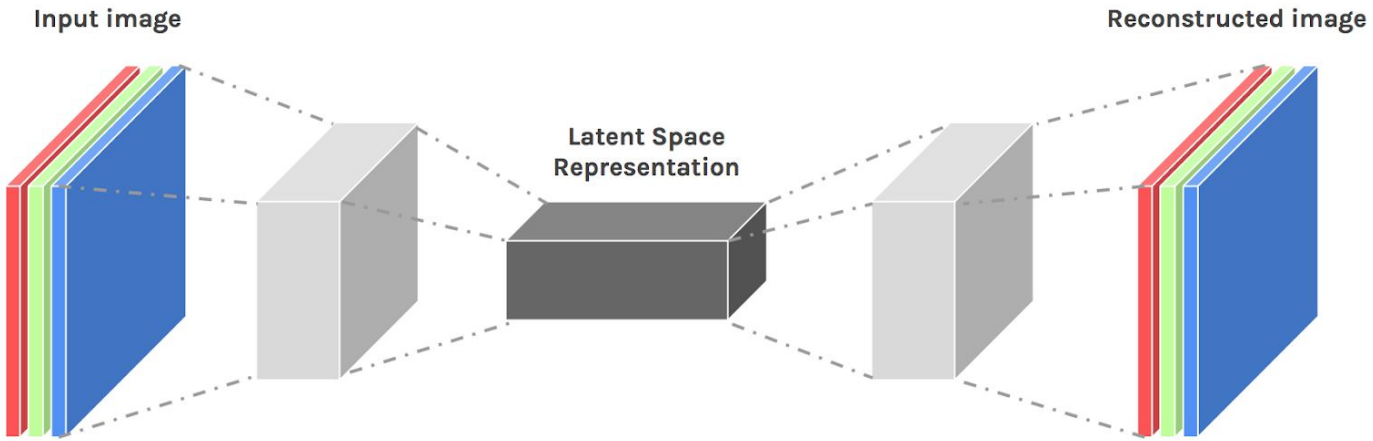
U-net





Fuente: Ronneberger, O., Fischer, P., & Brox, T. (2015, October). U-net: Convolutional networks for biomedical image segmentation. In International Conference on Medical image computing and computer assisted intervention.

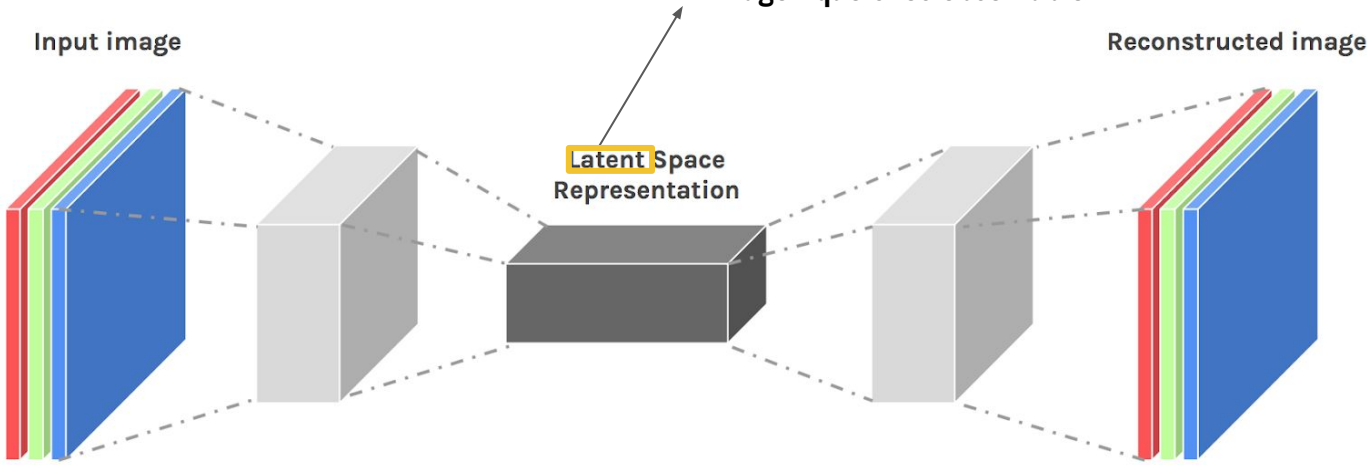
Encoder-decoder: Autoencoders



Fuente:https://becominghuman.ai/understanding-autoencoders-unsupervised-learning-technique-82fb3fbaec2

Encoder-decoder: Autoencoders

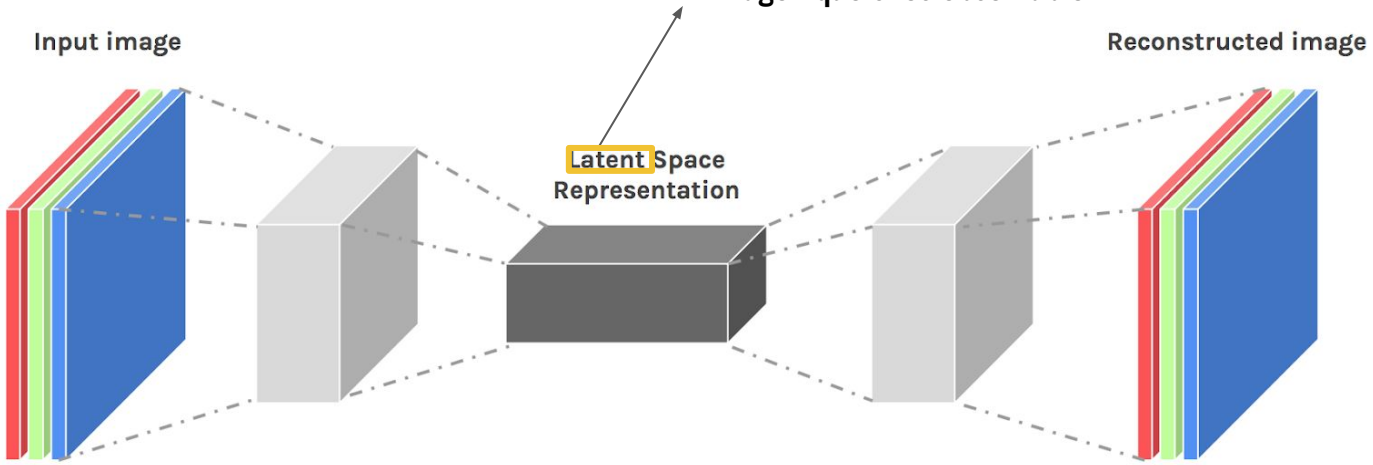
Quiere decir:
No-observadas, en contraste con la imagen que sí es observable



Fuente:
https://becominghuman.ai/understanding-autoencoders-unsupervised-learning-technique-82fb3fbaec2

Encoder-decoder: Autoencoders

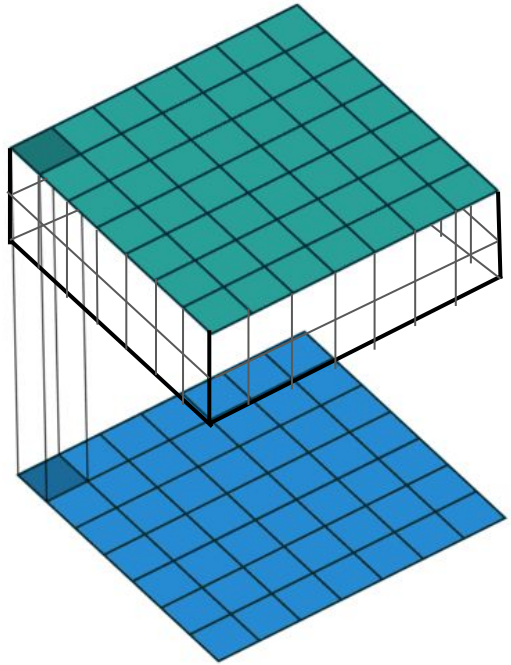
Quiere decir:
No-observadas, en contraste con la imagen que sí es observable



Diferente a nuestro caso, aquí se quiere reconstruir la imagen de entrada

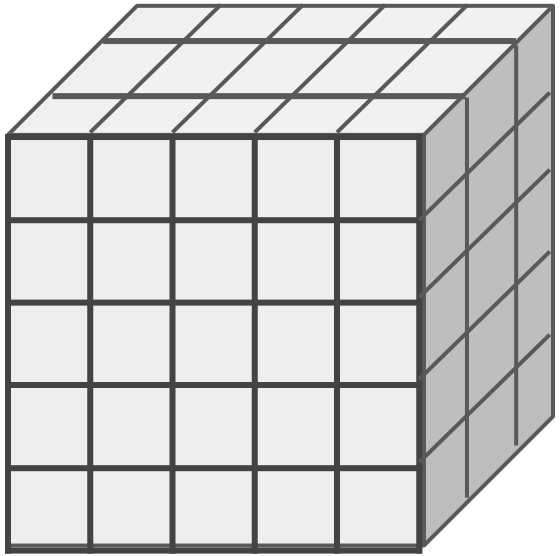
Fuente:
https://becominghuman.ai/understanding-autoencoders-unsupervised-learning-technique-82fb3fbaec2

Convolución 1x1



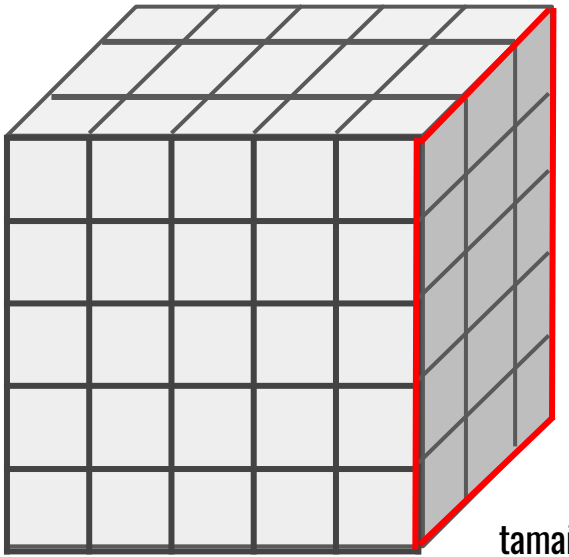
Fuente:https://iamaadity a.github.io/2016/03/one-by-one-convolution/

Convolución 1x1: Clasificación



Volumen Final

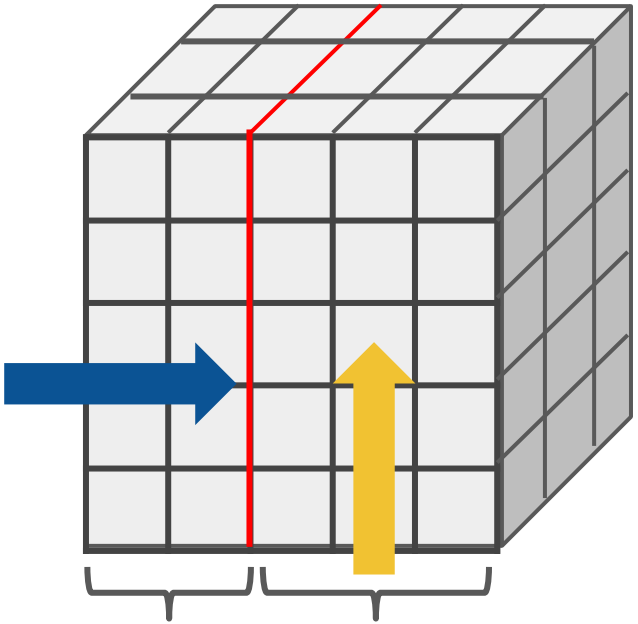
Convolución 1x1: Clasificación



Volumen Final

tamaño de la imagen
original (espacialmente)

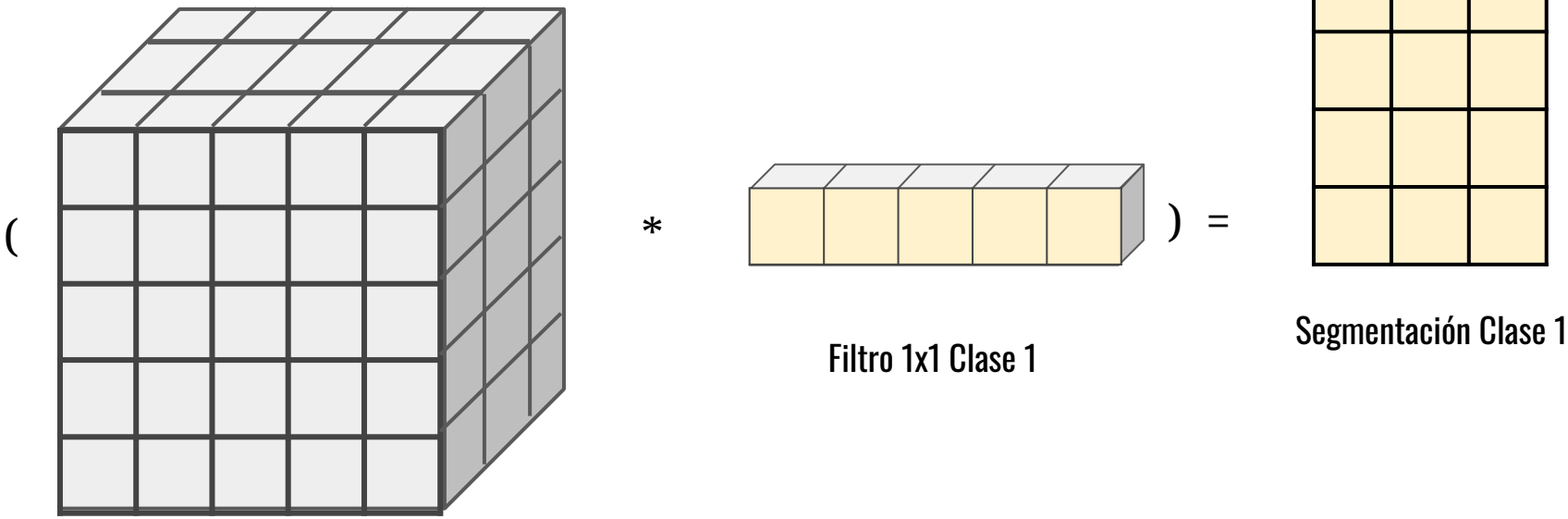
Convolución 1x1: Clasificación



skip-connection up-path

Volumen Final

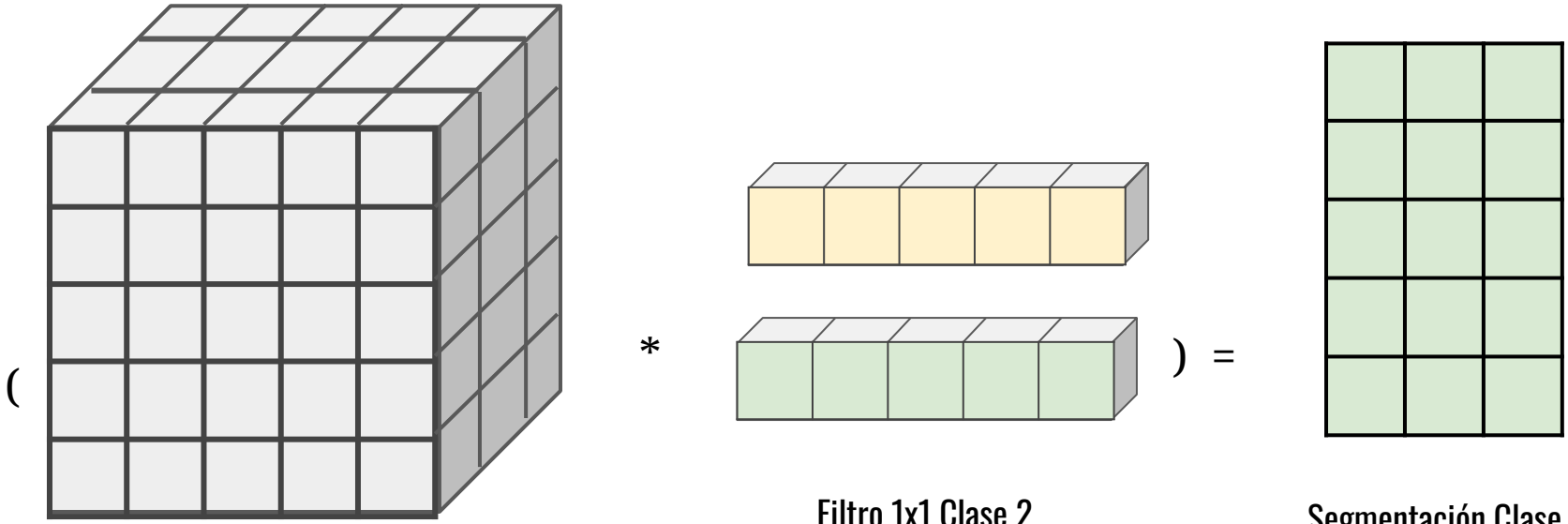
Convolución 1x1: Clasificación



Filtro 1x1 Clase 1

Segmentación Clase 1

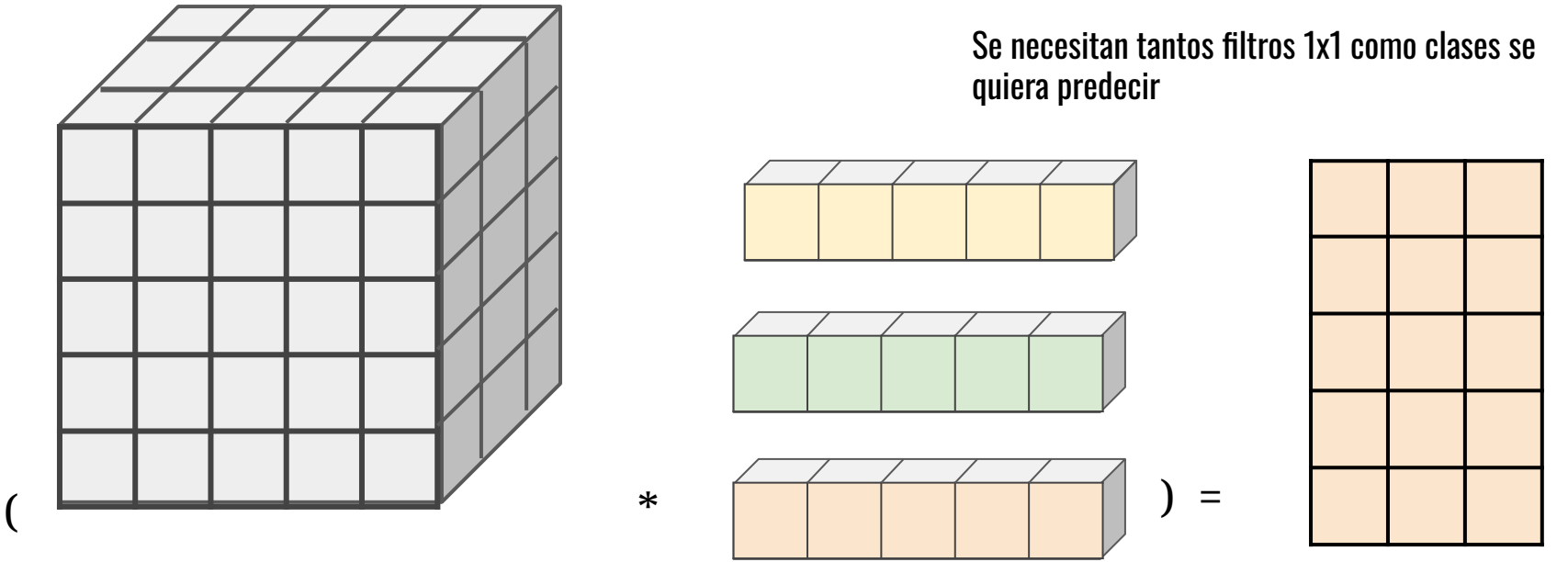
Convolución 1x1: Clasificación



Filtro 1x1 Clase 2

Segmentación Clase 2

Convolución 1x1: Clasificación

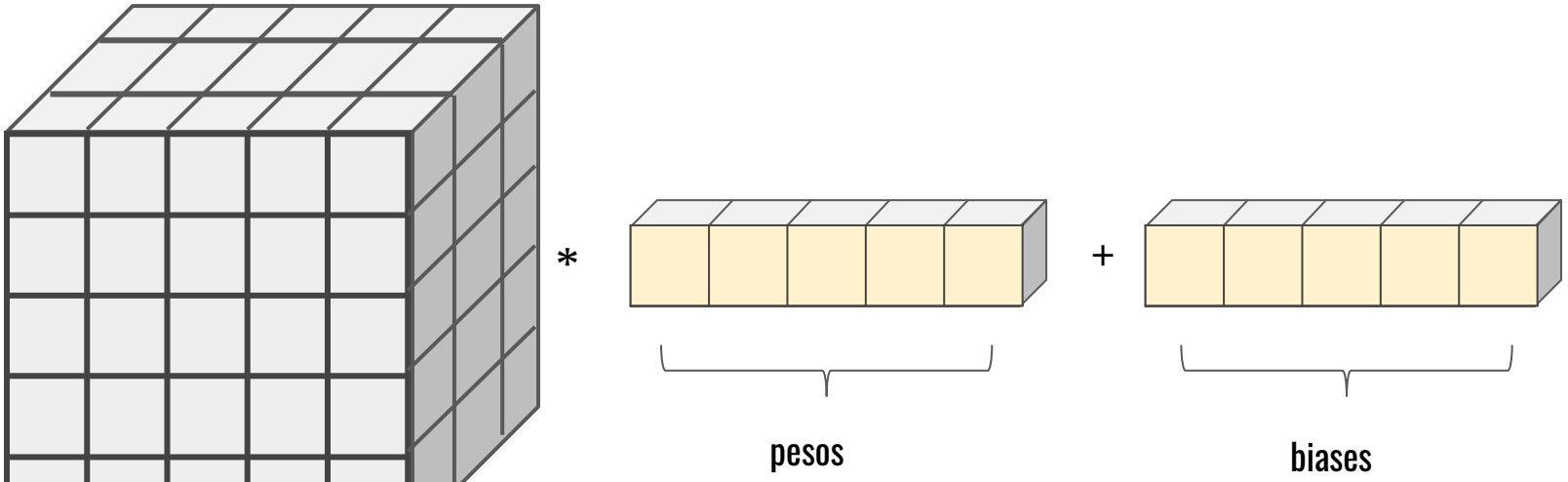


Se necesitan tantos filtros 1x1 como clases se quiera predecir

Filtro 1x1 Clase 3

Segmentación Clase 3

Convolución 1x1: No olvidar de los biases



Es posible (y muy común) incluir biases en la convolución, es un término aditivo que no es común en procesamiento de imágenes.

Métricas

Input



Fuente:
https://arxiv.org/abs/1611.09326 y
https://www.jeremyjordan.me/evaluating-image-segmentation-models/

Métricas

Etiquetas por cada pixel

Input



Ground truth



Fuente:
https://arxiv.org/abs/1611.09326 y
https://www.jeremyjordan.me/evaluating-image-segmentation-models/

Métricas

Etiquetas por cada pixel

Input



Ground truth



Prediction



Salida del algoritmo

Fuente:
https://arxiv.org/abs/1611.09326 y
https://www.jeremyjordan.me/evaluating-image-segmentation-models/

Métricas: Intersección sobre la Unión (IoU)

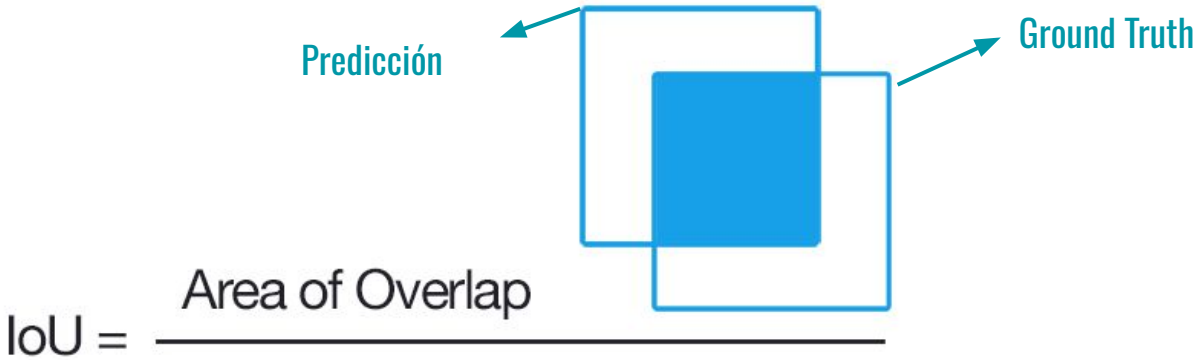
$$IoU =$$

Fuente:
https://www.pyimagesearch.com/2016/11/07/intersection-over-union-iou-for-object-detection/

Métricas: Intersección sobre la Unión (IoU)

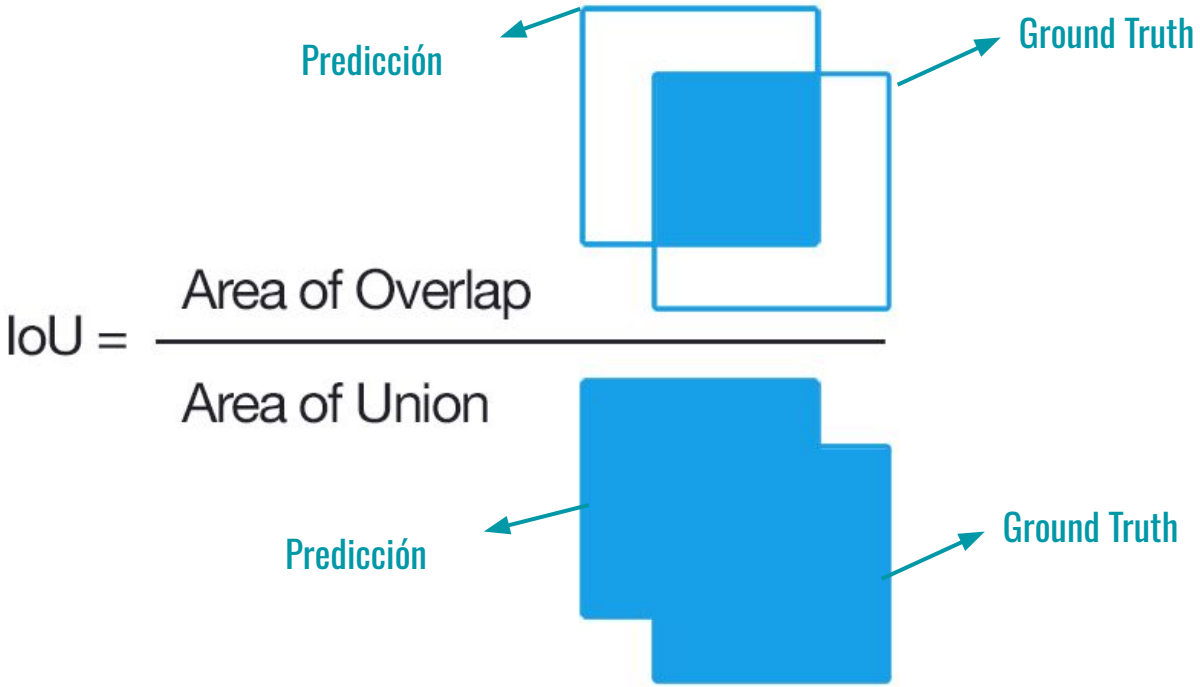
$$IoU = \frac{\quad\quad\quad

Métricas: Intersección sobre la Unión (IoU)



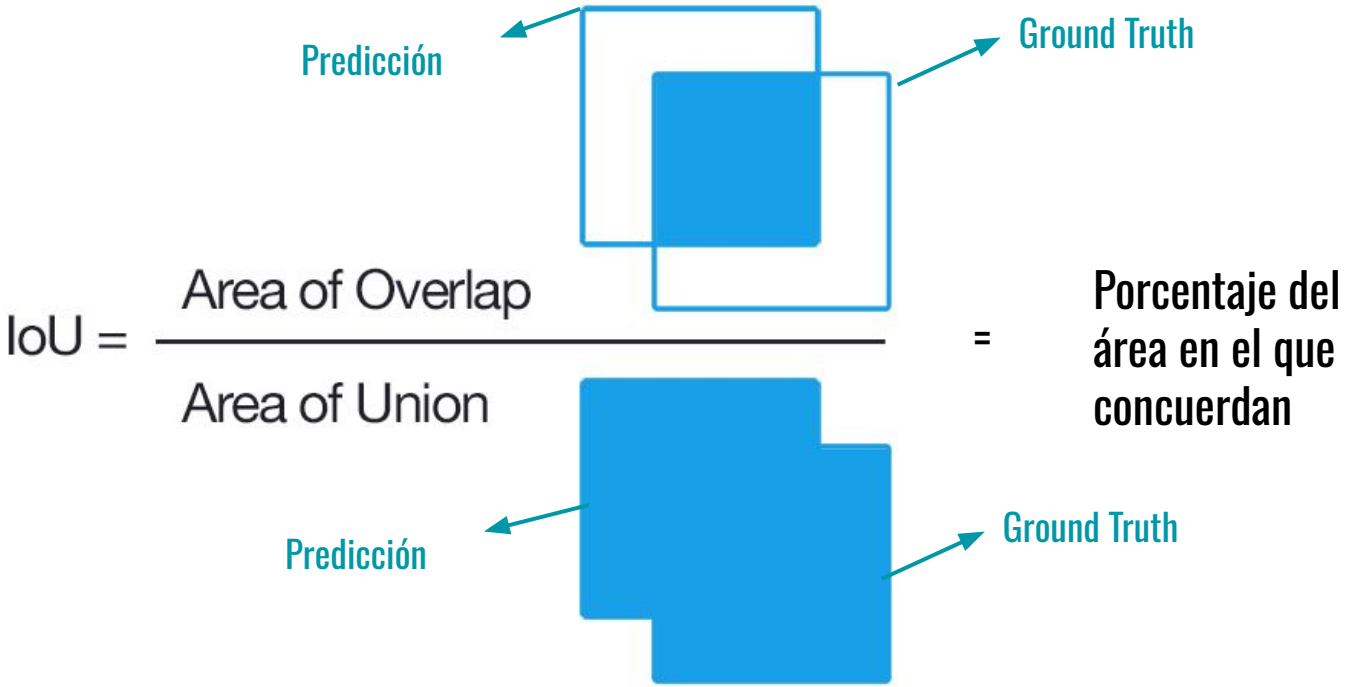
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Métricas: Intersección sobre la Unión (IoU)



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Métricas: Intersección sobre la Unión (IoU)



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Métricas: Intersección sobre la Unión (IoU)



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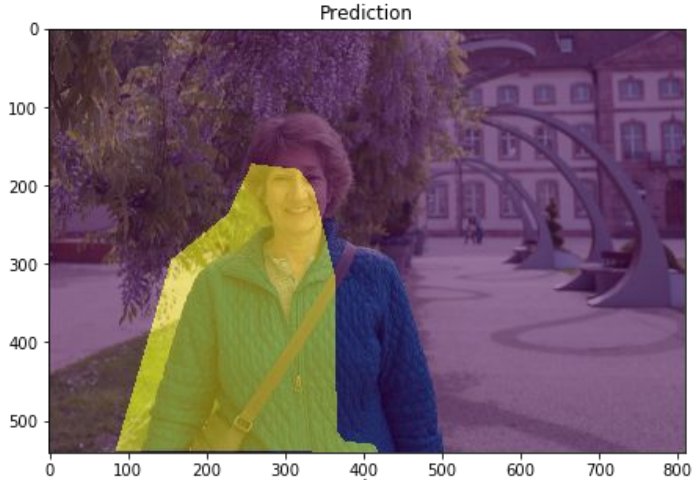
Métricas: Intersección sobre la Unión (IoU)



Fuente:
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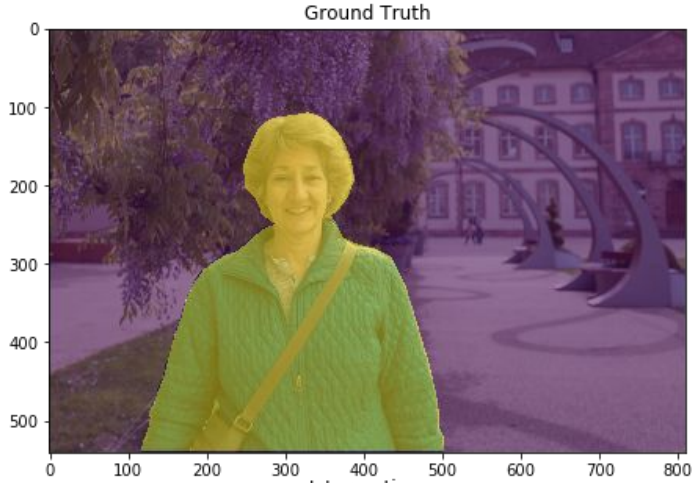
Métricas: Intersección sobre la Unión (IoU)

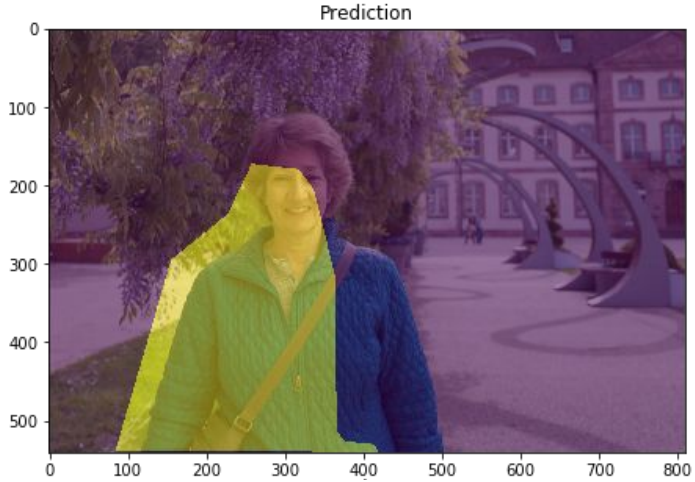


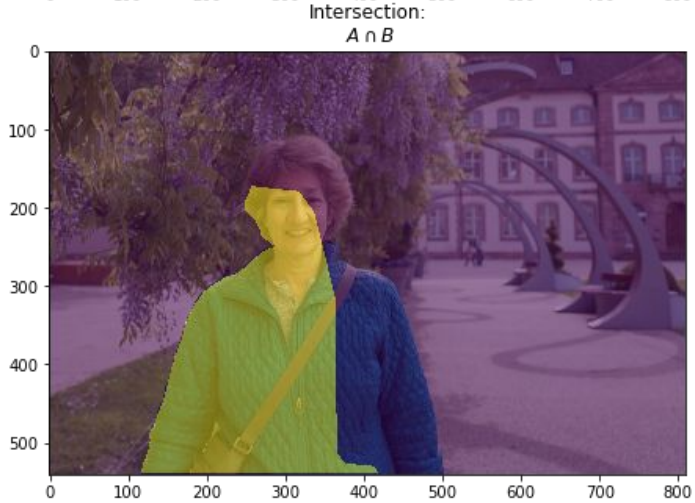


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Métricas: Intersección sobre la Unión (IoU)



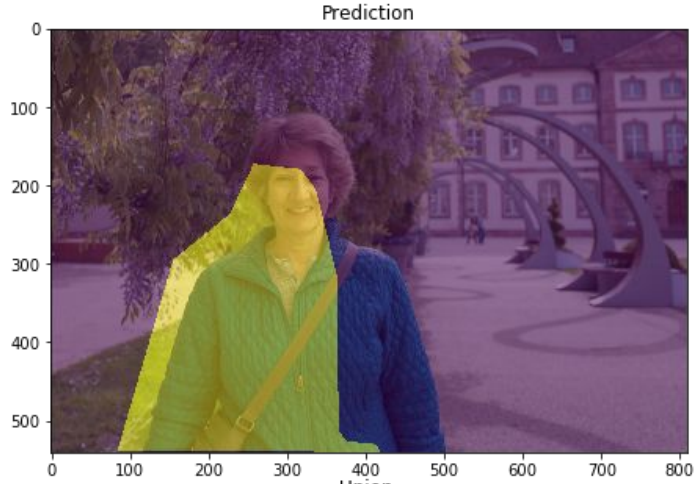


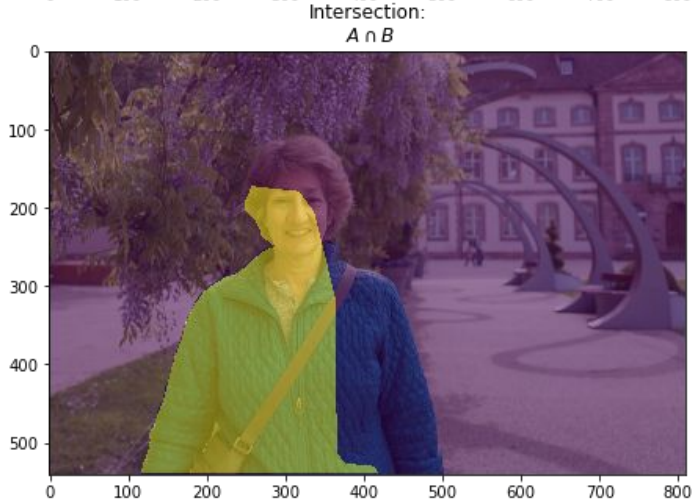


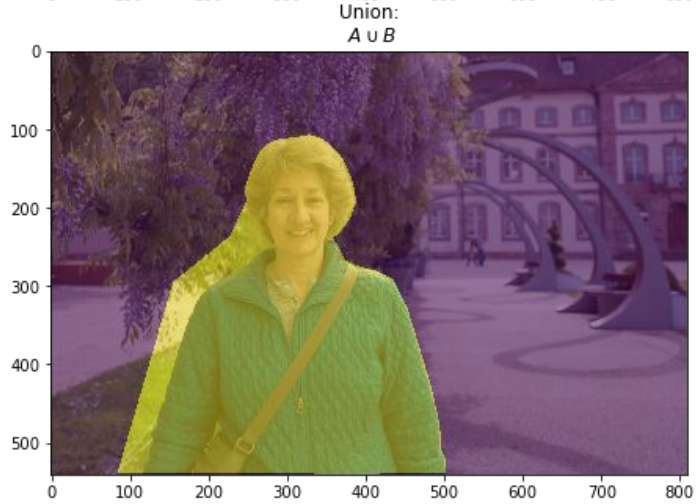
Fuente:
https://www.jeremyjordan.me/evaluating-image-segmentation-models/

Métricas: Intersección sobre la Unión (IoU)









Fuente:
https://www.jeremyjordan.me/evaluating-image-segmentation-models/

Datasets: MS COCO 2017





- Object segmentation
- Recognition in context
- Superpixel stuff segmentation
- 330K images (>200K labeled)
- 1.5 million object instances
- 80 object categories
- 91 stuff categories
- 5 captions per image
- 250,000 people with keypoints

https://cocodataset.org/#home

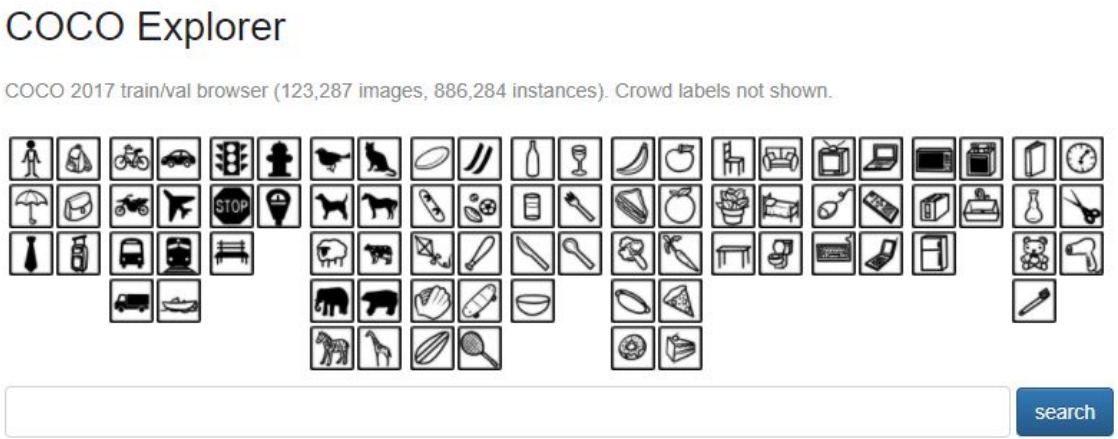
Datasets: MS COCO 2017

Tareas

- Detection
- DensePose
- Keypoints
- Stuff
- Panoptic
- Captions

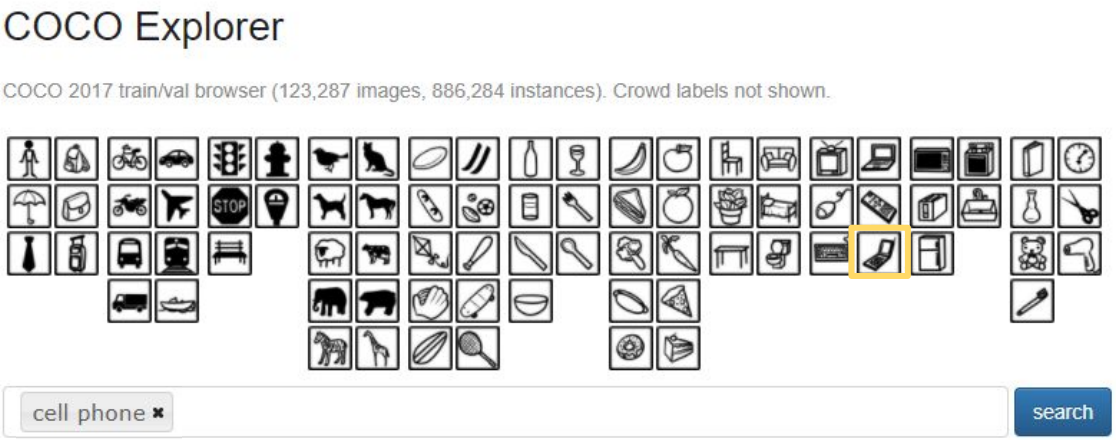
https://cocodataset.org/#home

Datasets: MS COCO 2017 https://cocodataset.org/#explore



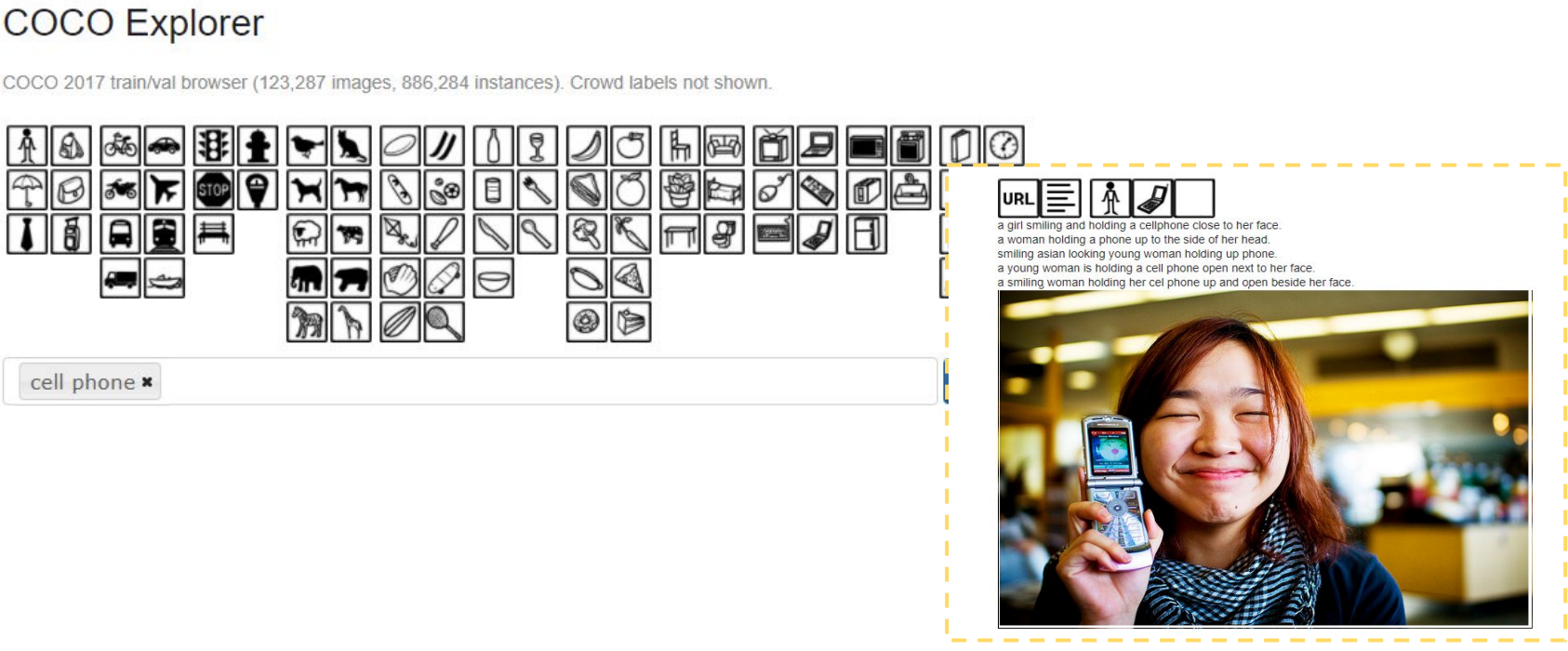
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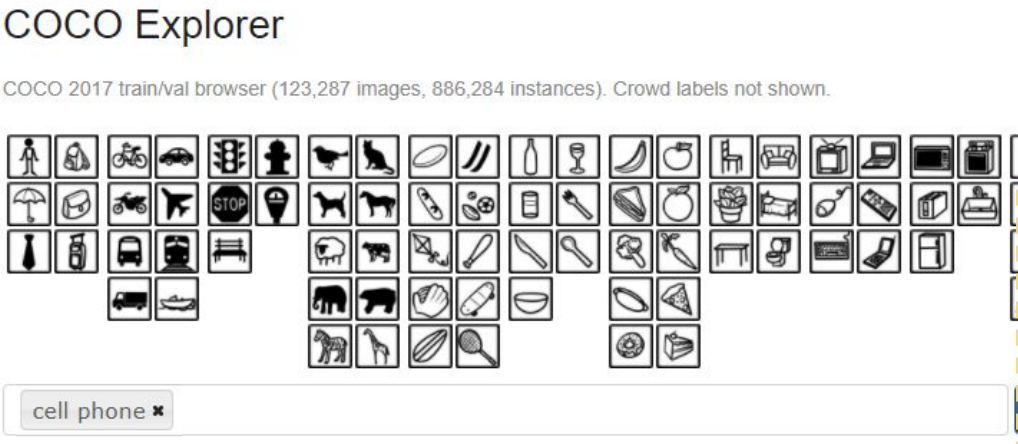
Datasets: MS COCO 2017 https://cocodataset.org/#explore



https://cocodataset.org/#home

http://cocodataset.org/#explore?id=52902
http://farm2.staticflickr.com/1172/3353023663_c3fb2b3350_z.jpg

Datasets: MS COCO 2017 https://cocodataset.org/#explore

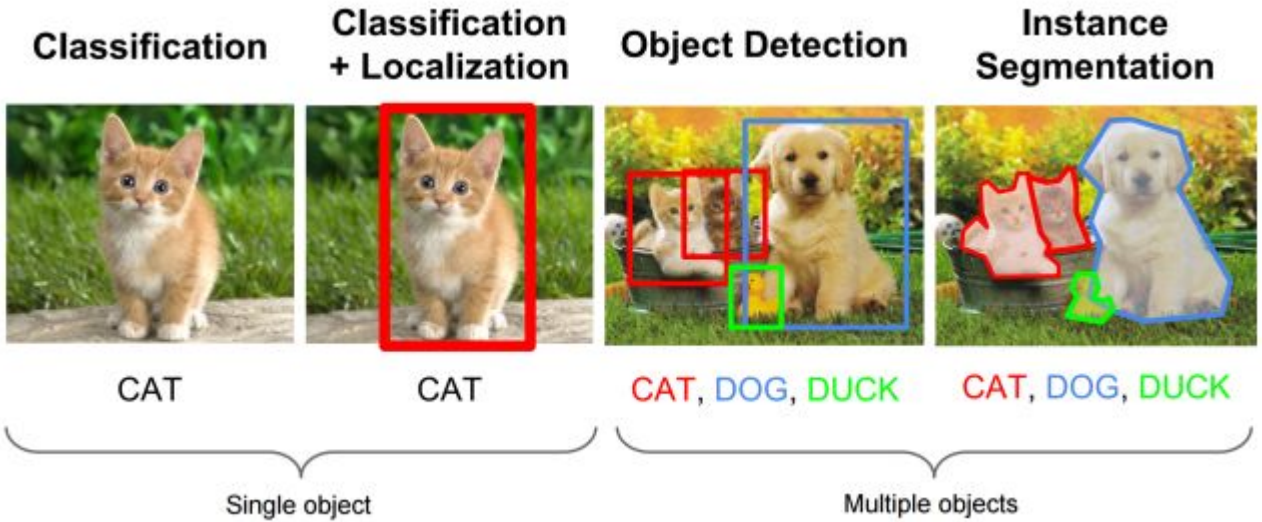




https://cocodataset.org/#home

http://cocodataset.org/#explore?id=52902
http://farm2.staticflickr.com/1172/3353023663_c3fb2b3350_z.jpg

Presentación



https://appsilon.com/object-detection-yolo-algorithm/

Detección de Objetos: Definición de la Tarea

Entrada



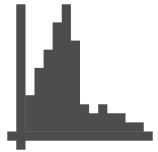
https://appsilon.com/object-detection-yolo-algorithm/

Detección de Objetos: Definición de la Tarea

Entrada



Clasificación



Distribución de probabilidades sobre las clases posibles, dada la entrada

https://appsilon.com/object-detection-yolo-algorithm/

Detección de Objetos: Definición de la Tarea

Entrada





https://appsilon.com/object-detection-yolo-algorithm/

Detección de Objetos: Definición de la Tarea

Entrada



Detección

Conjunto de objetos detectados

- Categoría de una lista de clases
- Bounding Box (x, y, ancho, alto)

https://appsilon.com/object-detection-yolo-algorithm/

Detección de Objetos: Definición de la Tarea

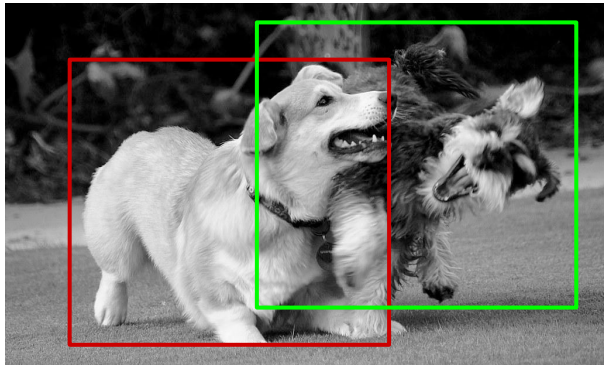
Entrada



Detección

Conjunto de objetos detectados

- Categoría de una lista de clases
- Bounding Box (x, y, ancho, alto)



https://appsilon.com/object-detection-yolo-algorithm/

Retos de la detección de objetos



- Necesidad de diferenciar objetos de la misma clase cuando existe oclusión
- Número variable de predicciones
- Se necesita mayor resolución, y con ello los problemas de memoria asociados (De acuerdo a Justin Johnson[+], la clasificación puede funcionar con 224x224, mientras que la detección necesita 800x600)

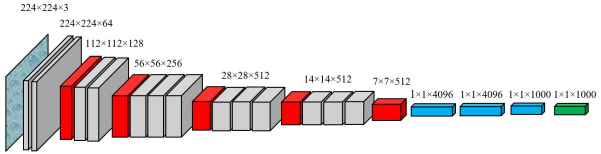
[+]https://www.youtube.com/watch?v=TB-fdlSzpHQ

https://appsilon.com/object-detection-yolo-algorithm/

Detección = Clasificación



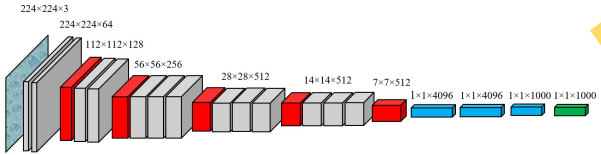
Foto: Pablo Fonseca



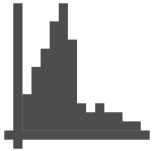
Detección = Clasificación



Foto: Pablo Fonseca



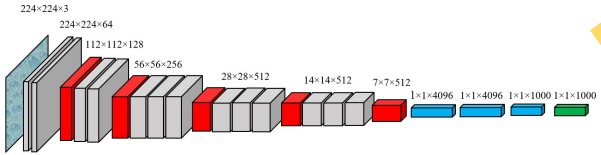
Clase



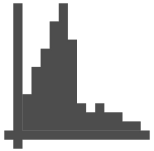
Detección = Clasificación + Localización



Foto: Pablo Fonseca



Clase

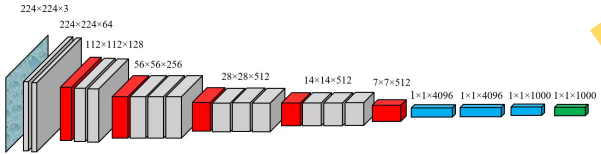


Detección = Clasificación + Localización

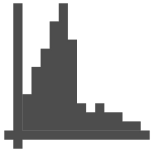
X



Foto: Pablo Fonseca



Clase



Localización

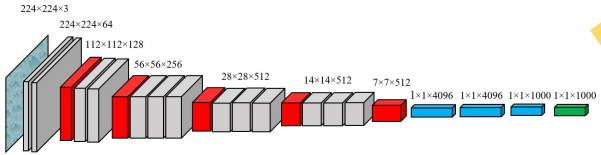
X			

Ahora tenemos una **red multiobjetivo**

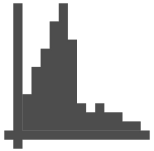
Detección = Clasificación + Localización



Foto: Pablo Fonseca



Clase



Localización

x	y		

Ahora tenemos una **red multiobjetivo**

Detección = Clasificación + Localización

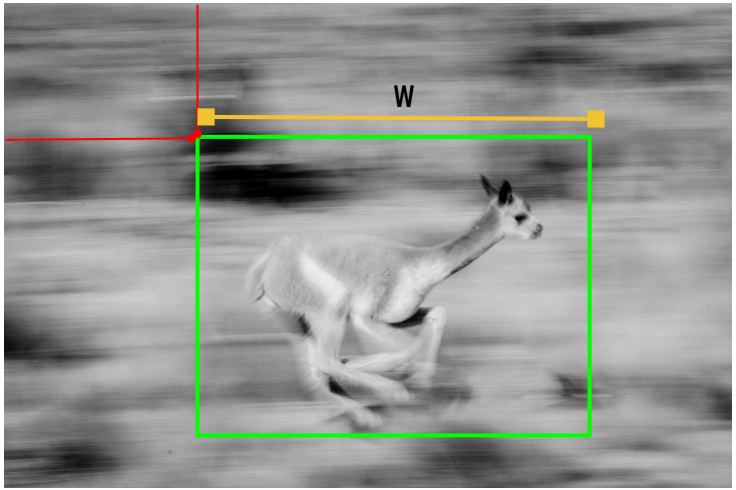
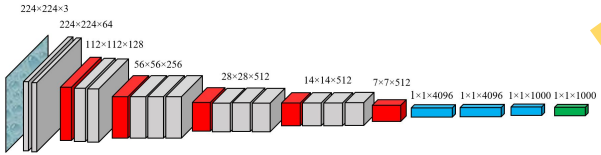
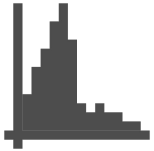


Foto: Pablo Fonseca



Clase



Localización

x	y	w	

Ahora tenemos una **red multiobjetivo**

Detección = Clasificación + Localización

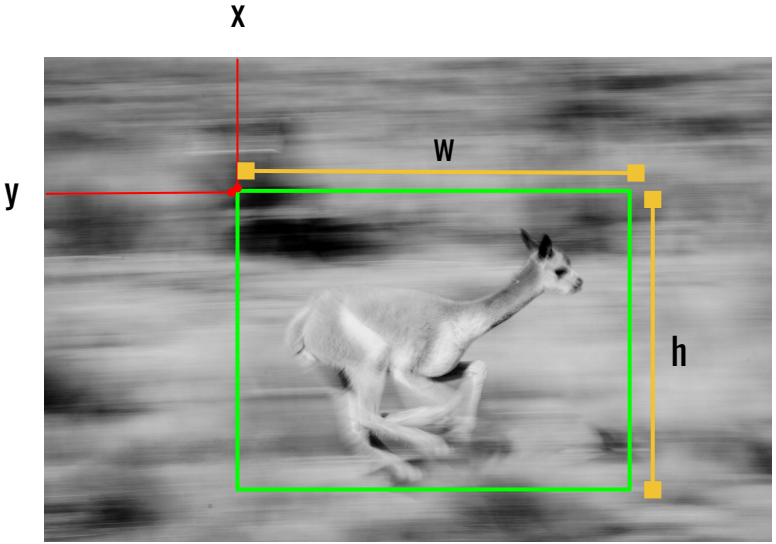
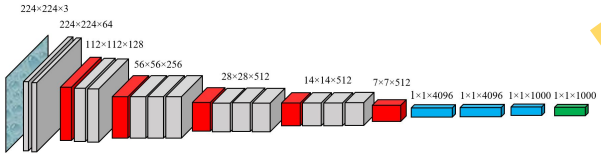
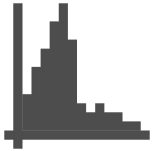


Foto: Pablo Fonseca



Clase



Localización

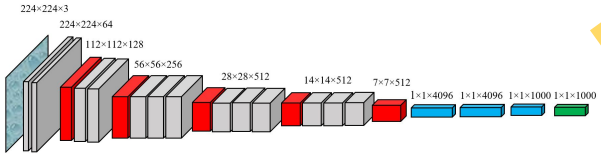
x	y	w	h

Ahora tenemos una **red multiobjetivo**

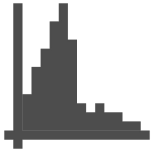
Detección = Clasificación + Localización



Foto: Pablo Fonseca



Clase



Localización

x	y	w	h

Ahora tenemos una **red multiobjetivo**

Puede ser entrenado por regresión

Detección = Clasificación + Localización

Solo funciona para un objeto dentro de la imagen

Detección = Clasificación + Localización



Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window

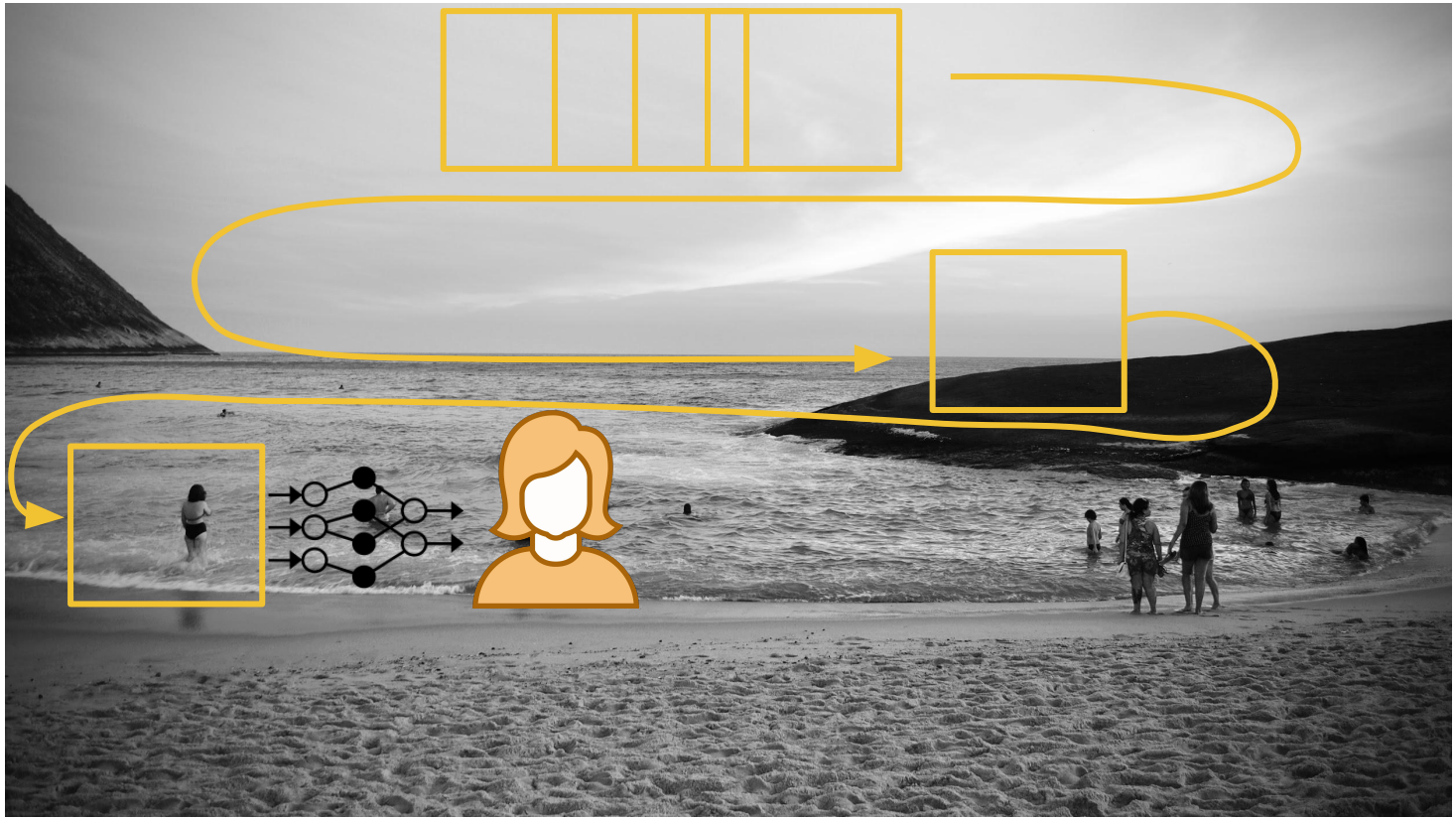


Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window

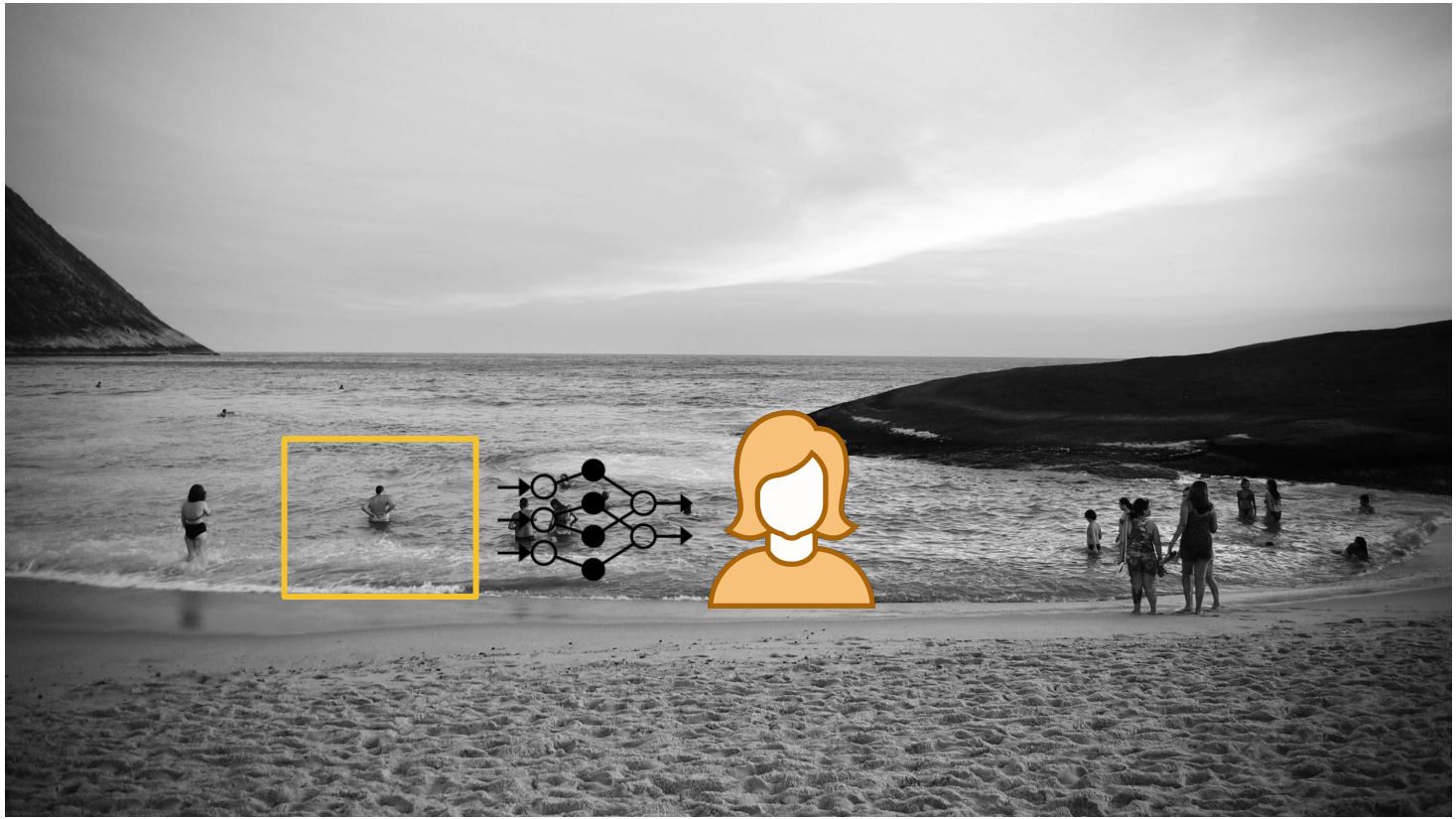


Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window

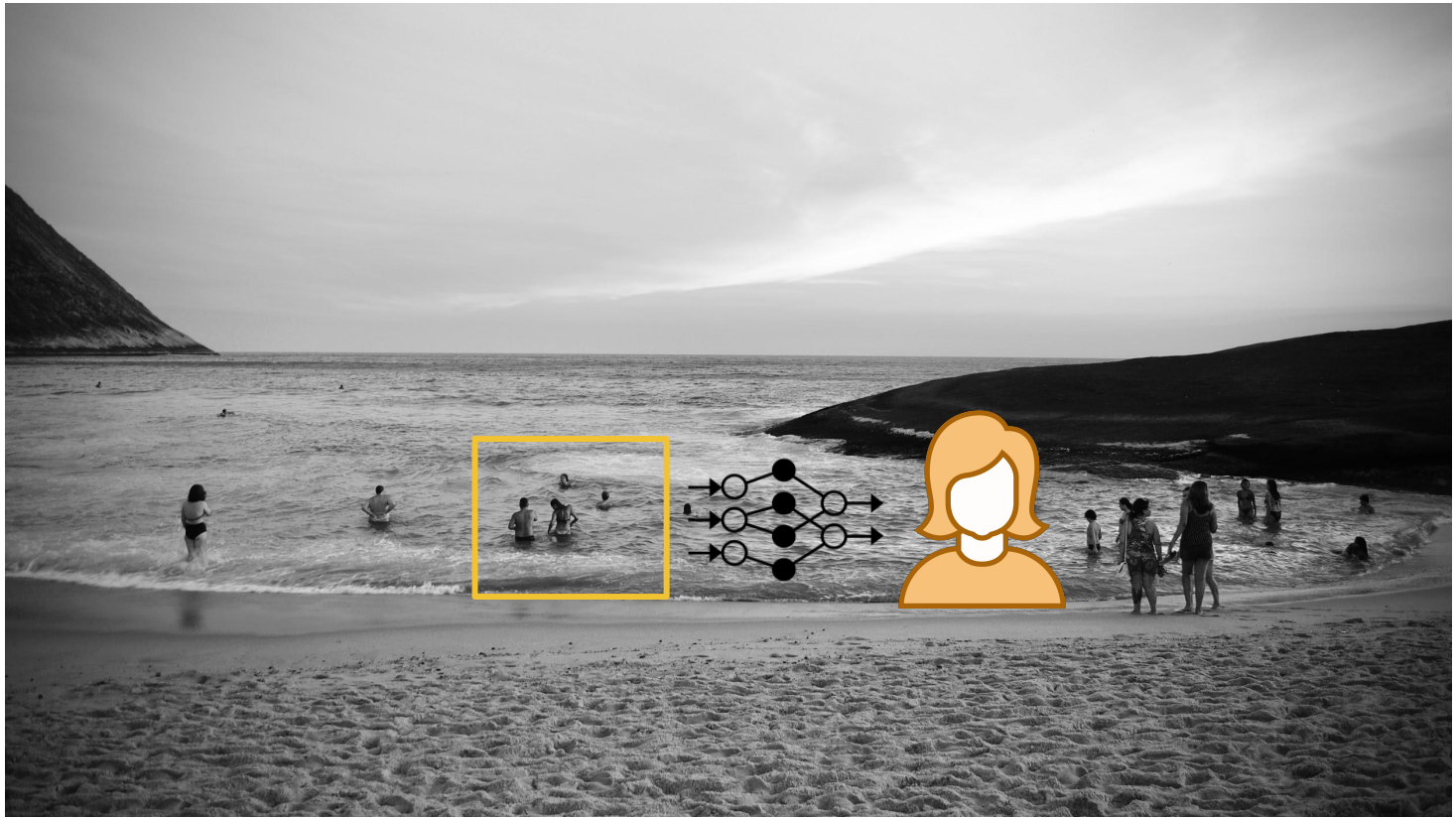


Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window



Foto: Pablo Fonseca

Detección = Clasificación + Sliding Window (más pequeños)



Cantidad de posibles posiciones si el **w** y **h** están definidos:

$(W-w+1)*(H-h+1)$

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

Detección = Clasificación + Sliding Window (más pequeños)



Cantidad de posibles posiciones si el **w** y **h** están definidos:

(W-w+1)*(H-h+1)

...pero si w y h pueden variar en todas las posiciones posibles

$$\sum_{h=1}^{H} \sum_{w=1}^{W} (W - w + 1)(H - h + 1)$$
$$= \frac{H(H+1)}{2} \frac{W(W+1)}{2}$$

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

Detección = Clasificación + Sliding Window (más pequeños)



Cantidad de posibles posiciones si el **w** y **h** están definidos:

(W-w+1)*(H-h+1)

...pero si w y h pueden variar en todas las posiciones posibles

$$\sum_{h=1}^{H}\sum_{w=1}^{W}(W-w+1)(H-h+1)$$
$$=\frac{H(H+1)}{2}\frac{W(W+1)}{2}$$

Para una imagen 800x600 equivaldría a llamar a la CNN 58 millones de veces

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

R-CNN

Podemos comenzar con un método de **region proposal**, que nos de alrededor de 2k propuestas



R-CNN

Podemos comenzar con un método de **region proposal**, que nos de alrededor de 2k propuestas



R-CNN

Haremos un resize de la imagen a 224, porque usaremos una CNN similar a la que imaginábamos al inicio de la clase



Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

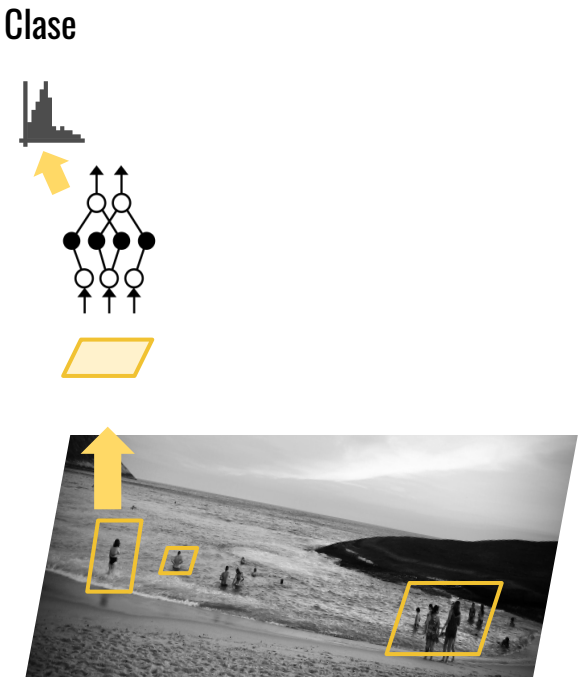
R-CNN



Haremos un resize de la imagen a 224, porque usaremos una CNN similar a la que imaginábamos al inicio de la clase

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

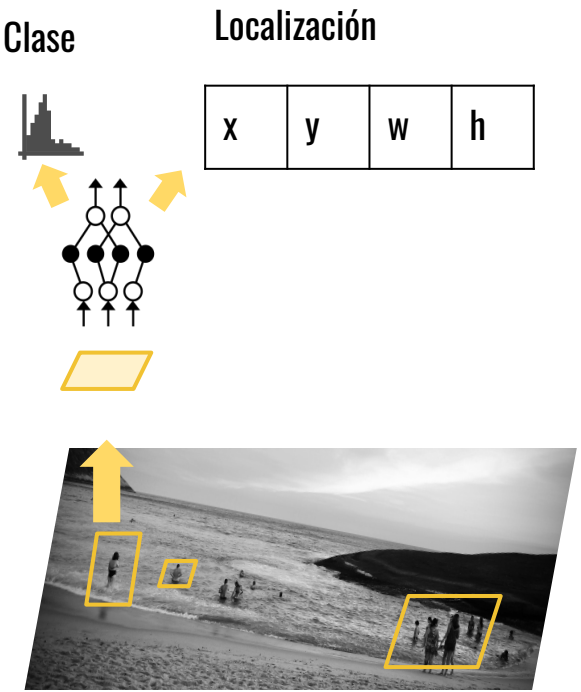
R-CNN



Haremos un resize de la imagen a 224, porque usaremos una CNN similar a la que imaginábamos al inicio de la clase… para predecir **clase**

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

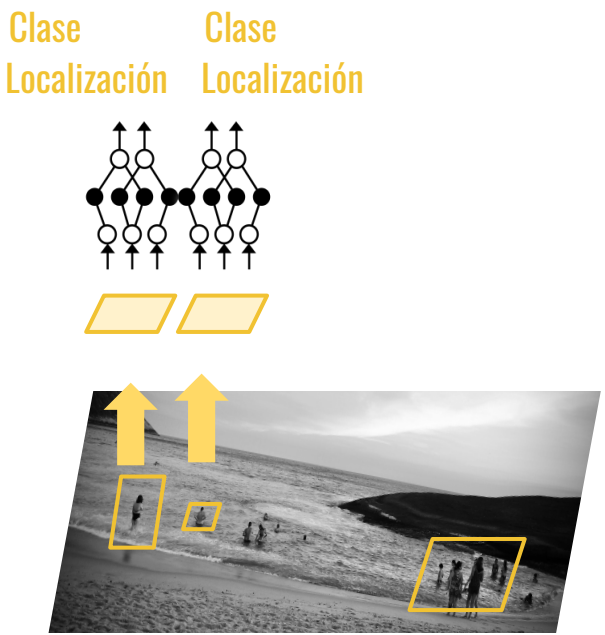
R-CNN



Haremos un resize de la imagen a 224, porque usaremos una CNN similar a la que imaginábamos al inicio de la clase… para predecir **clase y localización,** de modo que corrija la propuesta original

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

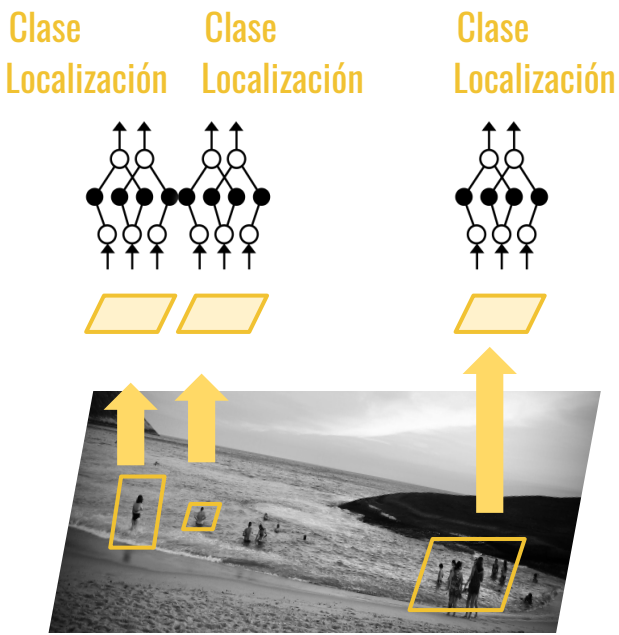
R-CNN



Haremos lo mismo para todas las propuestas

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

R-CNN



Haremos lo mismo para todas las propuestas, **compartiendo pesos**

Adaptado de: https://www.youtube.com/watch?v=TB-fdISzpHQ&list=PL5-TkQAfAZFbzxjBHtzdVCWE0Zbhomg7r

Selective Search

Uijlings, J. R., Van De Sande, K. E., Gevers, T., & Smeulders, A. W. (2013). **Selective search for object recognition**. *International journal of computer vision*, *104*(2), 154-171.



Figure 2: Two examples of our selective search showing the necessity of different scales. On the left we find many objects at different scales. On the right we necessarily find the objects at different scales as the girl is contained by the tv.

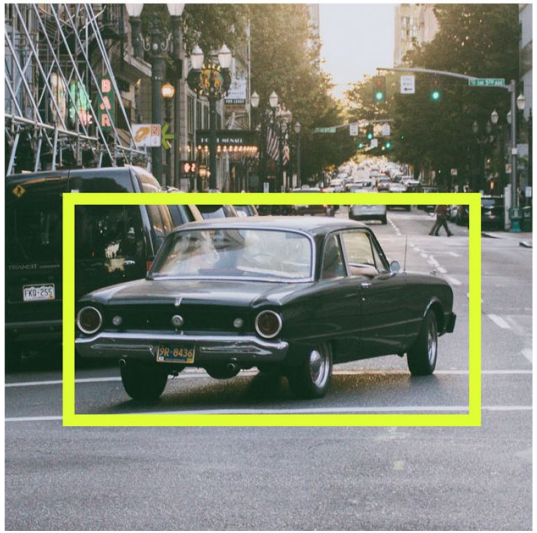
Non Max Suppression (técnica de posprocesamiento)

Before non-max suppression



Non-Max Suppression

After non-max suppression



https://appsilon.com/object-detection-yolo-algorithm/

Non Max Suppression (técnica de posprocesamiento)

Seleccionar la caja con mayor probabilidad

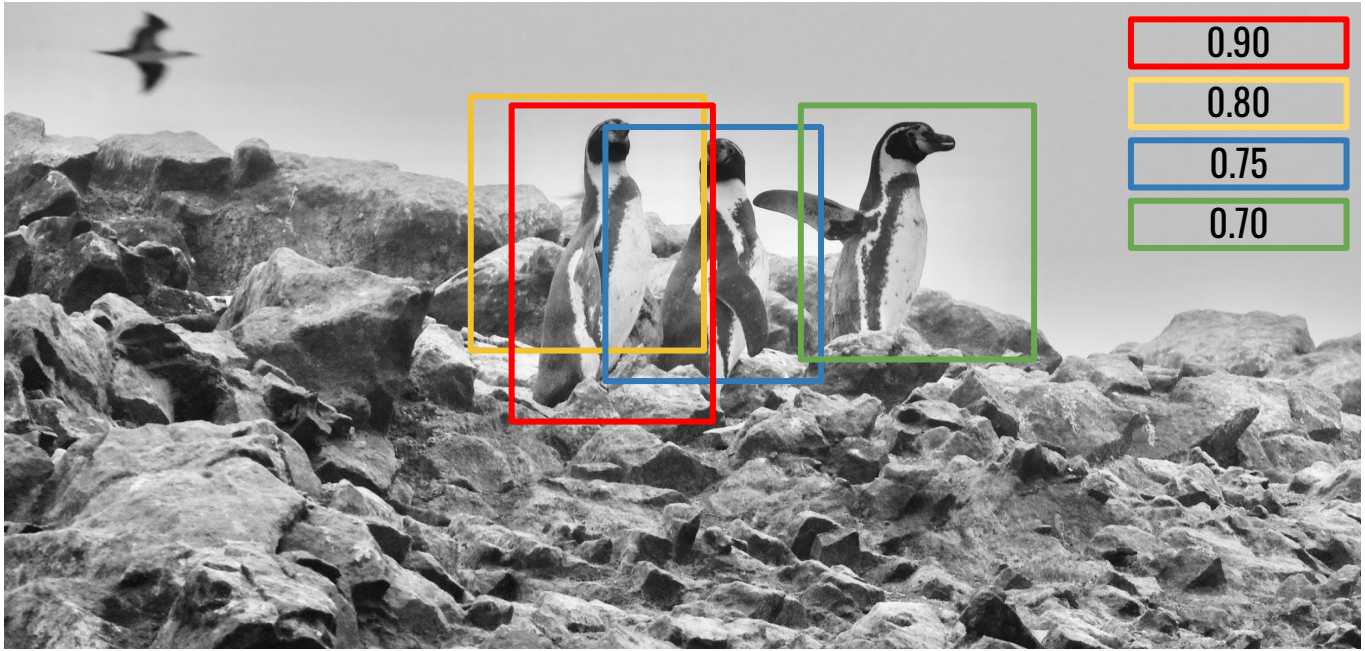


Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

Eliminar las cajas que tengan un IoU mayor que el threshold, por ejemplo 0.7

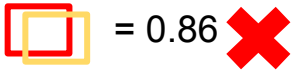
= 0.86



Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

Eliminar las cajas que tengan un IoU mayor que el threshold, por ejemplo 0.7



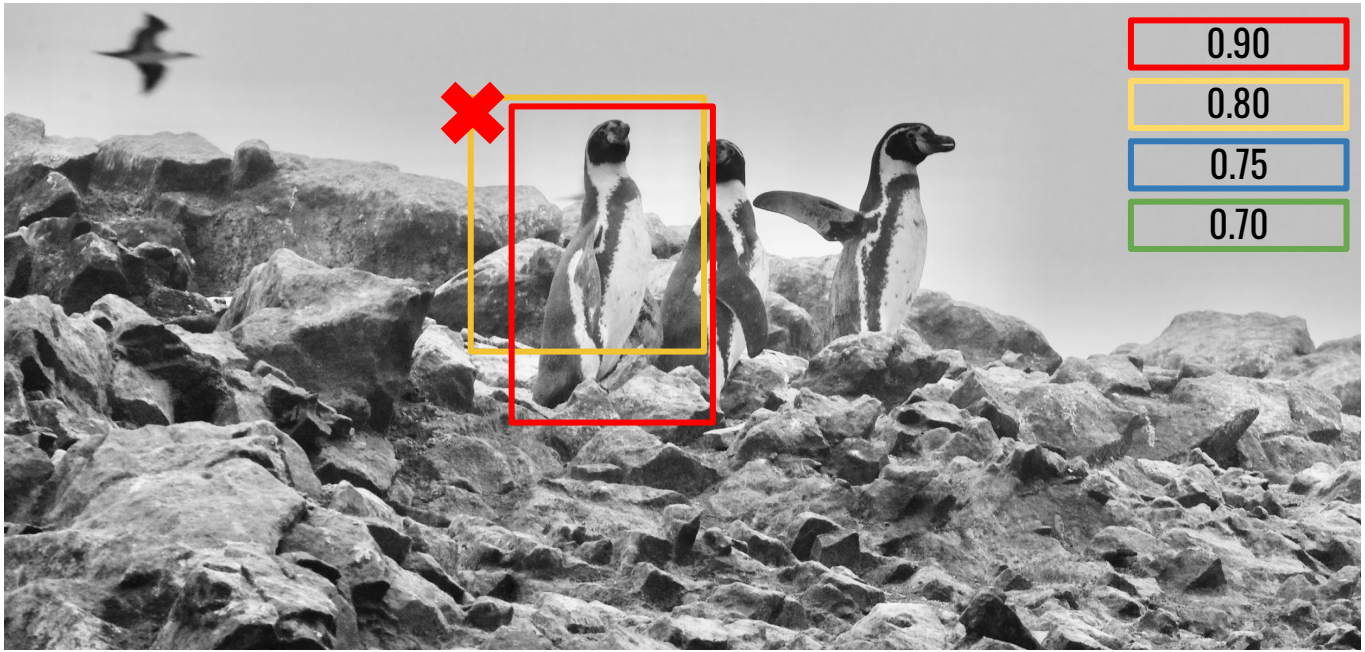


Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

Eliminar las cajas que tengan un IoU mayor que el threshold, por ejemplo 0.7



= 0.5

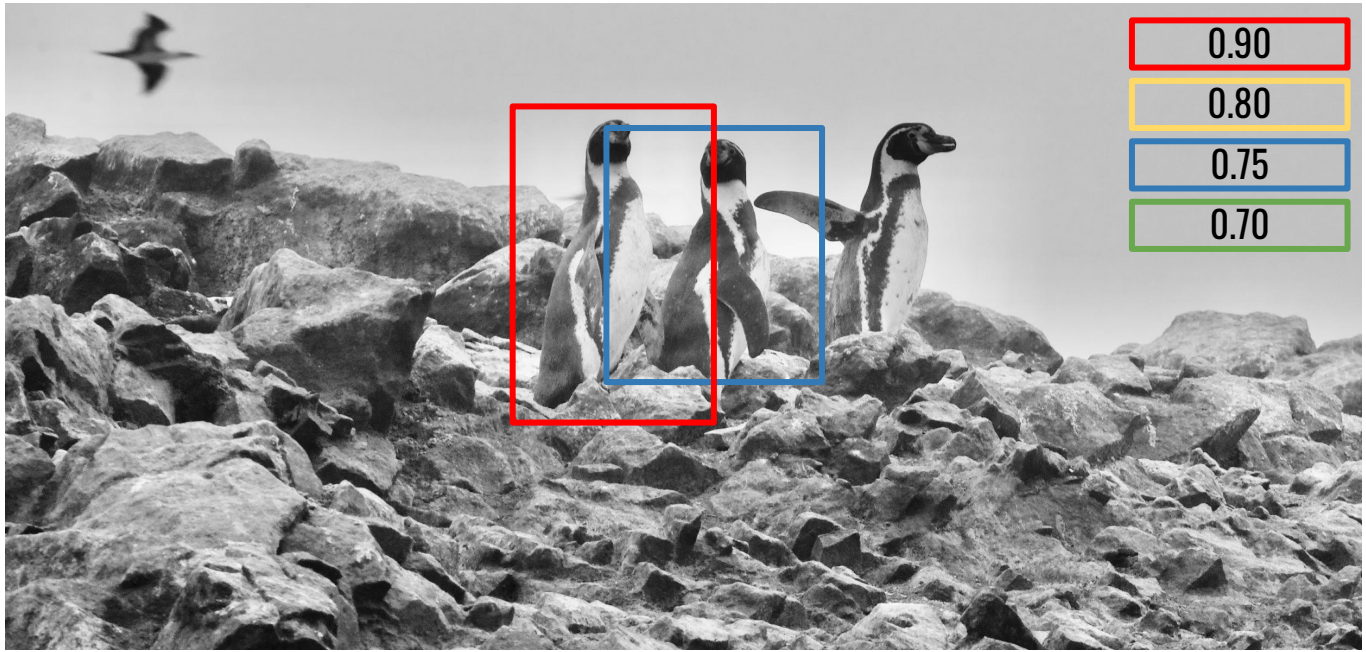
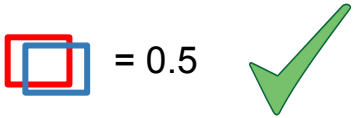


Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

Eliminar las cajas que tengan un IoU mayor que el threshold, por ejemplo 0.7



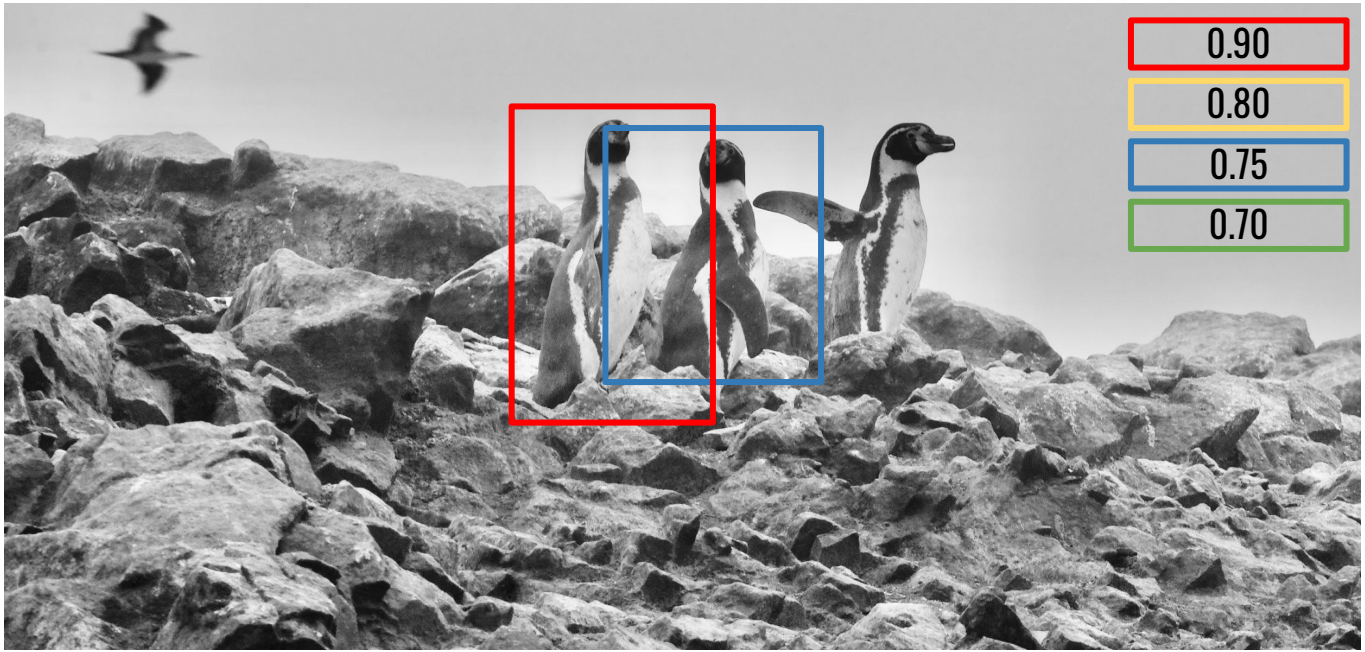


Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

Eliminar las cajas que tengan un IoU mayor que el threshold, por ejemplo 0.7

 = 0.00



Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

Eliminar las cajas que tengan un IoU mayor que el threshold, por ejemplo 0.7

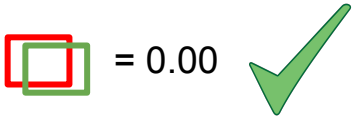




Foto: Pablo Fonseca

Non Max Suppression (técnica de posprocesamiento)

...finalmente

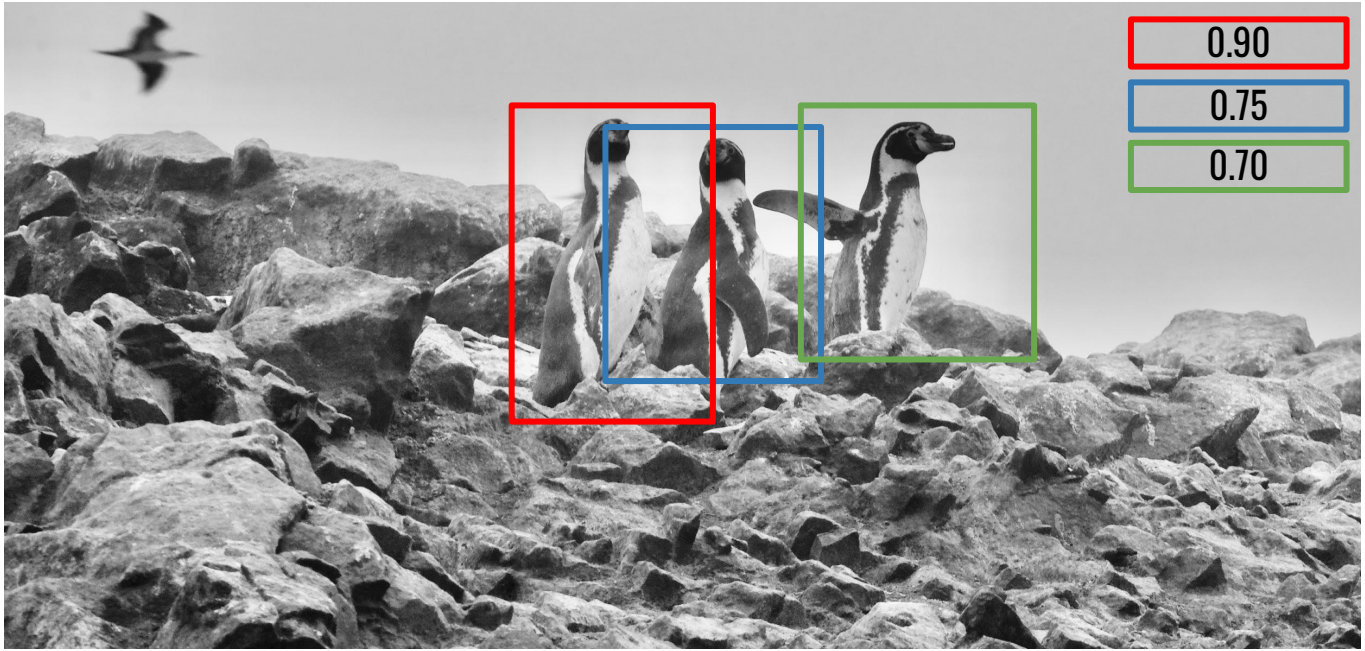
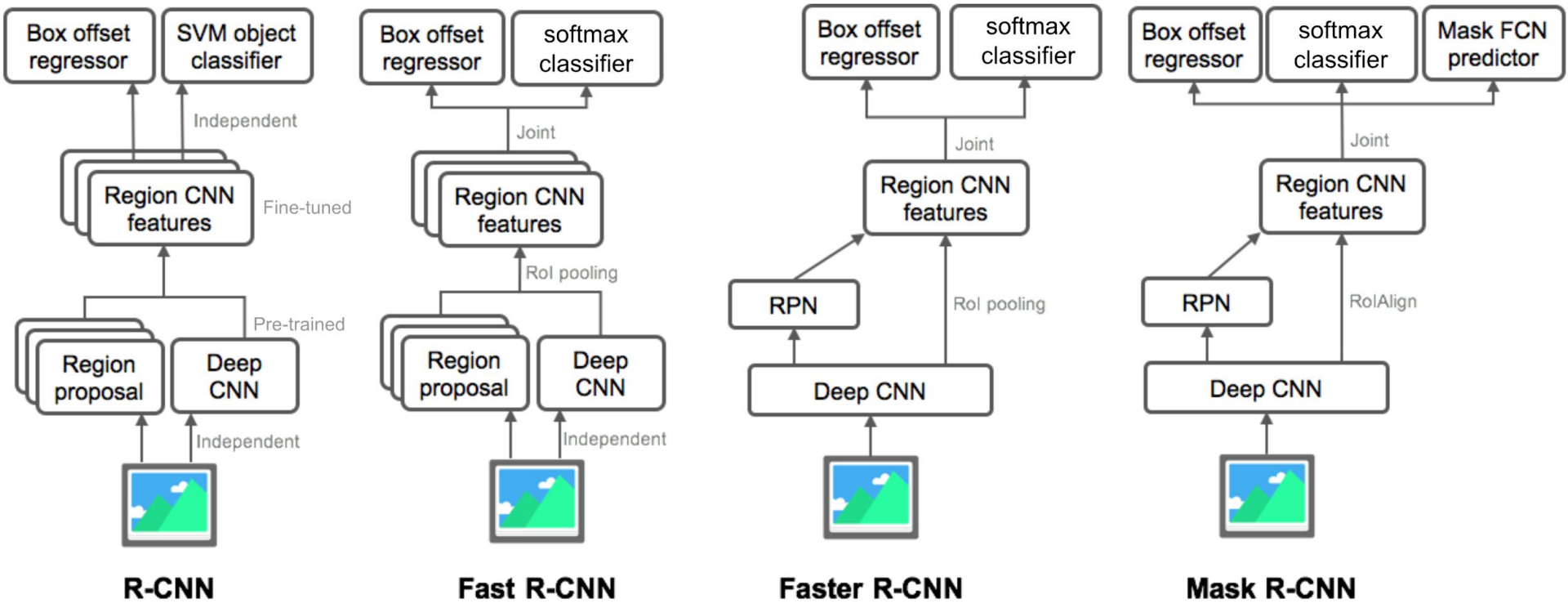


Foto: Pablo Fonseca

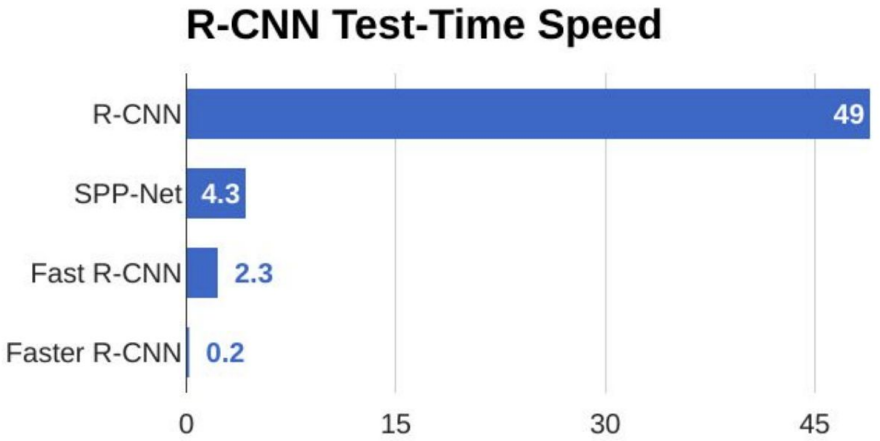
Familia R-CNN



https://lilianweng.github.io/lil-log/2017/12/31/object-recognition-for-dummies-part-3.html

Limitaciones de R-CNN

- No puede funcionar en tiempo real, por que es muy costoso hacer 2000 pasadas a la red convolucional
- Selective Search no tiene componente de aprendizaje



https://towardsdatascience.com/r-cnn-fast-r-cnn-faster-r-cnn-yolo-object-detection-algorithms-36d53571365e

Fast R-CNN

- Se sigue proponiendo las regiones de interés por el método de **Selective Search**
- Se reaprovechan los cálculos!
- La limitación sigue presente: Selective Search no tiene componente de aprendizaje

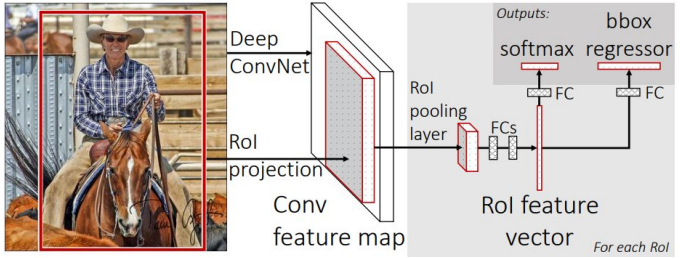
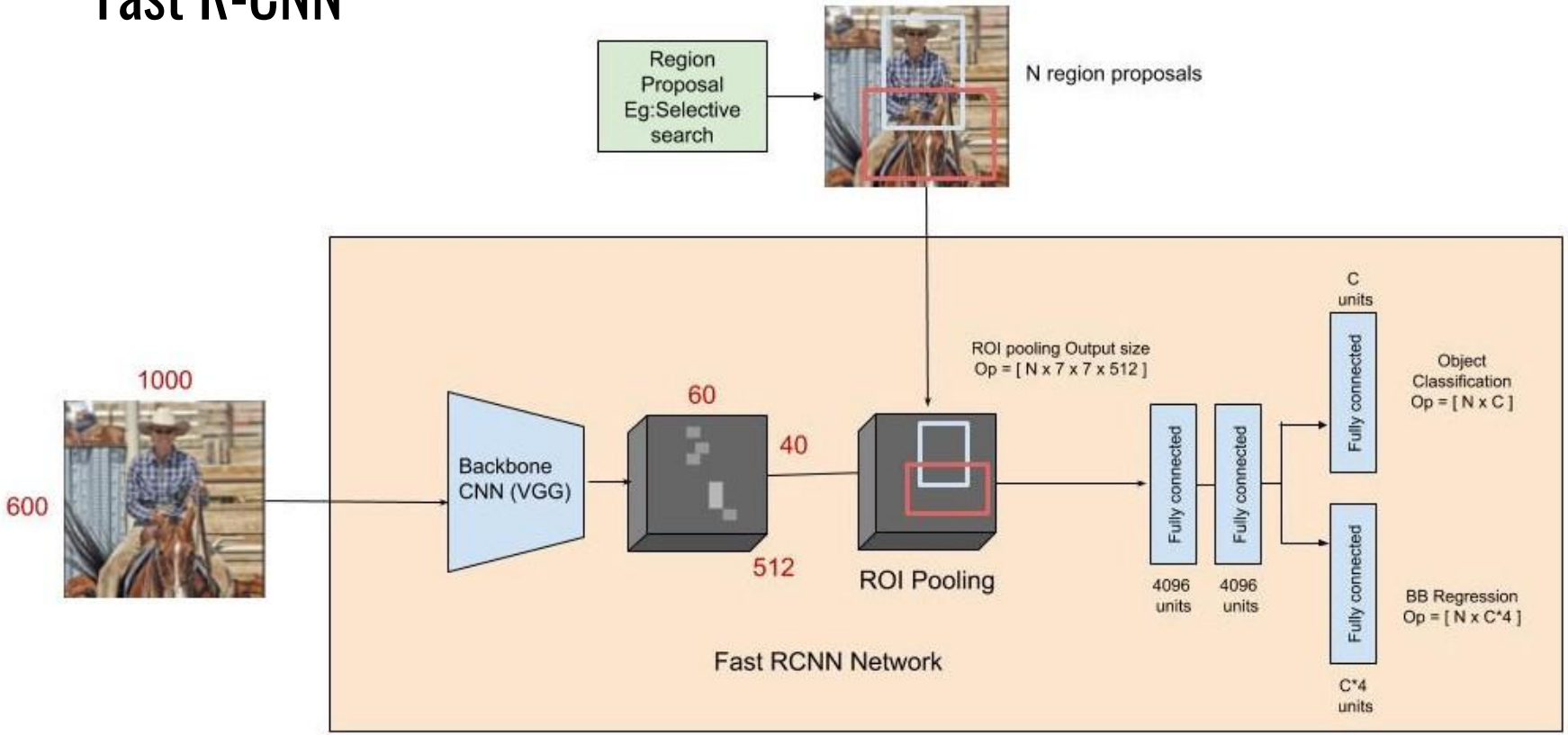


Figure 1. Fast R-CNN architecture. An input image and multiple regions of interest (RoIs) are input into a fully convolutional network. Each RoI is pooled into a fixed-size feature map and then mapped to a feature vector by fully connected layers (FCs). The network has two output vectors per RoI: softmax probabilities and per-class bounding-box regression offsets. The architecture is trained end-to-end with a multi-task loss.

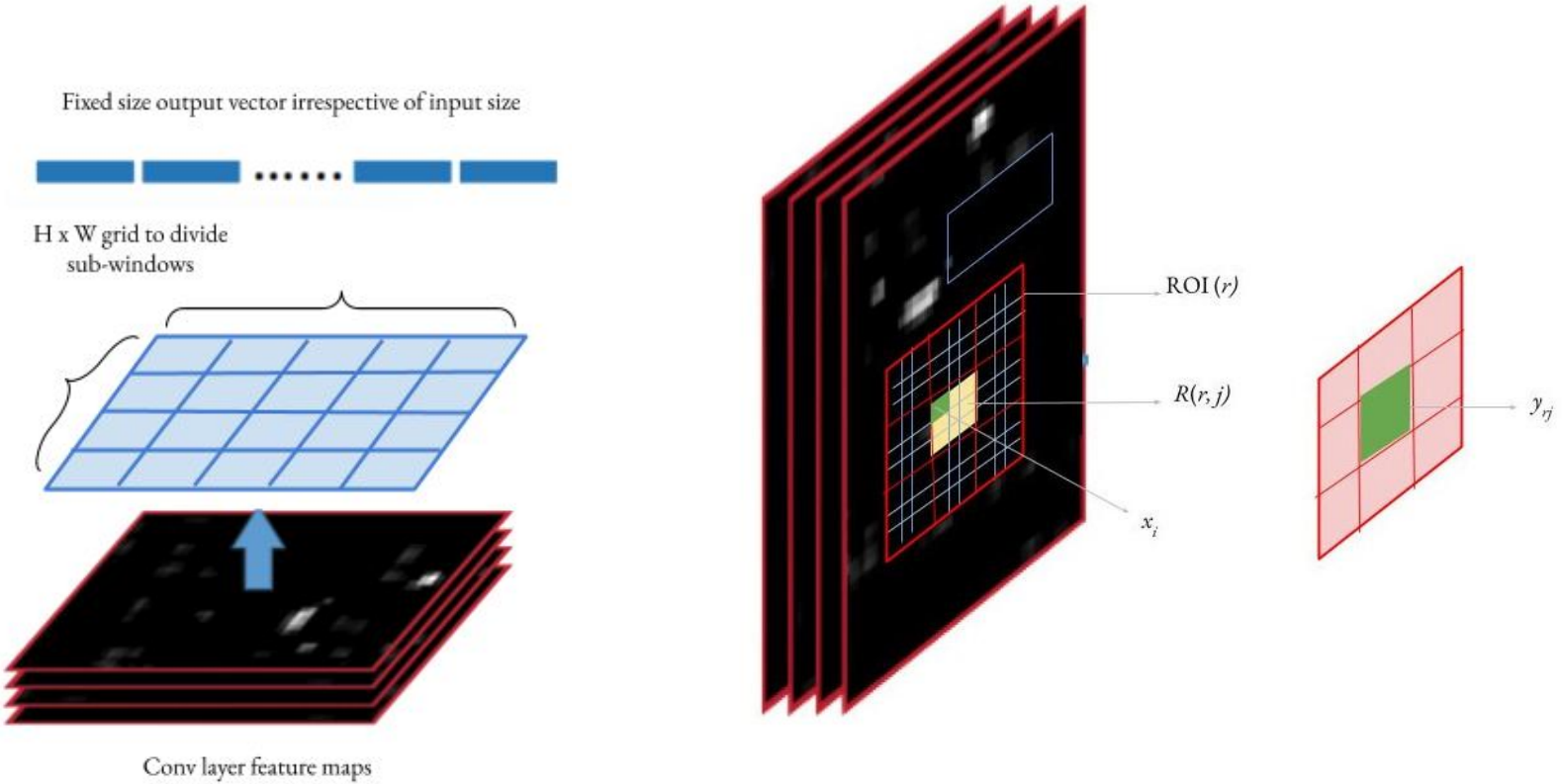
Ren, S., He, K., Girshick, R., & Sun, J. (2016). Faster R-CNN: towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, *39*(6), 1137-1149.

Fast R-CNN



https://towardsdatascience.com/fast-r-cnn-for-object-detection-a-technical-summary-a0ff94faa022

Fast R-CNN



https://towardsdatascience.com/fast-r-cnn-for-object-detection-a-technical-summary-a0ff94faa022

Faster R-CNN

- R-CNN y Fast R-CNN usan selective search para las propuestas de regiones de interés

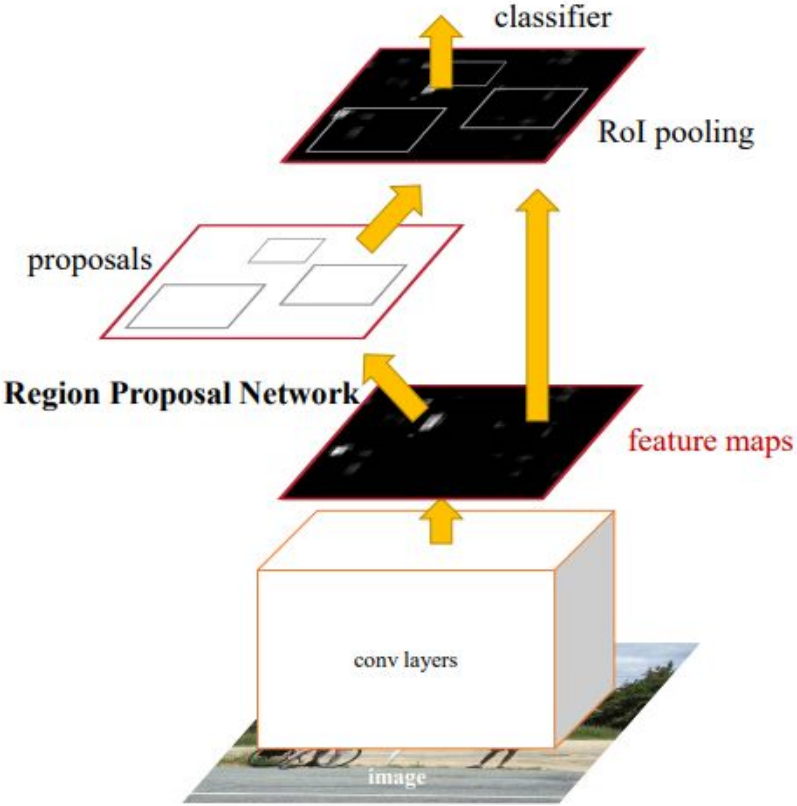


Figure 2: Faster R-CNN is a single, unified network for object detection. The RPN module serves as the 'attention' of this unified network.

Ren, S., He, K., Girshick, R., & Sun, J. (2016). Faster R-CNN: towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, *39*(6), 1137-1149.

Region Proposal Network

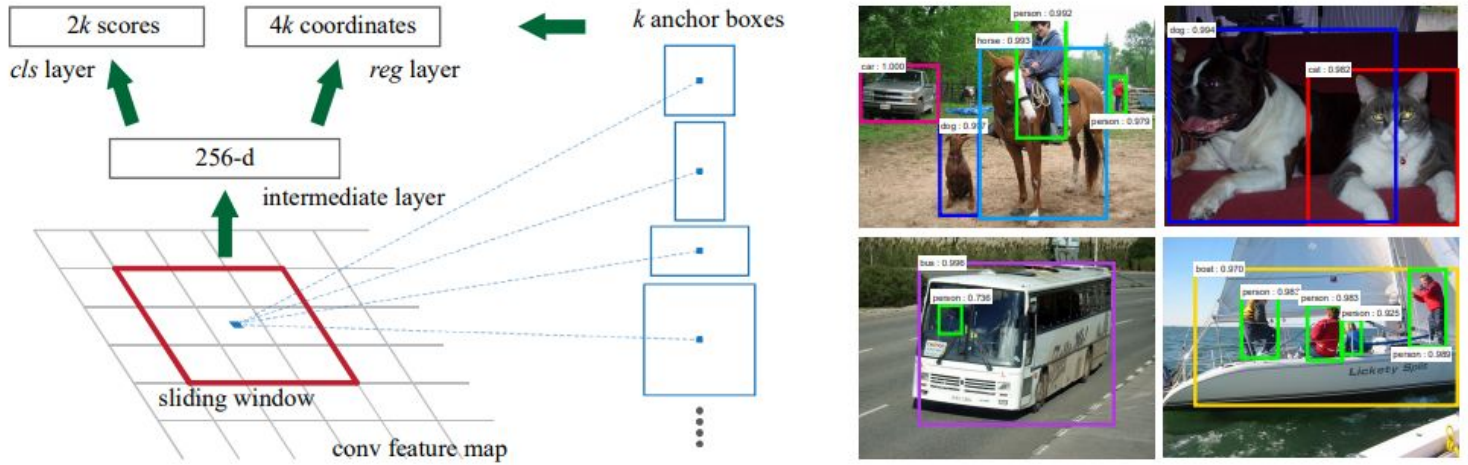
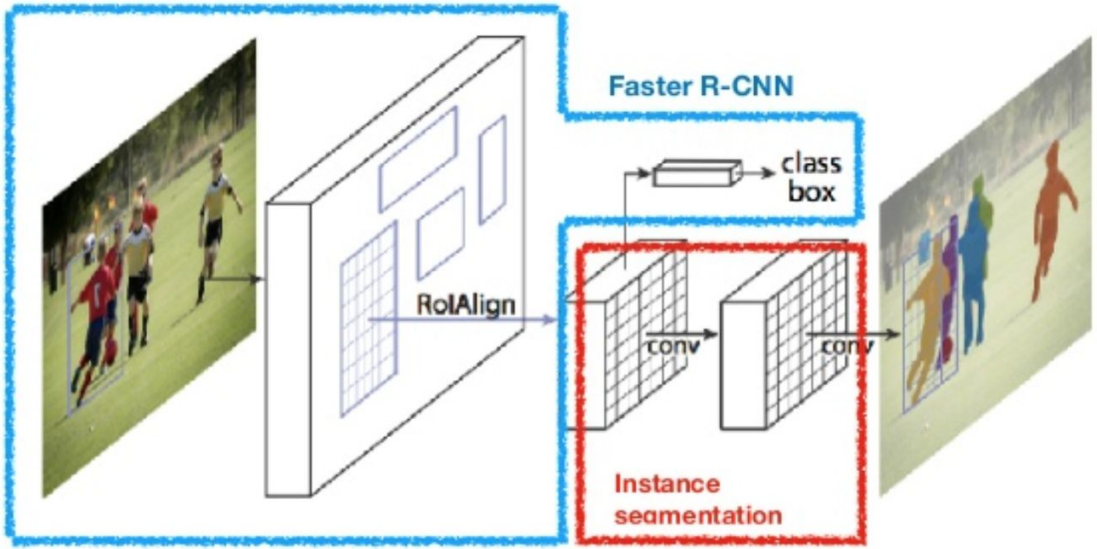


Figure 3: **Left**: Region Proposal Network (RPN). **Right**: Example detections using RPN proposals on PASCAI VOC 2007 test. Our method detects objects in a wide range of scales and aspect ratios.

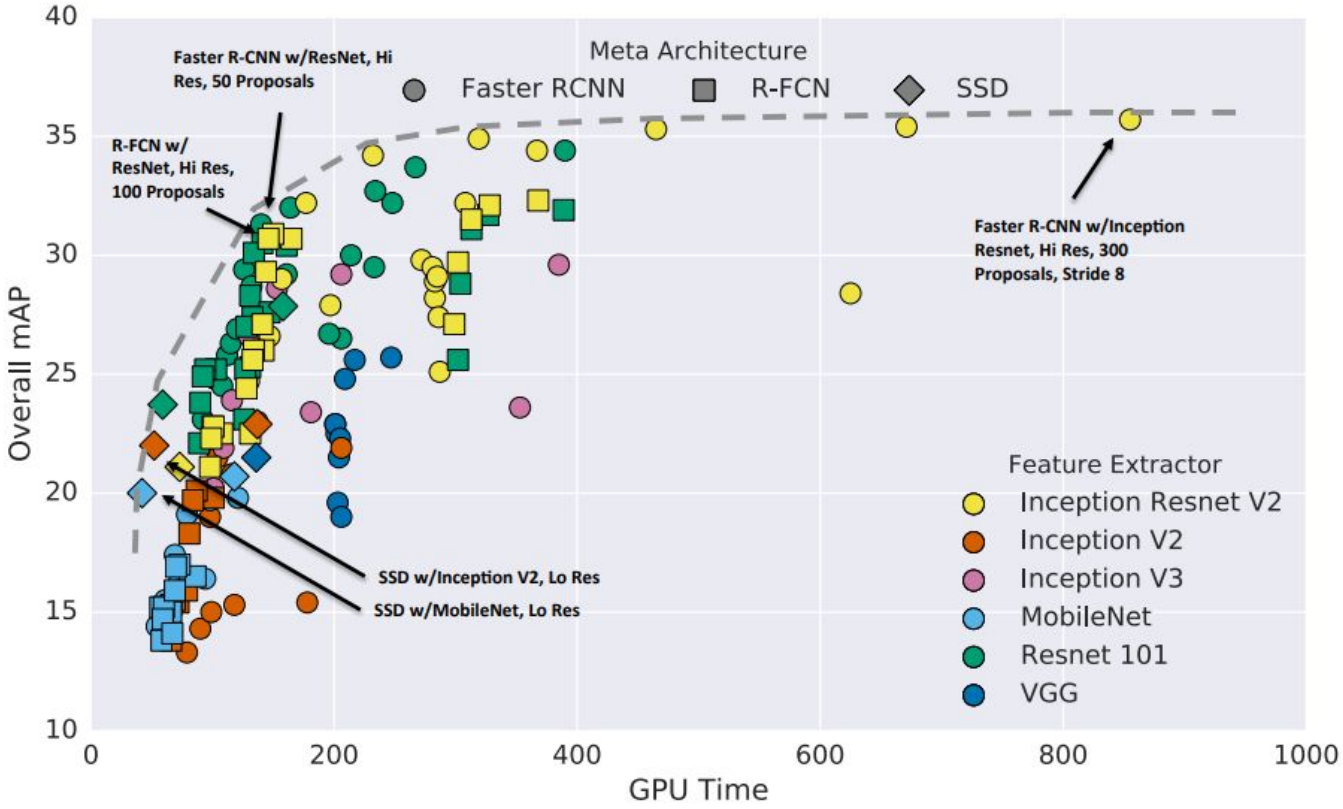
Ren, S., He, K., Girshick, R., & Sun, J. (2016). Faster R-CNN: towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*, *39*(6), 1137-1149.

Mask R-CNN



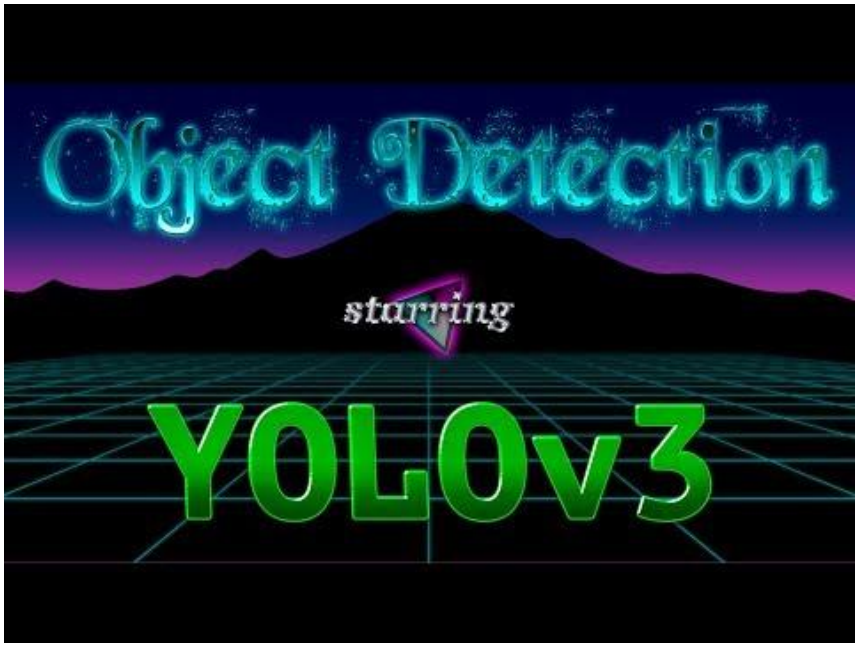
He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 2961-2969).

Object Detectors



Huang, J., Rathod, V., Sun, C., Zhu, M., Korattikara, A., Fathi, A., ... & Murphy, K. (2017). **Speed/accuracy trade-offs for modern convolutional object detectors**. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7310-7311).

You Only Look Once

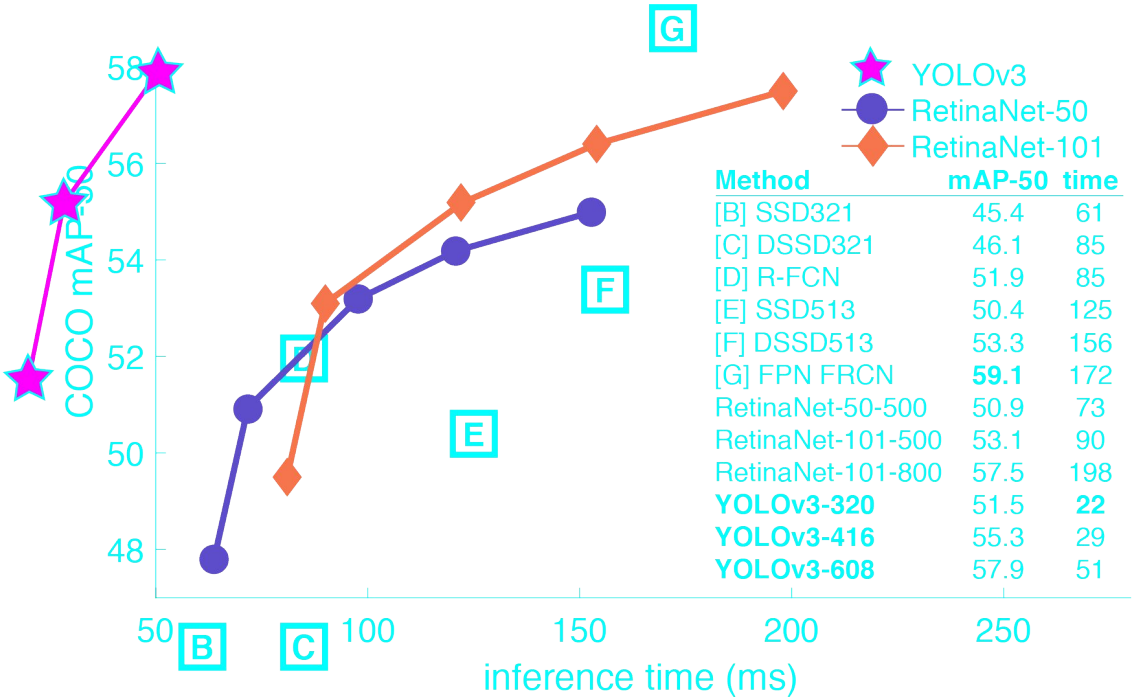




https://pjreddie.com/darknet/yolo/

https://www.youtube.com/watch?v=QOC6vgnWnYo

Presentación



Method	mAP-50	time
[B] SSD321	45.4	61
[C] DSSD321	46.1	85
[D] R-FCN	51.9	85
[E] SSD513	50.4	125
[F] DSSD513	53.3	156
[G] FPN FRCN	**59.1**	172
RetinaNet-50-500	50.9	73
RetinaNet-101-500	53.1	90
RetinaNet-101-800	57.5	198
YOLOv3-320	51.5	**22**
YOLOv3-416	55.3	29
YOLOv3-608	57.9	51

https://pjreddie.com/darknet/yolo/

YOLO Architecture

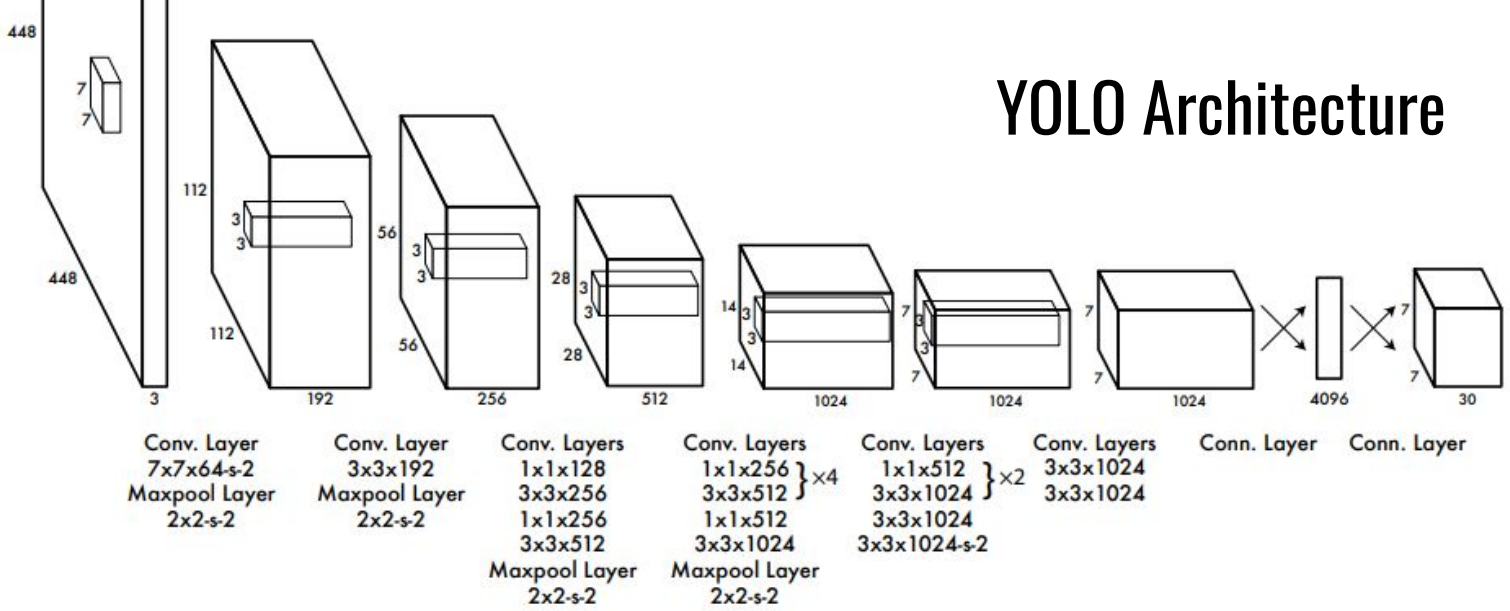
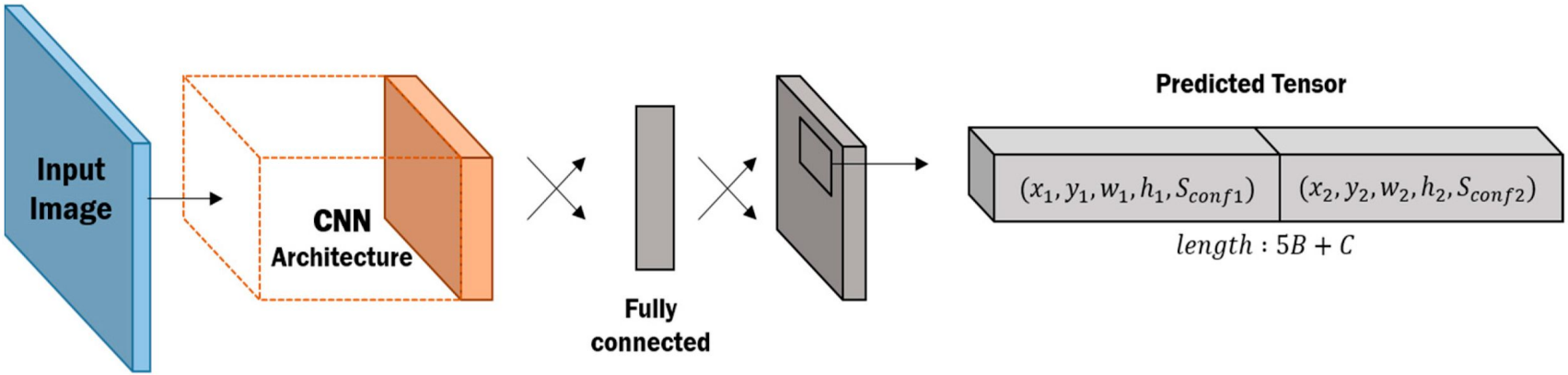


Figure 3: The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.

https://arxiv.org/pdf/1506.02640.pdf

YOLO Architecture



https://www.mdpi.com/2076-3417/10/2/612/htm

Referencias

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